SIEMENS

MOTION CONTROL

SINUMERIK ONE

Equipment for Machine Tools
Function overview

siemens.com/sinumerik

Catalog
NC 63

Edition
January
2026



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Related catalogs

Motion Control NC 82
SINUMERIK 828
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PDF (E86060-K4482-A101-B2-7600)



Motion Control Drives D 21.4
SINAMICS S120, SINAMICS S220 and SIMOTICS

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Motion Control Drives D 32
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eCatalog NC 63

For the complete eCatalog of SINUMERIK ONE with all major system components, please refer to SiePortal



The SINUMERIK ONE glossary is available in SiePortal at:

sieportal.siemens.com/en-ww/products-services/10356138

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SINUMERIK ONE

Equipment for Machine Tools

Motion Control



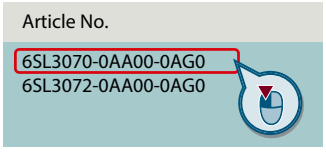
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Introduction

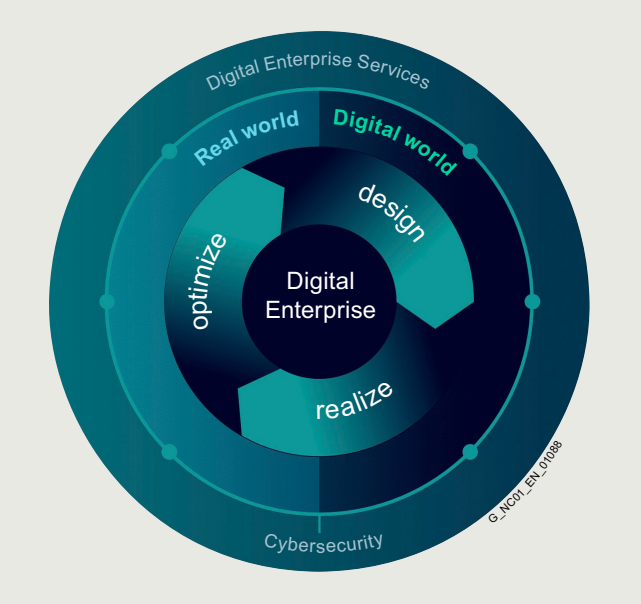


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Overview

Perfectly integrated virtual and real CNC systems



Siemens is introducing its first *digital native CNC*, SINUMERIK ONE, and thus driving forward the digital transformation of the machine tool industry.

SINUMERIK ONE is the pioneering CNC system for highly productive machine tools. But it is also much more than a powerful hardware innovation.
Using its digital twin, it helps to simulate and test work processes completely virtually.

With SINUMERIK ONE virtual processes become the guiding factor for real-world management. The digital twin lowers the non-productive times of real machine tools to a minimum. Thanks to the seamless interplay of virtual and real worlds, SINUMERIK ONE can be used to easily implement innovations and execute new business models. SINUMERIK ONE therefore sets new standards in terms of digitalization, productivity and performance. It is the pioneering CNC for machine tools in increasingly digital production processes.

Machine manufacturers and machine operators alike benefit from the end-to-end digital twin of products, production and performance. SINUMERIK ONE supports consistent machine development based on the digital twin.

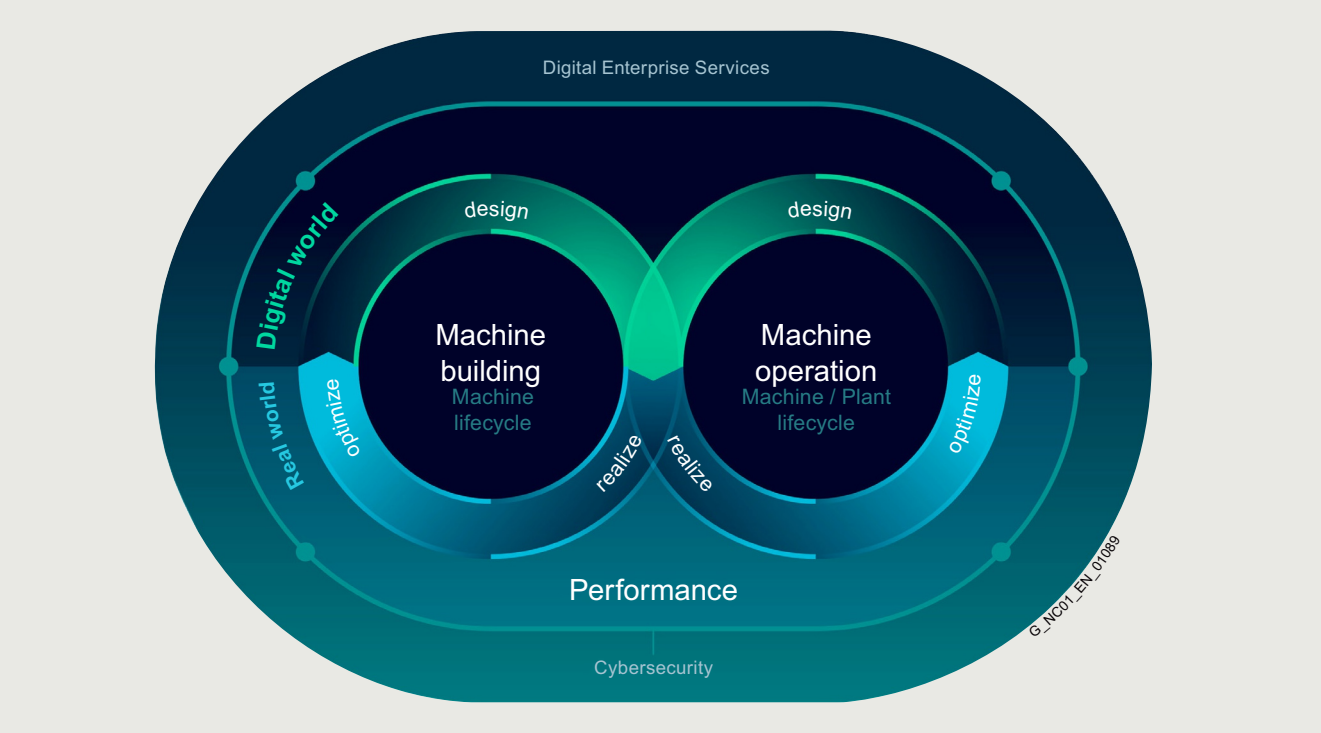
For machine manufacturers, closed-loop engineering has a significant advantage: The mechatronic development approach parallelizes mechanics, electronics and automation and combines these disciplines in a digital model.
The digital model then becomes the guiding factor in the development process. The digital twin is based on an end-to-end and comprehensive engineering tool landscape.
As a result, product development and hence time-to-market and commissioning times can be significantly reduced. Based on the virtual machine model, entirely new possibilities open up for machine manufacturers and machine operators alike. Accordingly, machine concepts and functions can be discussed in a targeted way even before the real hardware is available.

Thanks to realistic machining simulation, SINUMERIK ONE enables machine users to simulate the programming of workpieces and the setup and operation of machines entirely in a virtual environment. Personnel training can also take place on the digital twin, away from the real machine. The software and hardware innovations of SINUMERIK ONE and the possibility to create an end-to-end digital twin for products and production significantly accelerates machining processes.

The CNC system with integrated SIMATIC S7-1500F PLC is available as a cabinet-based and a panel-based version. SINUMERIK ONE can be fully integrated into the TIA Portal and opens up the possibilities of a highly efficient engineering framework for the machine manufacturer. The new control system also relies on Safety Integrated as a uniform Siemens industrial standard in the area of safety. SINUMERIK ONE implements the multi-stage defense-in-depth concept, the IT security already envisaged during the conception of the product.
Thanks to Create MyVirtual Machine and Run MyVirtual Machine, SINUMERIK ONE can be equipped with software for the digital twins of engineering and machining. Its end-to-end concept of the digital twin, its powerful hardware and integrated IT security make the SINUMERIK ONE the pioneering CNC that is driving the digital transformation of the machine tool industry.

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Benefits



The new SINUMERIK ONE is specifically designed for smart manufacturing.

- The digital twin is an integral part of the CNC system; virtual and real CNC merge and complement each other
- Significant reduction in product development and market launch times thanks to the **digital first** strategy
- Improved networking and data communication capabilities
- Significant reduction in the duration of real commissioning thanks to virtual commissioning preparation
- Significantly increased CNC performance
- Significantly shorter downtimes and complete integration into the TIA Portal thanks to the integrated SIMATIC S7-1500F PLC

This means that the SINUMERIK ONE can be seamlessly integrated into digital solutions and workflows.

More information

More information on SINUMERIK ONE is available at

www.siemens.com/sinumerik-one

You will find the complete eCatalog NC 63 of SINUMERIK ONE with all major system components in SiePortal in the section: Automation technology → Automation systems → SINUMERIK CNC automation systems → SINUMERIK ONE

sieportal.siemens.com

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Function overview



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Important ordering information:
Products with a product ID can be exclusively ordered using the product ID.
More information is available through your Siemens contact and on the internet at: www.dex.siemens.com/?selected=industrialsoftware

Click on the note See SiePortal in the catalog PDF to obtain more detailed information about each product.

The most important functions of the SINUMERIK ONE are listed in the function overview.

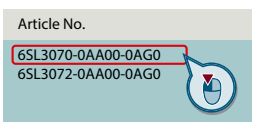
This gives you quick and selective access to the individual functions.

Refer to → SiePortal for current updates of this function overview.

The information in the function overview of the SINUMERIK controls is based on the following software version:

Control	SW version
SINUMERIK ONE	6.15
SINUMERIK ONE	6.25

Click on an Article No. in the catalog PDF to call it up in SiePortal and to obtain all the information.



Or directly on the internet, e.g.
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Control structure and configuration

Description ✓ Basic version I O Option I – Not available		Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Control structure and configuration								
SINUMERIK ONE compact for installation in the panel:								
• PPU 1740-1500 with PLC S7-1500F – 15" blackline plus		**6FC5317-4AA00-1CA0**		O	–	–	–	O
• PPU 1740-1900 with PLC S7-1500F – 19" blackline plus		**6FC5317-4AA00-1DA0**		O	–	–	–	O
SINUMERIK ONE For installation in the control cabinet:								
• NCU 1740 with PLC S7-1500F		**6FC5317-4AA00-0AA0**		–	O	–	–	O
• NCU 1750 with PLC S7-1500F		**6FC5317-5AA00-0AA1**		–	–	O	–	O
• NCU 1760 with PLC S7-1500F		**6FC5317-6AA00-0AA0**		–	–	–	O	O
Software for SINUMERIK ONE NCU/PPU:								
• CNC software 31-3 export version, 6 languages (en, de, fr, it, es, zh)								
- on SD card, single license, current SW version (CNC SW 6.25)		**6FC5852-1YG00-0YA0**		O	O	O	O	O
- on SD card, without license, current SW version (CNC SW 6.25)		**6FC5852-1YG00-0YA8**		O	O	O	O	O
- on SD card, single license, SW version 6.25 SP1	NEW	**6FC5852-1YG25-1YA0**		O	O	O	O	O
- on SD card, without license, SW version 6.25 SP1	NEW	**6FC5852-1YG25-1YA8**		O	O	O	O	O
- on SD card, single license, SW version 6.15 SP5		**6FC5852-1YG15-5YA0**		O	O	O	O	O
- on SD card, without license, SW version 6.15 SP5		**6FC5852-1YG15-5YA8**		O	O	O	O	O
- without data storage medium, single license, current SW version (CNC SW 6.25)		**6FC5852-1YF00-0YB0**		O	O	O	O	O
• CNC software 31-3 version with full functional scope, 6 languages (en, de, fr, it, es, zh)								
- on SD card, single license, current SW version (CNC SW 6.25)[1]		**6FC5852-1XG00-0YA0**		O	O	O	O	–
- on SD card, without license, current SW version (CNC SW 6.25)[1]		**6FC5852-1XG00-0YA8**		O	O	O	O	–
- on SD card, single license, SW version 6.25 SP1[1]	NEW	**6FC5852-1XG25-1YA0**		O	O	O	O	–
- on SD card, without license, SW version 6.25 SP1[1]	NEW	**6FC5852-1XG25-1YA8**		O	O	O	O	–
- on SD card, single license, SW version 6.15 SP5[1]		**6FC5852-1XG15-5YA0**		O	O	O	O	–
- on SD card, without license, SW version 6.15 SP5[1]		**6FC5852-1XG15-5YA8**		O	O	O	O	–
- without data storage medium, single license, current SW version (CNC SW 6.25)[1]		**6FC5852-1XF00-0YB0**		O	O	O	O	–
Software for SINUMERIK ONE NCU/PPU, SIMATIC IPC for SINUMERIK and PC:								
• CNC software export version toolbox and SINUMERIK Operate, on DVD-ROM, without license, update when ordering SW version 6.25 SP1	NEW	**6FC5852-1YC25-1YA8**		O	O	O	O	O
• CNC software export version toolbox and SINUMERIK Operate, on DVD-ROM, without license, update when ordering SW version 6.15 SP5		**6FC5852-1YC15-5YA8**		O	O	O	O	O
• CNC software version with full functional scope toolbox and SINUMERIK Operate, on DVD-ROM, without license, update when ordering SW version 6.25 SP1[1]	NEW	**6FC5852-1XC25-1YA8**		O	O	O	O	–
• CNC software version with full functional scope toolbox and SINUMERIK Operate, on DVD-ROM, without license, update when ordering SW version 6.15 SP5[1]		**6FC5852-1XC15-5YA8**		O	O	O	O	–

[1] Export approval required.

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Control structure and configuration (continued)							
Axes/spindles or positioning axes/auxiliary spindle CNC software			✓3	✓3	✓3	✓3	✓
• Maximum configuration of NC axes/spindles			12	12	31	31	✓
• Configuration of axes/spindles per channel Requirement: CNC software ≥ SW version 6.24	NEW		12	12	31	31	✓
• PLC-controlled axes/spindles			✓	✓	✓	✓	✓
• PLC positioning axis via PROFINET			✓	✓	✓	✓	✓
• Maximum configuration axes/spindles, CNC + PLC-controlled:			20	20	40	50	✓
• Maximum configuration axes/spindles, PLC-controlled (n = CNC-controlled axes/spindles)			20-n	20-n	40-n	50-n	✓
• **Axis/spindle**, each additional	**6FC5800-0BA00-0YB0**	**A01...A28**	O	O	O	O	O
• **Positioning axis/auxiliary spindle**, each additional	**6FC5800-0BB00-0YB0**	**B01...B28**	O	O	O	O	O
Expansion of the drive control:							
• Numeric Control Extension NX10.3	**6SL3040-1NC00-0AA0**		O	O	O	O	O
• Numeric Control Extension NX15.3	**6SL3040-1NB00-0AA0**		O	O	O	O	O
• DRIVE-CLiQ ports			4	4	6	6	✓
• Maximum configuration NX			2	2	5	5	✓
• Maximum configuration NCU/PPU + NX + CU3x0-2			9	9	13	15	✓
• Maximum configuration CU3x0-2 (n = number of NX)			8-n	8-n	12-n	14-n	✓
• Maximum configuration SINAMICS S120 Combi (per 1 × S120 Combi, 1 × NCU/PPU or 1 × NX is required)			3	3	3	3	✓
Machining channels/mode groups:			✓1	✓1	✓1	✓1	✓
• **Mode group**, each additional	**6FC5800-0BC00-0YB0**	**C01...C09**	O	O	O	O	O
• **Machining channel**, each additional	**6FC5800-0BC10-0YB0**	**C11...C19**	O	O	O	O	O
• Maximum configuration			4	4	10	10	✓
• **Multi-axis package** (expansion to 31 axes/spindles and 10 channels)	**6FC5800-0BM10-0YB0**	**M10**	O	O	O	O	O
CNC user memory for programs and OEM cycles, buffered internally on NCU/PPU:			✓ 10 MB	✓ 10 MB	✓ 10 MB	✓ 10 MB	✓
• **CNC user memory, expanded by 2 MB each for programs and OEM cycles**, buffered internally on NCU/PPU, to maximum configuration	**6FC5800-0BD00-0YB0**	**D01...D09**	O	O	O	O	O
- Maximum configuration			28 MB	28 MB	28 MB	28 MB	✓
PLC user memory: Program memory in MB / Data memory in MB			✓ 1.5 / 5	✓ 1.5 / 5	✓ 1.5 / 5	✓ 1.5 / 5	✓
• Expansion of the PLC memory by 1.5 MB program memory / 5 MB data memory	**6FC5800-0BD10-0YB0**	**D11...D13**	O 3 / 10	O 3 / 10	O 6 / 20	O 6 / 20	O
CNC user memory, expanded, for programs and OEM cycles, on SD card	**6FC5800-0BP77-0YB0**	**P77**	O 100 MB	O 100 MB	O 100 MB	O 100 MB	O
HMI user memory, expanded on SD card of the NCU/PPU Requirement: CNC user memory expanded (option P77) Note: Not in combination with SIMATIC IPC for SINUMERIK	**6FC5800-0BP12-0YB0**	**P12**	O 6 GB	O 6 GB	O 6 GB	O 6 GB	O

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Connectable motors

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Connectable motors							
Feed motors, synchronous motors:			O	O	O	O	O
• SIMOTICS S-1FK2 (2 cable connection) [1]	See SiePortal						
• SIMOTICS S-1FT2 (2 cable connection) [1]	See SiePortal						
• SIMOTICS S-1FK7	See SiePortal						
• SIMOTICS S-1FT7	See SiePortal						
Main spindle motors, asynchronous motors:			O	O	O	O	O
• SIMOTICS M-1PH3 [2]	See SiePortal						
• SIMOTICS M-1PH8	See SiePortal						
• SIMOTICS M-1FE2 asynchronous	See SiePortal						
Main spindle motors, synchronous motors:			O	O	O	O	O
• SIMOTICS M-1PH8	See SiePortal						
• SIMOTICS M-1FE1	See SiePortal						
• SIMOTICS M-1FE2	See SiePortal						
• 2SP1 motor spindles (Innomotics GmbH - Product Partner SINUMERIK Systems)	See SiePortal						
Linear motors, synchronous motors:			O	O	O	O	O
• SIMOTICS L-1FN3	See SiePortal						
Torque motors, synchronous motors:			O	O	O	O	O
• SIMOTICS T-1FW6 Standard and High Speed	See SiePortal						
• SIMOTICS T-1FW67 external rotor	See SiePortal						
• SIMOTICS T-1FW68 segment motor	See SiePortal						

[1] Operation of the SIMOTICS S-1FK2 and SIMOTICS S-1FT2 motor series with 1-connector technology on SINAMICS S210 in 1-cable technology possible from CNC software version ≥ 6.25.

[2] Operation of the motors is only permitted with SLM infeeds or in Smart Mode.

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Connectable measuring systems							
SINAMICS S120 DRIVE-CLiQ on SIMOTICS motor with resolver			✓	✓	✓	✓	✓
SINAMICS S120 DRIVE-CLiQ on SIMOTICS motor with sin/cos 1 V_{pp}			✓	✓	✓	✓	✓
SINAMICS S120 DRIVE-CLiQ on SIMOTICS motor with EnDat 2.1			✓	✓	✓	✓	✓
Measuring systems per axis, maximum number			2	2	2	2	✓
Absolute encoders, with DRIVE-CLiQ interface (via SINAMICS S120)			✓	✓	✓	✓	✓
Absolute encoders, integrated in the motors SIMOTICS S-1FK2/-1FT2/-1FT7/-1FK7, SIMOTICS M-1PH8/-1PH3, via SINAMICS Sensor Module motor-integrated			✓	✓	✓	✓	✓
Absolute encoders, with SSI interface, via SINAMICS SMC30 Sensor Module			✓	✓	✓	✓	✓
Incremental encoders, integrated in the motors SIMOTICS S-1FT7/-1FK7, SIMOTICS M-1PH8, via SINAMICS Sensor Module motor-integrated			✓	✓	✓	✓	✓
Resolvers, integrated in the motors SIMOTICS S-1FK7, via SINAMICS Sensor Module motor-integrated			✓	✓	✓	✓	✓
Resolvers, as external machine encoders, via SINAMICS SMC10 Sensor Module			✓	✓	✓	✓	✓
Rotary measuring systems with RS422 (TTL), via SINAMICS SMC30 Sensor Module			✓	✓	✓	✓	✓
Rotary measuring systems with sin/cos 1 V_{pp}, via SINAMICS SMC20/SME20 Sensor Module			✓	✓	✓	✓	✓
Rotary measuring systems with distance-coded reference marks, via SINAMICS SMC20/SME20 Sensor Module			✓	✓	✓	✓	✓
Rotary measuring systems with EnDat 2.1, via SINAMICS SMC20/SME25 Sensor Module			✓	✓	✓	✓	✓
Rotary measuring systems with EnDat 2.2, via SINAMICS SMC40 Sensor Module			✓	✓	✓	✓	✓
Linear scale LMS with sin/cos 1 V_{pp}, via SINAMICS SMC20/SME20 Sensor Module			✓	✓	✓	✓	✓
Linear scale LMS with distance-coded reference marks, via SINAMICS SMC20/SME20 Sensor Module			✓	✓	✓	✓	✓
Linear scale LMS with EnDat 2.1, via SINAMICS SMC20/SME25 Sensor Module			✓	✓	✓	✓	✓
Linear scale LMS with EnDat 2.2, via SINAMICS SMC40 Sensor Module			✓	✓	✓	✓	✓

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Drives

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Drives							
SINAMICS S120 booksize format Internal air cooling	See SiePortal						
• Single Motor Modules:	See SiePortal						
- C type			O	O	O	O	O
- D type			O	O	O	O	O
• Double Motor Modules:	See SiePortal						
- C type			O	O	O	O	O
- D type			O	O	O	O	O
SINAMICS S120 booksize format Active Line Modules Internal air cooling This product can be used to save energy.	See SiePortal		O	O	O	O	O
SINAMICS S120 booksize format Active Interface Modules to match the Active Line Modules	See SiePortal		O	O	O	O	O
SINAMICS S120 booksize format Smart Line Modules Internal air cooling	See SiePortal		O	O	O	O	O
SINAMICS S120 booksize format Basic Line Modules Internal air cooling	See SiePortal		O	O	O	O	O
SINAMICS S120 Hydraulic Linear Actor HLA module	**6SL3420-2HX00-0AA0**		O	O	O	O	O
SINAMICS S120 Combi Power Modules:	See SiePortal						
• 3-axis Power Modules	See SiePortal						
- Frame type B11 16 kW / 18 A + 2 × 9 A, C type	**6SL3111-3VE21-6FC1**		O	O	O	O	O
- Frame type B12 16 kW / 24 A + 2 × 9 A, C type	**6SL3111-3VE21-6EC1**		O	O	O	O	O
- Frame type B13 20 kW / 30 A + 2 × 9 A, C type	**6SL3111-3VE22-0HC1**		O	O	O	O	O
• 4-axis Power Modules	See SiePortal						
- Frame type B14 16 kW / 18 A + 3 × 9 A, C type	**6SL3111-4VE21-6FC1**		O	O	O	O	O
- Frame type B01 16 kW / 24 A + 12 A + 2 × 9 A, C type	**6SL3111-4VE21-6EC1**		O	O	O	O	O
- Frame type B02 16 kW / 24 A + 12 A + 2 × 9 A, D type	**6SL3111-4VE21-6ED1**		O	O	O	O	O
- Frame type B17 16 kW / 24 A + 12 A + 2 × 9 A, D type	**6SL3111-4VE21-6ED2**		O	O	O	O	O
- Frame type B16 20 kW / 30 A + 12 A + 2 × 9 A, C type	**6SL3111-4VE22-0HC1**		O	O	O	O	O
SINAMICS S120 chassis format	See SiePortal						
Motor Modules Internal air cooling up to 490 A rated current (rated pulse frequency 2 kHz)	**6SL3320-1TE3.-.AA3**		O	O	O	O	O
SINAMICS S120 chassis format Active Line Modules Internal air cooling up to 300 kW	**6SL3330-7TE3.-.AA.**		O	O	O	O	O
SINAMICS S120 chassis format Active Interface Modules to match the approved Active Line Modules	**6SL3330-7TE3.-.AA.**		O	O	O	O	O

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Drives (continued)							
Control Units							
SINAMICS S120 CU310-2 PN Control Unit for positioning tasks of single-axis drives via PROFINET on PLC S7-1500 of the SINUMERIK ONE (without CompactFlash card)	**6SL3040-1LA01-0AA0**		O	O	O	O	O
SINAMICS S120 CU320-2 PN Control Unit for positioning tasks via PROFINET on PLC S7-1500 of the SINUMERIK ONE (without CompactFlash card)	**6SL3040-1MA01-0AA0**		O	O	O	O	O
CompactFlash card with SINAMICS firmware, license for basic performance V5.2 Service Pack 3	**6SL3054-0FC30-1BA0**		O	O	O	O	O
CompactFlash card with SINAMICS firmware, license incl. performance expansion firmware option V5.2 Service Pack 3	**6SL3054-0FC31-1BA0**		O	O	O	O	O
CUA31 Control Unit Adapter	**6SL3040-0PA00-0AA1**		O	O	O	O	O
SINAMICS S220 booksize format Internal air cooling							
• Single Motor Modules							
- 24 A, D type, width 50 mm	**6SL5120-1UE22-4AD0**		O	O	O	O	O
- 30 A, C type, width 50 mm (slim)	**6SL5120-1UE23-0SC0**		O	O	O	O	O
- 30 A, D type, width 100 mm	**6SL5120-1UE23-0AD0**		O	O	O	O	O
- 30 A, D type, width 50 mm (slim)	**6SL5120-1UE23-0SD0**		O	O	O	O	O
- 60 A, C type, width100 mm	**6SL5120-1UE26-0AC0**		O	O	O	O	O
- 85 A, C type, width 150 mm	**6SL5120-1UE28-5AC0**		O	O	O	O	O
- 85 A, D type, width 150 mm	**6SL5120-1UE28-5AD0**		O	O	O	O	O
- 132 A, C type, width 150 mm	**6SL5120-1UE31-3AC0**		O	O	O	O	O
- 132 A, D type, width 150 mm	**6SL5120-1UE31-3AD0**		O	O	O	O	O
- 200 A, C type, width 200 mm	**6SL5120-1UE32-0AC0**		O	O	O	O	O
- 200 A, D type, width 200 mm	**6SL5120-1UE32-0AD0**		O	O	O	O	O
• Double Motor Modules							
- 2 × 9 A, D type, width 50 mm	**6SL5120-2UE21-0AD0**		O	O	O	O	O
- 2 × 18 A, C type, width 50 mm (slim)	**6SL5120-2UE21-8SC0**		O	O	O	O	O
- 2 × 18 A, D type, width 50 mm (slim)	**6SL5120-2UE21-8SD0**		O	O	O	O	O
• Smart Line Modules							
- 16 kW, C type	**6SL5130-6UE21-6AC0**		O	O	O	O	O
- 16 kW, D type	**6SL5130-6UE21-6AD0**		O	O	O	O	O
- 24 kW, C type	**6SL5130-6UE22-4AC0**		O	O	O	O	O
- 24 kW, D type	**6SL5130-6UE22-4AD0**		O	O	O	O	O
- 36 kW, C type	NEW **6SL5130-6UE23-6AC0**		O	O	O	O	O
- 36 kW, D type	NEW **6SL5130-6UE23-6AD0**		O	O	O	O	O
- 55 kW, C type	NEW **6SL5130-6UE25-5AC0**		O	O	O	O	O
- 55 kW, D type	NEW **6SL5130-6UE25-5AD0**		O	O	O	O	O

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Drives

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Drives (continued)							
SINAMICS S120 Sensor Modules Cabinet-Mounted:							
• SMC10 for resolvers	6SL3055-0AA00-5AA3		O[1]	O[1]	O[1]	O[1]	O
• SMC20 for sin/cos 1 V_{pp} encoders or absolute encoders: EnDat	6SL3055-0AA00-5BA3		O	O	O	O	O
• SMC30 for TTL/HTL encoders or SSI combi encoders	6SL3055-0AA00-5CA2		O[1]	O[1]	O[1]	O[1]	O
• SMC40 for EnDat 2.2 encoders	6SL3055-0AA00-5DA0		O	O	O	O	O
SINAMICS S120 Sensor Modules External:							
• SME20 for sin/cos 1 V_{pp} encoders, IP67 degree of protection	6SL3055-0AA00-5EA3		O	O	O	O	O
• SME25 for EnDat 2.1 absolute encoders, IP67 degree of protection	6SL3055-0AA00-5HA3		O	O	O	O	O
• SME120 for sin/cos 1 V_{pp} encoders; PTC and KTY inputs with safe electrical isolation, IP67 degree of protection	6SL3055-0AA00-5JA3		O	O	O	O	O
• SME125 for EnDat 2.1 absolute encoders; PTC and KTY inputs with safe electrical isolation; IP67 degree of protection	6SL3055-0AA00-5KA3		O	O	O	O	O
SINAMICS S120 TB30 Terminal Board	6SL3055-0AA00-2TA0		–	–	–	–	–
SINAMICS S120 TM Terminal Modules:							
• TM15 for connecting probes and cam outputs	6SL3055-0AA00-3FA0		O	O	O	O	O
• TM31 with digital and analog I/Os	6SL3055-0AA00-3AA1		O	O	O	O	O
• TM41-3 with digital and analog I/Os as well as incremental encoder emulation	6SL5155-0TM04-1AA0		O	O	O	O	O
• TM120 for connecting PTC and KTY inputs	6SL3055-0AA00-3KA0		O	O	O	O	O
SINAMICS S120 VSM10 Voltage Sensing Module (for SINAMICS S120 Chassis format integrated)	6SL3053-0AA00-3AA1		O	O	O	O	O
AM600 adapter module	6SL3555-2BC10-0AA0		O	O	O	O	O
SINAMICS S120 DRIVE-CLiQ Hub Modules:							
• DMC20-3 DRIVE-CLiQ switch with 6 ports	6SL5155-0DC02-0AA0		O	O	O	O	O
• DME20, IP67	6SL3055-0AA00-6AB0		O	O	O	O	O
DRIVE-CLiQ extension	6SL3555-0AA00-6AB0		O	O	O	O	O
Hybrid cabinet bushing	6SL3555-2DA00-0AA0		O	O	O	O	O
Voltage protection for motors SIMOTICS M-1PH8/-1FE1/-1FE2/-2SP1, external via VPM 120 Voltage Protection Module for rated motor current 120 A, max. (see also SIMOTICS motors)	6SN1113-1AA00-1JA1		O	O	O	O	O
Voltage protection for motors SIMOTICS M-1PH8/-1FE1/-1FE2/-2SP1, external via VPM 200 Voltage Protection Module for rated motor current 200 A, max. (see also SIMOTICS motors)	6SN1113-1AA00-1KA1		O	O	O	O	O
Voltage protection for motors SIMOTICS M-1PH8/-1FE1/-1FE2/-2SP1, external via VPM 200 DYNAMIC Voltage Protection Module for rated motor current 200 A, max. (see also SIMOTICS motors)	6SN1113-1AA00-1KC1		O	O	O	O	O

[1] No SINUMERIK Safety Integrated.

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Drives (continued)							
SINAMICS G120 PM240-2 Power Modules standard variant Internal air cooling (Power Modules > 90 kW are not permissible)	See SiePortal						
• Without integrated line filter:							
- 200 … 240 V 1 AC/3 AC	**6SL3210-1PB..-.UL0**		O	O	O	O	O
- 200 … 240 V 3 AC	**6SL3210-1PC..-.UL0**		O	O	O	O	O
- 380 … 480 V 3 AC	**6SL3210-1PE..-.UL.**		O	O	O	O	O
• With integrated line filter class A:							
- 200 … 240 V 1 AC/3 AC	**6SL3210-1PB..-.AL0**		O	O	O	O	O
- 200 … 240 V 3 AC	**6SL3210-1PC..-.AL0**		O	O	O	O	O
- 380 … 480 V 3 AC	**6SL3210-1PE..-.AL.**		O	O	O	O	O

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Drives

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Drives (continued)							
SINAMICS S210 Operation of the SINAMICS S210 on SINUMERIK ONE as an external PLC drive via PROFINET on X150 P1/P2, X160: On the motor side, SIMOTICS S-1FT2/1FK2 with 1-cable connection is required. The operation of the SINAMICS S210 (6SL5…) as an NC axis on SINUMERIK ONE is suitable for applications outside the classic machine tool up to shaft height 40. Operation as an NC axis is carried out via PROFINET on X150 P1/P2 (IRT). The properties stated in the operating instructions apply to the operation of the SINAMICS S210 (6SL5…)[1]. When operating a SINAMICS S210 (6SL5…) drive device with SINUMERIK ONE, the following special issues and restrictions must be taken into consideration: • Commissioning and diagnostics are carried out using STARTDRIVE Advanced or the web server (not supported in SINUMERIK Operate) • Mixing SINAMICS S210 (6SL5…) and SINAMICS S120/S220 in an interpolation group is not possible. • SINAMICS S210 (6SL5…) is not supported in SINUMERIK Operate: Limited diagnostics, no transfer and display of alarm texts, no expert list and drive archive. • Restriction when using SINAMICS S210 (6SL5…) as an NC axis As soon as the SINAMICS S210 (6SL5…) is operated as an NC axis, acyclic communications to the drive from the PLC is not permissible, i.e. drive parameters cannot be read/written to, not even for alarm evaluation or drive diagnostics. Otherwise, alarm 300406 may be output. • Safety Integrated functionalities available (Extended Safety), with the exception of component replacement. The Extended Safety license 6SL3074-0AA10-0AA0 must be ordered separately for SINAMICS S210 (6SL5…) (option K00, S61 cannot be used). Safety acceptance test is carried out via Startdrive (not via SINUMERIK Operate). • No Auto Servo Tuning (AST), no SINUMERIK Tuning Center (STC) and no optimization of gantry axes • No time synchronization between CNC and SINAMICS S210 (6SL5…) • No star-delta changeover • Drive, motor, encoder data sets, maximum number = 1 • No torque feedforward control (MD32620 $MA_FFW_MODE = 4) • No $AA_TORQUE from the drive • No SINAMICS Technology Extension • No Support of IdentSnapshot Requirements: • SINUMERIK ONE NCU 17x0 or SINUMERIK ONE PPU 1740 • SINAMICS S210 (6SL5…) with firmware version ≥ 6.4 • CNC software ≥ 6.25 • TIA Portal ≥ V20 • SINUMERIK STEP 7 Toolbox ≥ V20 • SINAMICS Startdrive ≥ V20	NEW See SiePortal		O [2]	O [2]	O [2]	O [2]	O [2]
SINAMICS S120M Distributed servo drive	See SiePortal		O	O	O	O	O

[1] The properties listed in the function overview under "Drive functions" apply exclusively to the operation of the SINAMICS S120 on SINUMERIK ONE.

[2] When operating SINAMICS S210 on SINUMERIK ONE, the CNC options D30, D50, K00, M01, M03, M12, M13, M41, M60, M61, S11, S23, S61 cannot be used.

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Description ✓ Basic version I O Option I – Not available	**Article No.** **Product ID**	**Order code**	**SINUMERIK ONE**				
			PPU 1740	**NCU 1740**	**NCU 1750**	**NCU 1760**	**in CNC SW export version**
Drive functions (applies to the operation of SINAMICS S120 on SINUMERIK ONE)							
Closed-loop control:							
• Servo control			✓	✓	✓	✓	✓
• Vector control			–	–	–	–	–
• *V/f* control			✓	✓	✓	✓	✓
• Ability to mix servo and *V/f* control on an NCU/PPU, NX or CU, for example			–	–	–	–	–
• Setting the pulse frequency grid in fine steps (3.2 kHz/4 kHz/5.33 kHz/6.4 kHz/8 kHz)			✓[1]	✓[1]	✓[1]	✓[1]	✓
• Sine-wave filter			–	–	–	–	–
• Unit switchover US/SI etc.			–	–	–	–	–
• Direction reversal without changing the setpoint			–	–	–	–	–
• Technology controller			–	–	–	–	–
• k_T estimator			–	–	–	–	–
• k_T-(iq) characteristic			✓	✓	✓	✓	✓
• Rotor/pole position identification saturation-based/motion-based			✓	✓	✓	✓	✓
• Edge modulation			–	–	–	–	–
• Motor data identification stationary/rotating			✓	✓	✓	✓	✓
• Flux reduction for asynchronous motors			✓	✓	✓	✓	✓
Modular machine concept (sub-topologies):							
• Parking axis/encoder			✓	✓	✓	✓	✓
Brakes:							
• Brake control:							
- Basic			✓	✓	✓	✓	✓
- Advanced			✓	✓	✓	✓	✓
• Armature short-circuit braking:							
- Internal			✓	✓	✓	✓	✓
- External			✓	✓	✓	✓	✓
• DC braking			✓	✓	✓	✓	✓
Internal motor winding switchover			✓	✓	✓	✓	✓
Motor changeover			✓[2]	✓[2]	✓[2]	✓[2]	✓
Suspended axis/electronic weight counterbalance			✓	✓	✓	✓	✓
Dynamic energy management (DC link voltage management)			✓	✓	✓	✓	✓
Operating hours counter			✓	✓	✓	✓	✓
I^2t motor monitoring			✓	✓	✓	✓	✓
Automatic restart mechanism (servo/infeed)			–	–	–	–	–
Technology function: Friction characteristic			–	–	–	–	–

[1] Requires current control cycle of 62.5 µs or 31.25 µs in some cases.
[2] Note general boundary conditions.

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Drive functions

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Drive functions (continued) (applies to the operation of SINAMICS S120 on SINUMERIK ONE)							
Star-delta changeover for spindle motors can be controlled via the PLC (FC17) The function can be used to save energy.			✓	✓	✓	✓	✓
Simulated motor encoder Operation of a motor without encoder on SINUMERIK ONE SW license for 1 axis Note: The achievable accuracies must be observed.	**6FC5800-0BD30-0YB0**	**D31...D39**	O	O	O	O	O
SINAMICS Technology Extension: CHATDTEC (Chatter Detection) [1] Chatter detection in the drive	**6SL3077-0AA03-3AH0**		O	O	O	O	O
SINAMICS Technology Extension: DCDCCONV (DC-DC Converter) Operation of a SINAMICS S120 Motor Module in booksize format as a DC-DC converter, e.g. for implementing DC power supply for external energy storage The function can be used to save energy.	**6SL3077-0AA02-0AH0**		O	O	O	O	O
SINAMICS Technology Extension: DYNEGMA (Dynamic Energy Management) Dynamic energy management during operation of SINAMICS S120 drives The function can be used to save energy.	**6SL3077-0AA03-0AH0**		O	O	O	O	O
SINAMICS Technology Extension: FASTBRK (Fast Brake Reaction for Coupled Axes) Fast braking and error response integrated in the drive for mechanically coupled axes	**6SL3077-0AA01-7AH0**		O	O	O	O	O
SINAMICS Technology Extension: POLYGON (Polygonal Line) Signal generator as value table, e.g. for test stands	**6SL3077-0AA00-7AH0**		O	O	O	O	O
SINAMICS Technology Extension: ROTDTEC (Rotation Detection) [1] Rotation detection to determine the speed	**6SL3077-0AA02-5AH0**		O	O	O	O	O
SINAMICS Technology Extension: SERVCOUP (Servo Coupling) Coupling of multiple SINAMICS S120 Motor Modules, e.g. allowing motors to be operated, which due to their size and/or type of construction, cannot be operated from a single SINAMICS power unit (e.g. segment motors).	**6SL3077-0AA00-8AH0**		O	O	O	O	O
SINAMICS Technology Extension: SETPGEN (Setpoint Generator) [1] Setpoint signal generator in drive cycles	**6SL3077-0AA00-3AH0**		O	O	O	O	O
SINAMICS Technology Extension: SIVALU (Signal Evaluation) [1] Fast signal evaluation in drive cycles	**6SL3077-0AA01-4AH0**		O	O	O	O	O
SINAMICS Technology Extensions In addition to the already available SINAMICS Technology Extensions, further drive-related functions can be developed in partnership with Siemens on request.	**On request**		O	O	O	O	O

[1] Software option that is not generally released. Requires machine-specific release process. Please contact your local Siemens sales office or Regional Company.

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Description ✓ Basic version I O Option I – Not available	**Article No.** **Product ID**	**Order code**	**SINUMERIK ONE**				
			PPU 1740	**NCU 1740**	**NCU 1750**	**NCU 1760**	**in CNC SW export version**
Drive functions (continued) (applies to the operation of SINAMICS S120 on SINUMERIK ONE)							
Basic positioner			–[1)]	–[1)]	–[1)]	–[1)]	–
2 command data sets			–	–	–	–	–
Parallel connection of SINAMICS S120 Motor Modules			–	–	–	–	–
Maximum configuration valid for standard PROFIBUS DP cycle for:							
• Current/speed controller cycle 125 µs:							
- Axes/spindles			12	12	31	31	✓
- Axes/spindles per NCU/PPU or NX			6	6	6	6	✓
- Motor measuring systems and direct measuring systems per NCU or NX			12	12	12	12	✓
• Current/speed controller cycle 62.5 µs:							✓
- Axes/spindles			11	11	11	11	✓
- Axes/spindles per NCU/PPU or NX			3	3	3	3	✓
• Current/speed controller cycle 31.25 µs:							✓
- Axes/spindles			1	1	1	1	✓
• SINAMICS S120 chassis format Motor Modules per NCU/PPU or NX			6	6	6	6	✓
Current/speed controller cycle:							
• SINAMICS S120 booksize format, minimum			31.25 µs	31.25 µs	31.25 µs	31.25 µs	✓
• SINAMICS S120 chassis format, minimum			125 µs	125 µs	125 µs	125 µs	✓
• Maximum			250 µs	250 µs	250 µs	250 µs	✓
Drive data sets, maximum number			32	32	32	32	✓
Motor data sets, maximum number			16	16	16	16	✓
Encoder data sets, maximum number			8	8	8	8	✓

[1)] Not possible for integrated drives. Included in basic scope on external SINAMICS S120 drives.

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Axis functions – Spindle functions

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Axis functions							
Traversing range			± 9 decades	± 9 decades	± 9 decades	± 9 decades	✓
Rotary axis, turning endlessly			✓	✓	✓	✓	✓
Velocity, maximum			300 m/s	300 m/s	300 m/s	300 m/s	✓
Acceleration with jerk limitation			✓	✓	✓	✓	✓
Programmable acceleration			✓	✓	✓	✓	✓
Follow-up mode			✓	✓	✓	✓	✓
Measuring systems 1 and 2, selectable			✓	✓	✓	✓	✓
Feedrate interpolation			✓	✓	✓	✓	✓
Separate feedrate for roundings and chamfers			✓	✓	✓	✓	✓
Jerk adaptation	**6FC5800-0BS22-0YB0**	**S22**	O	O	O	O	O
Travel to fixed stop			✓	✓	✓	✓	✓
Travel to fixed stop with Force Control	**6FC5800-0BM01-0YB0**	**M01**	O	O	O	O	O
Setpoint exchange	**6FC5800-0BM05-0YB0**	**M05**	O	O	O	O	O
Tangential control	**6FC5800-0BM06-0YB0**	**M06**	O	O	O	O	O
Position switching signals/cam sequencer • Pairs, maximum number	**6FC5800-0BM07-0YB0**	**M07**	O 16	O 16	O 16	O 16	O
Advanced Position Control ECO	**6FC5800-0BM12-0YB0**	**M12**	O	O	O	O	O
Advanced Position Control	**6FC5800-0BM13-0YB0**	**M13**	O	O	O	O	O
Run MyCC /AJET Setting of the jerk filter time Intelligent adaptation of the machine's dynamic response Requirement: Loadable compile cycle	**6FC5800-0BN82-0YB0**	**N82**	O	O	O	O	–
Axis container (within 31 axes)			✓	✓	✓	✓	✓
Link axes (axis container distributed among several NCUs):			–	–	–	–	–
Spindle functions							
Digital spindle speed			✓	✓	✓	✓	✓
Spindle speed, maximum programmable range of values (display ± 999 999 999.9999)			10^6 ... 10^{-4}	10^6 ... 10^{-4}	10^6 ... 10^{-4}	10^6 ... 10^{-4}	✓
Gear stages			5	5	5	5	✓
Gear stage selection, automatic			✓	✓	✓	✓	✓
Oriented spindle stop			✓	✓	✓	✓	✓
Spindle speed limitation, minimum/maximum			✓	✓	✓	✓	✓
Constant cutting rate			✓	✓	✓	✓	✓
Spindle control via PLC positioning, oscillation			✓	✓	✓	✓	✓
Changeover to axis mode			✓	✓	✓	✓	✓
Axis synchronization on-the-fly			✓	✓	✓	✓	✓
Thread run-in and run-out, programmable			✓	✓	✓	✓	✓
Thread cutting with constant or variable pitch			✓	✓	✓	✓	✓
Tapping with compensating chuck/rigid tapping			✓	✓	✓	✓	✓
Run MyCC /SANS [1] Scalable analog setpoint Requirement: Loadable compile cycle	**6FC5800-0BN48-0YB0**	**N48**	O	O	O	O	–

[1] Compile cycle that is not generally released. Requires machine-specific release process.
Please contact your local Siemens sales office or Regional Company.

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Interpolations							
Typical block change times (block processing times) Requirement: Use of the compressor Note: The planned utilization of the system is under the control of the machine manufacturer. When various CNC options/functions are combined, the CNC can reach its performance limit. The remedy here is to use the next largest hardware. The machine manufacturer must ensure by configuration that the functions/options used can be executed on the NCU.			0.7 ms	0.7 ms	0.3 ms	0.2 ms	✓
Calculation accuracy, can be set to min. 0.1 nm Note: This is a theoretical value that assumes a suitable measuring system and suitable mechanics.			✓	✓	✓	✓	✓
Linear interpolating axes:			4	4	4	4	✓
• Maximum number			12	12	20	20	✓[1]
Circle via center point and end point			✓	✓	✓	✓	✓
Circle via interpolation point			✓	✓	✓	✓	✓
Helical interpolation			2D+6	2D+6	2D+6	2D+6	✓[2]
Universal interpolator NURBS (non-uniform rational B splines)			✓	✓	✓	✓	✓
Continuous-path mode with programmable rounding clearance			✓	✓	✓	✓	✓
Compressor for 3-axis/5-axis machining			✓	✓	✓	✓	✓
Continuous-path mode (G64) extended Requirement: CNC software ≥ SW version 6.22	6FC5800-0BS37-0YB0	S37	O	O	O	O	O
Spline interpolation (A, B and C splines)	6FC5800-0BS16-0YB0	S16	O	O	O	O	O
Polynomial interpolation	6FC5800-0BM18-0YB0	M18	O	O	O	O	O
Involute interpolation	6FC5800-0BM21-0YB0	M21	O	O	O	O	O
Multi-axis interpolation > 4 interpolating axes	6FC5800-0BM15-0YB0	M15	O	O	O	O	–
Run MyCC /RESU Continue machining at the contour, e.g. for laser machining (REtrace SUpport) Requirement: Loadable compile cycle, cross-mode actions ASUB and synchronized actions (option M43) Note: To increase the accuracy, Run MyCC /RESU can also be optionally operated with polynomial interpolation (option M18).	6FC5800-0BM24-0YB0	M24	O	O	O	O	O
Motion control:							
Advanced Surface	6FC5800-0BS07-0YB0	S07	O	O	O	O	O
Top Surface [5] Requirement: Advanced Surface (option S07)	6FC5800-0BS17-0YB0	S17	O[3]	O[3]	O	O	O
Top Speed plus	6FC5800-0BS62-0YB0	S62	O[3]	O[3]	O	O	O
Path acceleration limitation Limitation of the path acceleration	6FC5800-0BP26-0YB0	P26	O	O	O	O	O

[1] Restricted functionality of the export version:
The number of simultaneously traversing axes is restricted to 4.

[2] Restricted functionality of the export version: 2D+2

[3] As of CNC SW version 6.21, 5-axis function is possible (otherwise only 3-axis).

[4] As of CNC SW version 6.24, the combination with Protect MyMachine /3D STL (option S02) is enabled.
Not enabled in combination with Protect MyMachine /3D STL (option S02) up to CNC SW version 6.23.

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Interpolations

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Interpolations (continued)							
Advanced Rapid Movement (COMPPATH) Intelligent adaptation of the machine paths outside of the workpiece Requirement: CNC software ≥ SW version 6.20	**6FC5800-0BS64-0YB0**	**S64**	O	O	O	O	O
Reduced dynamics / gentle mode Intelligent adaptation of the machine's dynamic response The function can be used to save energy. Requirement: CNC software ≥ SW version 6.20	**6FC5800-0BS65-0YB0**	**S65**	O	O	O	O	O
Run MyCC /CRIP [1)] Crank interpolation Requirement: Loadable compile cycle	**6FC5800-0BN04-0YB0**	**N04**	O	O	O	O	–

[1)] Compile cycle that is not generally released. Requires machine-specific release process. Please contact your local Siemens sales office or Regional Company.

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Couplings							
Pair of synchronized axes (gantry axes):	**6FC5800-0BM02-0YB0**	**M02**	O	O	O	O	O
• Maximum number			8	8	8	8	✓
Main axis/sub axis for drives	**6FC5800-0BM03-0YB0**	**M03**	O	O	O	O	O
Generic couplings (Electronic gear):							
• **CP-Standard:** - 4 axis pairs in simultaneous coupled motion			✓	✓	✓	✓	✓
• **CP-Static:** - 1 × simple synchronous spindle, coupling ratio 1:1, no multi-edge machining	**6FC5800-0BM75-0YB0**	**M75**	O	O	O	O	O
• **CP-Basic:** - 4 axis pairs in simultaneous coupled motion and - 1 × synchronous spindle/multi-edge turning and/or master value coupling/curve table interpolation and/or axial coupling in the machine coordinate system	**6FC5800-0BM72-0YB0**	**M72**	O	O	O	O	O[1)]
• **CP-Comfort:** - 4 axis pairs in simultaneous coupled motion and - 4 × synchronous spindle/multi-edge turning and/or master value coupling/curve table interpolation and/or axial coupling in the machine coordinate system and - 1 × electronic gear for 3 leading axes without curve table interpolation, without cascading	**6FC5800-0BM73-0YB0**	**M73**	O	O	O	O	O[1)]
• **CP-Expert:** - 8 axis pairs in simultaneous coupled motion and - 8 × synchronous spindle/multi-edge turning and/or master value coupling/curve table interpolation and/or axial coupling in the machine coordinate system and - 8 × electronic gears for 3 leading axes with curve table interpolation, with cascading and - 5 × electronic gears for 5 leading axes with curve table interpolation, with cascading	**6FC5800-0BM74-0YB0**	**M74**	O	O	O	O	O[1)]
Electronic transfer CP Contains the function: I/O interfacing via PROFIBUS DP Contains the options: • Position switching signals/cam sequencer (option M07) • Polynomial interpolation (option M18) • Generic coupling CP-Comfort (option M73) • Cross-mode actions ASUB and synchronized actions (option M43) • Synchronized actions stage 2 (option M36) • Pairs of synchronized axes (gantry axes) (option M02) Note: No tool offsets or spindles possible.	**6FC5800-0BM76-0YB0**	**M76**	–	O	O	O	O[1)]
Run MyCC /AMOV [2)] Variable-based axis movement Requirement: Loadable compile cycle	**6FC5800-0BN62-0YB0**	**N62**	O	O	O	O	–

[1)] Restricted functionality of the export version:
The number of simultaneously traversing axes is restricted to 4.

[2)] Compile cycle that is not generally released. Requires machine-specific release process. Please contact your local Siemens sales office or Regional Company.

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Transformations

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Transformations							
Cartesian Point-to-Point travel PTP			✓	✓	✓	✓	✓
Concatenated transformations (inclined axis TRAANG after TRAORI/ cardanic milling head/TRANSMIT/TRACYL)			✓	✓	✓	✓	✓
Generic transformation Requirement: Machining package 5 axes (option M30) or SINUMERIK ONE Dynamics 5 axes (option S43)			✓	✓	✓	✓	–
Oblique angle transformation (inclined axis) TRAANG Enables programming in the Cartesian workpiece coordinate system (WCS) with "programmable angle" on machines with obliquely arranged machine axes, which is a typical axis arrangement on grinding machines	**6FC5800-0BM28-0YB0**	**M28**	O	O	O	O	O
TRANSMIT/cylinder surface transformation	**6FC5800-0BM27-0YB0**	**M27**	O	O	O	O	O
Run MyCC /2TRA [1] Transformation: DOUBLETRANSMIT Requirement: Loadable compile cycle	**6FC5800-0BM25-0YB0**	**M25**	O	O	O	O	–
Run MyCC /RDCC [1] Transformation: Redundant and spatially distributed axes at workpiece in Cartesian coordinate system Requirement: Loadable compile cycle	**6FC5800-0BN26-0YB0**	**N26**	O	O	O	O	–
Run MyCC /2RPT [1] Transformation: Rotating workpiece and tool Requirement: Loadable compile cycle	**6FC5800-0BN43-0YB0**	**N43**	O	O	O	O	–
Run MyCC /SW2A [1] Transformation: Swivel by 2 linear axes Requirement: Loadable compile cycle	**6FC5800-0BN45-0YB0**	**N45**	O	O	O	O	–
Run MyCC /ECCA [1] Transformation: Eccentric axis Requirement: Loadable compile cycle	**6FC5800-0BN44-0YB0**	**N44**	O	O	O	O	–
Run MyCC /ECCE [1] Transformation: Eccentric Requirement: Loadable compile cycle	**6FC5800-0BN41-0YB0**	**N41**	O	O	O	O	–
Run MyCC /DSTT [1] Transformation: Dynamic Swivel TRIPOD, Basis 5 axes Requirement: Loadable compile cycle	**6FC5800-0BM84-0YB0**	**M84**	O	O	O	O	–
Run MyCC /HEX [1] Transformation: HEXAPOD 6-axis transformation Requirement: Loadable compile cycle	**6FC5800-0BM71-0YB0**	**M71**	O	O	O	O	–
Run MyCC /PACO Transformation: PARACOP 3 axes Requirement: Loadable compile cycle	**6FC5800-0BM44-0YB0**	**M44**	O	O	O	O	–
Run MyCC /RCTRA Transformation: Handling Requirement: Loadable compile cycle	**6FC5800-0BM31-0YB0**	**M31**	O	O	O	O	–
Run MyRobot /Direct control ROBX AR Transformation for articulated arm robot Can only be used together with Run MyRobot /Direct Control Contains the options: • Lock MyCycles (option P54) • Run MyCC /AXCO (option M81)	**6FC5800-0BR05-0YB0**	**R05**	O	O	O	O	–
Run MyCC /ROBX [1] Transformation robotic extended Requirement: Loadable compile cycle	**6FC5800-0BN54-0YB0**	**N54**	O	O	O	O	–

[1] Compile cycle that is not generally released. Requires machine-specific release process.
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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Transformations (continued)							
Run MyCC /SCRA [1] Transformation SCARA 2/3 axes Requirement: Loadable compile cycle	6FC5800-0BM68-0YB0	M68	O	O	O	O	–
Run MyCC /ROTE [1] Transformation: Rotating eccentric Requirement: Loadable compile cycle	6FC5800-0BN37-0YB0	N37	O	O	O	O	–
Run MyCC /SCIS [1] Transformation: Pantograph kinematics Requirement: Loadable compile cycle	6FC5800-0BM51-0YB0	M51	O	O	O	O	–
Run MyCC /THYK [1] Transformation: TRIPOD hybrid kinematics Requirement: Loadable compile cycle Note: Function can only be executed on the 1st channel.	6FC5800-0BN36-0YB0	N36	O	O	O	O	–
Run MyCC /TRIC [1] Transformation: TRICEPT 5-axis transformation Requirement: Loadable compile cycle	6FC5800-0BM46-0YB0	M46	–	–	O	O	–
Run MyCC /DTOW [1] Differential Tower Transformation Requirement: Loadable compile cycle	6FC5800-0BN68-0YB0	N68	O	O	O	O	–
Run MyCC /SKID [1] Transformation: Double slide Requirement: Loadable compile cycle	6FC5800-0BM80-0YB0	M80	O	O	O	O	–
Run MyCC /WZTR [1] Transformation: W-Z Tracking Requirement: Loadable compile cycle	6FC5800-0BN22-0YB0	N22	O	O	O	O	–
Run MyCC /SCTR SC [1] Run MyCC /SCTR SC transformation Requirement: Loadable compile cycle	6FC5800-0BR03-0YB0	R03	O	O	O	O	–
Run MyCC /DIFA+FASTIPO [1] Drive-interpolating following axis Requirement: Loadable compile cycle	6FC5800-0BR04-0YB0	R04	O	O	O	O	–
Run MyCC /TPWA [1] Transformation with programmable wrist axes Requirement: Loadable compile cycle	6FC5800-0BR10-0YB0	R10	O	O	O	O	–

[1] Compile cycle that is not generally released. Requires machine-specific release process.
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Measuring – Technologies

Description ✓ Basic version I O Option I – Not available		Article No. Product ID	Order code	SINUMERIK ONE				
				PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Measuring								
Logging of measurement results in automatic mode				✓	✓	✓	✓	✓
Measuring stage 1 4 probes (switching) with/without deletion of distance-to-go Requirement: CNC software ≥ SW version 6.25 For CNC SW version ≤ 6.24 the maximum number is 2 probes	NEW			✓	✓	✓	✓	✓
Measuring stage 2: • Axial measurement • Measurements from synchronized actions • Cyclic measurement • Fast measurement MEASF (SW version 6.20 and higher)		**6FC5800-0BM32-0YB0**	**M32**	O	O	O	O	O
Measuring cycles for drilling/milling and turning: • Calibrating workpiece probes • Workpiece measurement • Tool measurement		**6FC5800-0BP28-0YB0**	**P28**	O	O	O	O	O
Measure kinematics (Cycle 996)		**6FC5800-0BP18-0YB0**	**P18**	O	O	O	O	O
Run MyCC /COPA [1] Reading of actual positions correlated with output signal Requirement: Loadable compile cycle		**6FC5800-0BN61-0YB0**	**N61**	O	O	O	O	O
Technologies								
Handwheel override				✓	✓	✓	✓	✓
Contour handwheel		**6FC5800-0BM08-0YB0**	**M08**	O	O	O	O	O
Punching/nibbling		**6FC5800-0BM33-0YB0**	**M33**	O	O	O	O	O
Slot shape recognition (recognition of slot shape)		**6FC5800-0BS18-0YB0**	**S18**	O	O	O	O	O
Interpolation turning		**6FC5800-0BP57-0YB0**	**P57**	O	O	O	O	O
Prime Turning™ Automatically generates the path, which is matched to the tool geometry Requirement: CNC software ≥ SW version 6.22		**6FC5800-0BR56-0YB0**	**R56**	O	O	O	O	O
Y-Turning Turning with Y axis Requirement: CNC software ≥ SW version 6.20		**6FC5800-0BS56-0YB0**	**S56**	O	O	O	O	O
Balance cutting, 4-axis roughing Requirement: additional channel (options C11 ... C19)		**6FC5800-0BS05-0YB0**	**S05**	O	O	O	O	O
Comb grooving Grooving in a comb pattern		**6FC5800-0BS58-0YB0**	**S58**	O	O	O	O	O
SMTE cone turning [2] Requirement: CNC software ≥ SW version 6.15 SP1		**6FC5800-0BR52-0YB0**	**R52**	O	O	O	O	O
SMTE QR code [2] Cycle for generating a QR code on the workpiece Requirement: CNC software ≥ SW version 6.15 SP1		**6FC5800-0BR53-0YB0**	**R53**	O	O	O	O	O
SMTE engraving [2] Engraving cycle Requirement: CNC software ≥ SW version 6.15 SP1		**6FC5800-0BR54-0YB0**	**R54**	O	O	O	O	O
SMTE chatter avoidance [2] Requirement: CNC software ≥ SW version 6.15 SP3 or ≥ SW version 6.22 (not SW versions 6.20, 6.21)	NEW	**6FC5800-0BR57-0YB0**	**R57**	O	O	O	O	O
SMTE Trochoidal Basic [2] Machining cycle for trochodial milling Requirement: CNC software ≥ SW version 6.15 SP1	NEW	**6FC5800-0BR58-0YB0**	**R58**	O	O	O	O	O

[1] Compile cycle that is not generally released. Requires machine-specific release process.
Please contact your local Siemens sales office or Regional Company.

[2] Regarding delivery, please contact your local Siemens sales office or Regional Company.

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Technologies (continued)							
Machining package 5 axes Contains the option: Multi-axis interpolation > 4 interpolating axes (option M15)	**6FC5800-0BM30-0YB0**	**M30**	O	O	O	O	–
Machining package 5 axes **Additional function 7th axis** Requirement: Machining package 5 axes (option M30)	**6FC5800-0BS01-0YB0**	**S01**	O	O	O	O	–
Orientation offset static/dynamic	**6FC5800-0BS14-0YB0**	**S14**	O	O	O	O	O
SINUMERIK ONE Dynamics Operate Milling/turning technology package for job shop manufacturing Contains the options: • ShopTurn/ShopMill (option P17) • DXF Reader (option P56) • Residual material detection and machining for contour pockets and stock removal (option P13) • 3D simulation 1 (finished part) (option P25) • Simultaneous recording (option P22) • TRANSMIT/cylinder surface transformation (option M27) • Execution from External Storage (EES) (option P75)	**6FC5800-0BS41-0YB0**	**S41**	O	O	O	O	O
SINUMERIK ONE Dynamics 3 Axis Milling Milling technology package Contains the options: • ShopTurn/ShopMill (option P17) • DXF Reader (option P56) • Residual material detection and machining for contour pockets and stock removal (option P13) • 3D simulation 1 (finished part) (option P25) • Simultaneous recording (option P22) • TRANSMIT/cylinder surface transformation (option M27) • Advanced Surface motion control (option S07) [1] • Top Surface (option S17) [1] [2] • Top Speed plus (option S62) [1] • Measuring cycles for drilling/milling and turning (option P28) • HMI user memory, expanded on SD card of the NCU/PPU (option P12) • Execution from External Storage (EES) (option P75)	**6FC5800-0BS42-0YB0**	**S42**	O	O	O	O	O
SINUMERIK ONE Dynamics 5 Axis Milling Milling technology package Contains the options: • Machining package 5 axes (option M30) • 3D tool radius compensation (option M48) • Measure kinematics (option P18) • ShopTurn/ShopMill (option P17) • DXF Reader (option P56) • Residual material detection and machining for contour pockets and stock removal (option P13) • 3D simulation 1 (finished part) (option P25) • Simultaneous recording (option P22) • TRANSMIT/cylinder surface transformation (option M27) • Advanced Surface motion control (option S07) • Top Surface (option S17) [2] • Top Speed plus (option S62) • Measuring cycles for drilling/milling and turning (option P28) • HMI user memory, expanded on SD card of the NCU/PPU (option P12) • Execution from External Storage (EES) (option P75)	**6FC5800-0BS43-0YB0**	**S43**	–	–	O[3]	O	–

[1] Maximum 3+2 interpolating axes.

[2] Can be combined with Protect MyMachine /3D STL (option S02) as of CNC SW version 6.24.
Not enabled in combination with Protect MyMachine /3D STL (option S02) up to CNC SW version 6.23.

[3] An NCU 1760 is recommended for maximum dynamic response and accuracy in mold making or in the high-speed cutting sector (HSC).

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Technologies

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Technologies (continued)							
Path-referred channel synchronization NEW Cross-channel synchronization of path motions Requirement: CNC software ≥ SW version 6.25	6FC5800-0BS66-0YB0	S66	–	–	–	O	O
Grinding:							
• Multiple feedrates in one block, e.g. for calipers			✓	✓	✓	✓	✓
• Continuous dressing, parallel dressing online modification of the tool offset			✓	✓	✓	✓	✓
• **Oscillation functions** block-related, modal and asynchronous	6FC5800-0BM34-0YB0	M34	O	O	O	O	O
• **Grinding Advanced** extended functions for grinding such as axis-parallel dressing/pre-profiling cycle	6FC5800-0BS35-0YB0	S35	O	O	O	O	O
Optimize MyMachining /CCG compiler For programming any closed non-circular contours as DLL for SIMATIC IPC for SINUMERIK (SINUMERIK PCU) Requirements: TRANSMIT/cylinder surface transformation (option M27), polynomial interpolation (option M18)	6FC5800-0BP10-0YB0		O	O	O	O	O
Run MyCC /SUTI [1] Utility programs for the servo Requirement: Loadable compile cycle	6FC5800-0BN10-0YB0	N10	O	O	O	O	–
Run MyCC /FXSC [1] Cartesian travel to fixed stop Requirement: Loadable compile cycle	6FC5800-0BN38-0YB0	N38	O	O	O	O	–
Run MyCC /MSPZ [1] Metal spinning protection zone Requirement: Loadable compile cycle	6FC5800-0BN42-0YB0	N42	O	O	O	O	–
Run MyCC /SYMO [1] Synchronous motion Requirement: Loadable compile cycle	6FC5800-0BN66-0YB0	N66	O	O	O	O	–
Handling package Contains the options: • 3 additional axes (option A03) • 3 additional machining channels (option C13) • Transformation: Handling (option M31) • Synchronized actions stage 2 (option M36) Requirement: Loadable compile cycle Note: No tool offsets or spindles possible.	6FC5800-0BS31-0YB0	S31	O	O	O	O	–

[1] Compile cycle that is not generally released. Requires machine-specific release process.
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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Technologies (continued)							
Run MyRobot /Direct Control Package for SINUMERIK ONE for selected types of robot Contains: Engineering documents by means of SIZER/TST Requirement: Robot equipped with SINUMERIK ONE CNC control with CNC-SW ≥ 6.15, SINAMICS S120 drives and SIMOTICS S-1FT/-1FK motors (specific to robot), Create MyConfig commissioning tool (required once). The following options are required for operation of the robot: • 3 additional axes/spindles (option A03) • 6 Safety Integrated axes/spindles (option K06) • Multi-axis interpolation (> 4 interpolating axes) (option M15) • Cross-mode actions ASUB and synchronized actions (option M43) • Transformation for articulated arm robot Run MyRobot /Direct control ROBX AR (option R05) • SINUMERIK Operate /NCU (option S00) • Safety Integrated F-PLC (option S60)	**On request**		O	O	O	O	–
SINUMERIK Run MyRobot /Direct Handling Handling package for Run MyRobot /Direct Control with SINUMERIK ONE Contains the options: • 4 additional axes (option A04) • 7 Safety Integrated axes/spindles (option K07) • Multi-axis interpolation (> 4 interpolating axes) (option M15) • Cross-mode actions ASUB and synchronized actions (option M43) • Lock MyCycles (option P54) • Run MyCC /AXCO (option M81) • Transformation for articulated arm robot Run MyRobot /Direct control ROBX AR (option R05) • Safety Integrated F-PLC (option S60) Requirements: Engineering documents by means of SIZER, and commissioning tool Create MyConfig (required once), robot equipped with SINUMERIK ONE CNC control with CNC-SW ≥ 6.15, SINAMICS S120 drives and SIMOTICS S-1FT/-1FK motors (specific to robot) The following option is additionally required for operation of the robot: SINUMERIK Operate /NCU (option S00) Note: No operation of spindles, no functionality for tool corrections and COMPCAD/COMPSURF	**6FC5800-0BP84-0YB0**		O	O	O	O	–
Electrical equipment for robotics Panel building for machine tools at the Siemens Systems Engineering Plant Chemnitz (WKC).	NEW See SiePortal		O	O	O	O	O

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Motion-synchronous actions

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Motion-synchronous actions							
High-speed inputs/outputs Number onboard Drive I/O (X122, X132) CNC I/O: NCU (X141), PPU (X242, X252):			36	28	28	28	✓
• Digital inputs on-board			20	16	16	16	✓
• Digital outputs on-board			8[1]	4	4	4	✓
• Digital inputs or outputs on-board			8	8	8	8	✓
• Analog outputs on-board			1	–	–	–	–
• High-speed inputs/outputs, expansion via SIMATIC S7 I/O:	See SiePortal		O	O	O	O	O
- Digital inputs			32	32	32	32	✓
- Digital outputs			32	32	32	32	✓
- Analog inputs			4	4	4	4	✓
- Analog outputs			4	4	4	4	✓
Synchronized actions and fast auxiliary function output incl. 3 synchronous functions (max. 159 elements for synchronized actions)			✓	✓	✓	✓	✓[2]
• Number of simultaneously active synchronous functions SYNFC			24	24	24	24	✓
Positioning axes and spindles via synchronized actions (command axes)			✓	✓	✓	✓	✓
Asynchronous subprograms ASUB High-speed CNC inputs/outputs			✓	✓	✓	✓	✓
Analog value control in interpolation cycle (IPO cycle) Requirement: Analog input			✓	✓	✓	✓	✓
Clearance control 1D in interpolation cycle via synchronized actions			✓	✓	✓	✓	✓
Synchronized actions stage 2	**6FC5800-0BM36-0YB0**	**M36**	O	O	O	O	O[3]
Path velocity-dependent analog output (laser power control)	**6FC5800-0BM37-0YB0**	**M37**	O	O	O	O	O
Run MyCC /HSLC Laser switching signal, high-speed Requirement: Loadable compile cycle	**6FC5800-0BM38-0YB0**	**M38**	O	O	O	O	O
Evaluation of internal drive variables	**6FC5800-0BM41-0YB0**	**M41**	O	O	O	O	O
Interrupt routines with fast retraction from the contour	**6FC5800-0BM42-0YB0**	**M42**	O	O	O	O	O
ASUB Cross-mode actions and synchronized actions in all control operating modes	**6FC5800-0BM43-0YB0**	**M43**	O	O	O	O	O
Run MyCC /CLC-FD 1D/3D clearance control in position control cycle with free direction Requirement: Loadable compile cycle and clearance control Run MyCC /CLCX (option N78)	**6FC5800-0BM65-0YB0**	**M65**	O	O	O	O	O[4]
Run MyCC /XOUT [5] Extrapolated switching signals, max. number 64 Requirement: Loadable compile cycle	**6FC5800-0BN51-0YB0**	**N51**	O	O	O	O	–

[1] 2 of 8 DO depending on the analog output used on board.

[2] Restricted functionality of the export version:
The number of simultaneously traversing axes is restricted to 4.

[3] Restricted functionality of the export version:
The number of simultaneously traversing path and positioning axes is restricted to 4.

[4] Restricted functionality of export version:
Clearance control can be applied to only one axis because 5-axis transformation is not available. The number of interpolating axes is restricted to 4 without Run MyCC /CLC and to 3 with Run MyCC /CLC.

[5] Compile cycle that is not generally released. Requires machine-specific release process.
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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Motion-synchronous actions (continued)							
Run MyCC /CLCX Clearance control Requirement: Loadable compile cycle	**6FC5800-0BN78-0YB0**	**N78**	O	O	O	O	O
Run MyCC /PRIO [1] Path-related switching signals Requirement: Loadable compile cycle	**6FC5800-0BN88-0YB0**	**N88**	O	O	O	O	O
Run MyCC /SDD [1] Special Device Driver. Enhances CLCX and HSLC with functions for communication with specific PROFINET devices. Requirements: Loadable compile cycle Run MyCC /HSLC (option M38) or Run MyCC /CLCX (option N78)	NEW **6FC5800-0BR08-0YB0**	**R08**	O	O	O	O	O
CNC programming language							
Programming language DIN 66025 and high-level language expansion			✓	✓	✓	✓	✓
Main program call from main program and subprogram			✓	✓	✓	✓	✓
Subroutine levels, maximum			16	16	16	16	✓
Interrupt routines, maximum			2	2	2	2	✓
Number of subprogram passes			9999	9999	9999	9999	✓
Number of levels for skip blocks			0 ... 8	0 ... 8	0 ... 8	0 ... 8	✓
Polar coordinates			✓	✓	✓	✓	✓
1/2/3-point contours			✓	✓	✓	✓	✓
Dimensions metric/inch, changeover via operator action or program			✓	✓	✓	✓	✓
Inverse-time feedrate			✓	✓	✓	✓	✓
Auxiliary function output via M word, maximum programmable range of values			INT 2^{31}-1	INT 2^{31}-1	INT 2^{31}-1	INT 2^{31}-1	✓
Auxiliary function output H word, maximum programmable range of values REAL ± 3.4028 ex 38 (display ± 999 999 999.9999)			INT -2^{31}…2^{31}-1	INT -2^{31}…2^{31}-1	INT -2^{31}…2^{31}-1	INT -2^{31}…2^{31}-1	✓
CNC high-level language with:							
• User variables, configurable			✓	✓	✓	✓	✓
• Predefined user variables (arithmetic parameters)			✓	✓	✓	✓	✓
• Predefined user variables (arithmetic parameters), configurable			✓	✓	✓	✓	✓
• Read/write system variables			✓	✓	✓	✓	✓
• Indirect programming			✓	✓	✓	✓	✓
• Program jumps and branches			✓	✓	✓	✓	✓
• Program coordination with WAIT, START, INIT			✓	✓	✓	✓	✓
• Arithmetic and trigonometric functions			✓	✓	✓	✓	✓
• Comparison operations and logic combinations			✓	✓	✓	✓	✓
• Macro techniques			✓	✓	✓	✓	✓
• Control structures IF-ELSE-ENDIF			✓	✓	✓	✓	✓
• Control structures WHILE, FOR, REPEAT, LOOP			✓	✓	✓	✓	✓
• Commands to HMI			✓	✓	✓	✓	✓
• STRING functions			✓	✓	✓	✓	✓

[1] Compile cycle that is not generally released. Requires machine-specific release process.
Please contact your local Siemens sales office or Regional Company.

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CNC programming language

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
CNC programming language (continued)							
Program functions:							
• Preprocessing memory, dynamic FIFO			✓	✓	✓	✓	✓
• Look Ahead, no. of recorded part program blocks (ONE Dynamics, MDynamics, Top Surface or COMP-SURF active)			3000	3000	4000	6000	✓
• Look Ahead, IPO blocks, buffered			1000	1000	1000	1000	✓
• Frame concept			✓	✓	✓	✓	✓
• Inclined-surface machining with frames			✓	✓	✓	✓	✓
• Axis/spindle interchange			✓	✓	✓	✓	✓
• Geometry axes, switchable online in the CNC program			✓	✓	✓	✓	✓
• Program preprocessing			✓	✓	✓	✓	✓
Online ISO dialect interpreter			✓	✓	✓	✓	✓
Program/workpiece management Part programs, maximum number (in total a maximum of 512 files per directory)			1000	1000	1000	1000	✓
Program/workpiece management Workpieces, maximum number (in total a maximum of 256 files per directory)			250	250	250	250	✓
Program/workpiece management Workpieces on the SSD of the SIMATIC IPC for SINUMERIK, maximum number (in total a maximum of 10000 user files)			10000	10000	10000	10000	✓
Templates for programs and workpieces			✓	✓	✓	✓	✓
Job lists			✓	✓	✓	✓	✓
Program/workpiece management on additional HMI user memory on the SD card of the SINUMERIK NCU/PPU (in total a maximum of 10000 user files and directories) Requirement: HMI user memory, expanded on SD card of the NCU/PPU			✓	✓	✓	✓	✓
Program/workpiece management on CompactFlash card to be plugged in additionally in the SIMATIC IPC for SINUMERIK Requirement: CF card in the PCU			✓	✓	✓	✓	✓
Program/workpiece management on integrated SSD in the SIMATIC IPC for SINUMERIK			✓	✓	✓	✓	✓
Program/workpiece management on USB storage medium, e.g. flash drive Requirement: USB storage medium			✓	✓	✓	✓	✓
Program/workpiece management on the network drive			✓	✓	✓	✓	✓
Basic frames, maximum number			16	16	16	16	✓
Settable offsets, maximum number			100	100	100	100	✓
Work offsets, programmable (frames)			✓	✓	✓	✓	✓
Scratching, determining work offset			✓	✓	✓	✓	✓
Work offsets, external via PLC			✓	✓	✓	✓	✓
Global and local user data			✓	✓	✓	✓	✓
Global program user data			✓	✓	✓	✓	✓
Display system variables, including in online configurable display, and log them			✓[1)]	✓[1)]	✓[1)]	✓[1)]	✓

[1)] Function depends on operating software.

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Programming support							
Program editor in SINUMERIK Operate:							
• Text editor: selecting, copying, deleting			✓	✓	✓	✓	✓
• Dual editor			✓	✓	✓	✓	✓
• Multi-editor, maximum			4	4	4	4	✓
• Write protection for lines			✓	✓	✓	✓	✓
• Suppression of lines in the display			✓	✓	✓	✓	✓
Technology cycles in SINUMERIK Operate:							
• Drilling			✓	✓	✓	✓	✓
• Milling			✓	✓	✓	✓	✓
• Pocket milling with free contour definition and islands			✓	✓	✓	✓	✓
• Turning			✓	✓	✓	✓	✓
• Grinding			✓	✓	✓	✓	✓
programGUIDE in SINUMERIK Operate:							
• Programming support for cycles			✓	✓	✓	✓	✓
• Dynamic programming graphics			✓	✓	✓	✓	✓
• Animated elements			✓	✓	✓	✓	✓
Programming support can be expanded, e.g. user cycles, See Engineering, Run MyScreens			✓	✓	✓	✓	✓
ShopTurn/ShopMill: • Machining step programming • Multiple clamping of identical/different workpieces • Manual machine	**6FC5800-0BP17-0YB0**	**P17**	O	O	O	O	O
DXF Reader integrated in SINUMERIK Operate: • Accepting contours • Accepting point patterns	**6FC5800-0BP56-0YB0**	**P56**	O	O	O	O	O
programSYNC	**6FC5800-0BP05-0YB0**	**P05**	O	O	O	O	O
Residual material detection and machining for contour pockets and stock removal Both cycles can be used in programGUIDE or in ShopMill (option P17).	**6FC5800-0BP13-0YB0**	**P13**	O	O	O	O	O

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Simulation

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Simulation							
Machining channels capable of simulation, maximum; Requirement: programSYNC (option P05)			4	4	4	4	✓
Simulation of program X, while program Y is being executed			–	–	✓	✓	✓
Quickview for mold-making programs			✓	✓	✓	✓	✓
Turning/drilling/milling:							
• Turning			✓	✓	✓	✓	✓
• Counterspindle			✓	✓	✓	✓	✓
• Turn-milling			✓	✓	✓	✓	✓
• Mill-turning with support for kinematics (application-specific by machine manufacturer)			✓	✓	✓	✓	✓
• Milling up to 5-axis machining with TRAORI			✓	✓	✓	✓	–
• 2D simulation (finished part)			✓	✓	✓	✓	✓
• **3D simulation 1** (finished part) Stock removal simulation of milling, drilling and turning operations in 3D representation	**6FC5800-0BP25-0YB0**	**P25**	O	O	O	O	O
• **Simultaneous recording** Real-time simulation of current machining operation	**6FC5800-0BP22-0YB0**	**P22**	O	O	O	O	O
Simulation of virtual machine tools with SINUMERIK ONE during production							
Run MyVirtual Machine /Operate Control-identical programming station for NC programs or for training and apprenticeship purposes	**MCS4220** [1] **MCS4220n** [2] **MCS4220a** [3]		O	O	O	O	O
Run MyVirtual Machine /Open Extension with an open communications interface for integration into a simulation system	**MCS4230** [1] **MCS4230n** [2] **MCS4230a** [3]		O	O	O	O	O
Run MyVirtual Machine /3D Extension to the virtual machine during work preparation	**MCS4240** [1] **MCS4240n** [2] **MCS4240a** [3]		O	O	O	O	O

[1] Floating: Server-based licenses, can be used on multiple workstations (not simultaneously); order recommendation.

[2] Node-locked Server: Server license that is connected to particular clients (work station/PC)
Node-locked Standalone: Local license file that is connected to a particular client (work station/PC)

[3] Academics: For non-commercial use.

For more information, request the detailed description of licenses.

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Operating modes							
JOG CNC operating mode:			✓	✓	✓	✓	✓
• Handwheel selection			✓	✓	✓	✓	✓
• Inch/metric changeover			✓	✓	✓	✓	✓
• Manual measurement of work offset			✓	✓	✓	✓	✓
• Manual measurement of tool offset			✓	✓	✓	✓	✓
• Automatic tool/workpiece measurement			✓	✓	✓	✓	✓
• Reference point approach automatic/via CNC program			✓	✓	✓	✓	✓
• Repos (repositioning on the contour) via operator action/semiautomatically/program:			✓	✓	✓	✓	✓
• Changing the velocity (feedrate override)			✓	✓	✓	✓	✓
• Positioning with electronic handwheel			✓	✓	✓	✓	✓
MDI CNC operating mode (Manual Data Automatic/Input):			✓	✓	✓	✓	✓
• Input in text editor			✓	✓	✓	✓	✓
• Save MDI program			✓	✓	✓	✓	✓
• Teach positions in MDI buffer			✓	✓	✓	✓	✓
• Teach-in function Handling			✓	✓	✓	✓	✓
AUTO CNC operating mode:			✓	✓	✓	✓	✓
• Execute directly: - from real CNC user memory (in line with basic scope)			✓ (10 MB)	✓ (10 MB)	✓ (10 MB)	✓ (10 MB)	✓ (10 MB)
- from CNC user memory, expanded Requirement: CNC user memory expanded (option P77)			O	O	O	O	O
- from CNC user memory on SD card of the NCU Requirement: HMI user memory, expanded on SD card of the NCU/PPU (option P12)			O	O	O	O	O
• Process from external source (EXTCALL): - Data storage medium on USB interface - User memory on the SSD of the SIMATIC IPC for SINUMERIK - from a network drive Requirement: SINUMERIK TCU or SIMATIC IPC for SINUMERIK			✓	✓	✓	✓	✓
• **Execution from External Storage (EES):** - Execution from external drives and memories e.g. server, local area network, USB flash drive, … - One part program memory for several NCUs - Backward jumps, far jumps, long program loops (GOTOF/GOTOB) Contains: CNC user memory, expanded, for programs and OEM cycles, on SD card (option P77)	**6FC5800-0BP75-0YB0**	**P75**	O	O	O	O	O
• DRF offset			✓	✓	✓	✓	✓
• Program control			✓	✓	✓	✓	✓
• Program editing			✓	✓	✓	✓	✓
• Block search with/without calculation			✓	✓	✓	✓	✓
• Overstore			✓	✓	✓	✓	✓
• **Configured Stop**	**6FC5800-0BS24-0YB0**	**S24**	O	O	O	O	O

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Tools

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Tools							
Tool types:							
• Drilling/milling			✓	✓	✓	✓	✓
• Turning			✓	✓	✓	✓	✓
• Groove sawing			✓	✓	✓	✓	✓
• Grinding			✓	✓	✓	✓	✓
• Nibbling (not in SINUMERIK Operate)			–	–	–	–	–
Angle head adapter	**6FC5800-0BM56-0YB0**	**M56**	O	O	O	O	O
Tool radius compensations in plane with:							
• Approach and retract strategies			✓	✓	✓	✓	✓
• Transition circle or transition ellipse at outside corners			✓	✓	✓	✓	✓
Configurable intermediate blocks with tool radius compensation active			✓	✓	✓	✓	✓
Tool change via T number			✓	✓	✓	✓	✓
Tool carrier with orientation capability			✓	✓	✓	✓	✓
Look-ahead detection of contour violations			✓	✓	✓	✓	✓
Grinding wheel peripheral speed, programmable			✓	✓	✓	✓	✓
Tool orientation interpolation Requirement: Machining package 5 axes (option M30) or SINUMERIK ONE Dynamics 5 axes (option S43)			✓	✓	✓	✓	–
Tool length compensation, online			✓	✓	✓	✓	✓
3D tool radius compensation (3D-WRK)	**6FC5800-0BM48-0YB0**	**M48**	O[1)]	O[1)]	O	O	O

[1)] Cannot be used in combination with TRAORI and Top Surface.

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Tools (continued)							
Operation without tool management:			✓	✓	✓	✓	✓
• Tool offset selection via D number without T assignment (flat D number)			–	–	–	–	–
• Editing of tool data			✓	✓	✓	✓	✓
• Tool offset selection via T and D numbers			✓	✓	✓	✓	✓
• Tools in the tool list:			1500	1500	1500	1500	✓
• Cutting edges in the tool list:			3000	3000	3000	3000	✓
Operation with tool management:							
• Operation with tool management, up to 4 magazines			✓	✓	✓	✓	✓
• **Operation with tool management, with more than 4 magazines**	**6FC5800-0BM88-0YB0**	**M88**	O	O	O	O	O
• **Monitoring for maximum tool speed/ acceleration**	**6FC5800-0BS08-0YB0**	**S08**	O	O	O	O	O
• System displays in standard software			✓	✓	✓	✓	✓
• User-friendly commissioning via system displays			✓	✓	✓	✓	✓
• Tool list			✓	✓	✓	✓	✓
• Configurable tool list			✓	✓	✓	✓	✓
• Tools in the tool list, maximum:			1500	1500	1500	1500	✓
• Cutting edges in the tool list, maximum:			3000	3000	3000	3000	✓
• Magazine list			✓	✓	✓	✓	✓
• Configurable magazine list			✓	✓	✓	✓	✓
• Magazines, maximum			64	64	64	64	✓
• Magazine locations, maximum			600	600	600	1500	✓
• Magazine data			✓	✓	✓	✓	✓
• Empty location search and place positioning			✓	✓	✓	✓	✓
• Easy empty location search using softkeys			✓	✓	✓	✓	✓
• Loading and unloading of tools			✓	✓	✓	✓	✓
• More than one loading and unloading point per magazine			✓	✓	✓	✓	✓
• Tool life monitoring and workpiece count			✓	✓	✓	✓	✓
• Multi-tool with tools, maximum			72	72	72	72	✓
• Adapter data			✓	✓	✓	✓	✓
• Location-dependent offsets			✓	✓	✓	✓	✓
Tool identification for loading/unloading tools with code carrier	See SiePortal						
• **Tool Ident Connection**	**6FC5800-0BP52-0YB0**	**P52**	O	O	O	O	O
Identify tool demand	**6FC5800-0BM77-0YB0**	**M77**	O	O	O	O	O

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Communication and data management

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Communication and data management							
User memory for CNC programs, OEM cycles and HMI on SSD of the SIMATIC IPC for SINUMERIK Requirement: CNC user memory expanded (option P77)			–	see memory IPC	see memory IPC	see memory IPC	✓
• On external storage medium (e.g. local network, server, USB flash drive, ...) Requirement: Execution from external storage EES (option P75)			O Unlim-ited	O Unlim-ited	O Unlim-ited	O Unlim-ited	O
Data on storage medium on the rear USB interface of the operator panel, e.g. memory stick Requirement: USB connection socket			✓	✓	✓	✓	✓
Manage additional drives via:							
• Ethernet			4	4	4	4	✓
• USB			✓	✓	✓	✓	✓
COM (RS232C) serial interface			–	–	–	–	–
Data backup:							
• SD card of the NCU/PPU (backup/restore) on flash drive or on network			✓	✓	✓	✓	✓
• On the integrated SSD of the SIMATIC IPC for SINUMERIK			✓	✓	✓	✓	✓
• With Ghost (backup/restore) on SIMATIC IPC for SINUMERIK or on network			✓	✓	✓	✓	✓
Company network interface (X130):			✓	✓	✓	✓	✓
Number of interfaces / Number of ports (RJ45)			1 / 1	1 / 1	1 / 1	1 / 1	✓
• Data transfer rate X130, max.			1 Gbps	1 Gbps	1 Gbps	1 Gbps	✓
System network interface (X120):			✓	✓	✓	✓	✓
Number of interfaces / Number of ports (RJ45)			1 / 1	1 / 1	1 / 1	1 / 1	✓
• Data transfer rate X120, max.			1 Gbps	1 Gbps	1 Gbps	1 Gbps	✓
Service network interface (X127):			✓	✓	✓	✓	✓
Number of interfaces / Number of ports (RJ45)			1 / 1	1 / 1	1 / 1	1 / 1	✓
• Data transfer rate X127, max.			1 Gbps	1 Gbps	1 Gbps	1 Gbps	✓

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Communication and data management (continued)							
Networking components for supply systems ≤ 100 Mbps:	See SiePortal						
Switch SCALANCE X005 unmanaged, IP30	**6GK5005-0BA10-1AA3**		O	O	O	O	
Switch SCALANCE XC108 unmanaged, IP30	**6GK5108-0BA00-2AC2**		O	O	O	O	
Switch SCALANCE XC208 managed, IP30	**6GK5208-0BA00-2AC2**		O	O	O	O	
Switch SCALANCE X208 PRO managed, IP65	**6GK5208-0HA10-2AA6**		O	O	O	O	
Industrial Ethernet Fast Connect 2 × 2 cables	See SiePortal						
IE FC TP standard cable, GP 2×2 (PROFINET type A), TP installation cable for connection to IE FC RJ45 2×2, for universal use, 4-wire, shielded CAT 5e, by the meter	**6XV1840-2AH10**		O	O	O	O	
IE FC TP trailing cable, 2×2 (PROFINET type C), TP installation cable for connection to FC Outlet RJ 45, for trailing cable use (4 million bending cycles), 4-wire, shielded, CAT 5e, by the meter	**6XV1840-3AH10**		O	O	O	O	
FastConnect RJ45 plug 180 2×2, RJ45 plug connector, CAT 5e (10/100 Mbps) with rugged metal housing and FC connecting method, for IE FC cable 2×2	**6GK1901-1BB10-2AA0**		O	O	O	O	
IE FC trailing cable GP pre-assembled with 2 × IE FC RJ45 plug 180, length 1 m	**6XV1871-5BH10**		O	O	O	O	
IE FC trailing cable GP pre-assembled with 2 × IE FC RJ45 plug 180, length 2 m	**6XV1871-5BH20**		O	O	O	O	
IE FC trailing cable GP pre-assembled with 2 × IE FC RJ45 plug 180, length 3 m	**6XV1871-5BH30**		O	O	O	O	
IE FC trailing cable GP pre-assembled with 2 × IE FC RJ45 plug 180, length 10 m	**6XV1871-5BN10**		O	O	O	O	
IE FC trailing cable GP pre-assembled with 2 × IE FC RJ45 plug 180, length 15 m	**6XV1871-5BN15**		O	O	O	O	
IE FC trailing cable GP pre-assembled with 2 × IE FC RJ45 plug 180, length 20 m	**6XV1871-5BN20**		O	O	O	O	

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Communication and data management

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Communication and data management (continued)							
Networking components for supply systems ≤ 1000 Mbps:	See SiePortal						
SCALANCE XB005G, 5×10/100/1000 Mbps, unmanaged, IP20	**6GK5005-0GA10-1AB2**		O	O	O	O	
SCALANCE XB008G, 8×10/100/1000 Mbps, unmanaged, IP20	**6GK5008-0GA10-1AB2**		O	O	O	O	
SCALANCE XCM108PoE, 8×100/1000 Mbps, unmanaged	**6GK5108-0RA00-2AC2**		O	O	O	O	
SCALANCE XC208G, 8×10/100/1000 Mbps, managed	**6GK5208-0GA00-2AC2**		O	O	O	O	
SCALANCE XP208PoE EEC, 8 × 100 Mbps, 4 × 1000 Mbps, managed IP65	**6GK5208-0UA00-5ES6**		O	O	O	O	
Industrial Ethernet Fast Connect 4 × 2 cables	See SiePortal						
IE FC TP standard cable GP 4×2, TP installation cable CAT 6A for connection to IE FC RJ45 plug 4×2, AWG24, by the meter	**6XV1878-2A**		O	O	O	O	
IE FC TP flexible Cable GP 4×2, TP-Installationsleitung CAT 6A for connection to IE FC RJ45 plug 4×2, AWG24, by the meter	**6XV1878-2B**		O	O	O	O	
FastConnect RJ45 plug 180 4×2, RJ45 plug connector, CAT 6A (10/100/1000/10000 Mbps) with rugged metal housing and FC connection method, for IE FC cable 4×2 (AWG24)	**6GK1901-1BB12-2AA0**		O	O	O	O	
Industrial Ethernet TP cord RJ45/RJ45, CAT 6A, TP cable 4×2, pre-assembled with 2 RJ45 connectors, length 1 m	**6XV1870-3QH10**		O	O	O	O	
Industrial Ethernet TP cord RJ45/RJ45, CAT 6A, TP cable 4×2, pre-assembled with 2 RJ45 connectors, length 2 m	**6XV1870-3QH20**		O	O	O	O	
Industrial Ethernet TP cord RJ45/RJ45, CAT 6A, TP cable 4×2, pre-assembled with 2 RJ45 connectors, length 3 m	**6XV1870-3QH30**		O	O	O	O	
Industrial Ethernet TP cord RJ45/RJ45, CAT 6A, TP cable 4×2, pre-assembled with 2 RJ45 connectors, length 10 m	**6XV1870-3QN10**		O	O	O	O	
Industrial Ethernet TP cord RJ45/RJ45, CAT 6A, TP cable 4×2, pre-assembled with 2 RJ45 connectors, length 15 m	**6XV1870-3QN15**		O	O	O	O	
Industrial Ethernet TP cord RJ45/RJ45, CAT 6A, TP cable 4×2, pre-assembled with 2 RJ45 connectors, length 20 m	**6XV1870-3QN20**		O	O	O	O	

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Communication and data management (continued)							
Access MyMachine /OPC UA OPC UA server for SINUMERIK ONE	**6FC5800-0BP67-0YB0**	**P67**	O	O	O	O	O
Access MyMachine /OPC UA – OEM Second server instance of the SINUMERIK OPC UA Server for machine manufacturers and WinCC (separate user management with its own management of rights)	**6FC5800-0BP83-0YB0**	**P83**	O	O	O	O	O
Access MyMachine /OPC UA – Advanced Expansion of the connection to production management systems with tool management and pallet handling Note: Can be used as an add-on license (in addition to OPC UA standard license option P67 or option P83) or as a potential successor solution to the Create MyInterface functionality (CMI version > 3.1 SP1). Requirements: Access MyMachine /OPC UA (option P67) or Access MyMachine /OPC UA - OEM (option P83)	**6FC5800-0BP85-0YB0**	**P85**	O	O	O	O	O
Access MyMachine /OPC UA – PLC OPC UA server/client license of the PLC S7-1500 integrated in SINUMERIK ONE	**6FC5800-0BP82-0YB0**	**P82**	O	O	O	O	O
Industrial cybersecurity	See Siemens Industry Online Support		✓	✓	✓	✓	✓
SINUMERIK PLC know-how protection			✓	✓	✓	✓	–
Data at rest protection Bitlocker (for PCU), Encryption of SD cards (for PPU/NCU) Requirement: CNC software ≥ SW version 6.25			✓	✓	✓	✓	–
Security Event Log Requirement: CNC software ≥ SW version 6.24			✓	✓	✓	✓	–
Local user management in SINUMERIK ONE Define access authorization to the machine Connection to central user management (User Management Component) Requirement: CNC software ≥ SW version 6.24			✓	✓	✓	✓	–
Totally Integrated Automation User Management Component NEW User Management Server for the central administration of users and groups (optional connection to Active Directory) Requirement: CNC software ≥ SW version 6.24							
• 365 days/100 users	**6ES7823-1UE30-0YA0**		O	O	O	O	–
• 365 days/4000 users	**6ES7823-1UE10-0YA0**		O	O	O	O	–
Lock MyCycles Access protection for cycles	**6FC5800-0BP54-0YB0**	**P54**	O	O	O	O	O

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Communication and data management

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Communication and data management (continued)							
Switch SCALANCE SC622-2C Industrial security appliance for protecting equipment and networks in discrete manufacturing and the process industry for protection of industrial communication with a firewall Additional functions: Address translation (NAT/NAPT); Connection to SINEMA RC; 2 × combo port, electrical or optical; 10/100/1000 Mbps RJ45 or 1000 Mbps SFP Degree of protection IP20	**6GK5622-2GS00-2AC2**		O	O	O	O	
Switch SCALANCE SC626-2C NEW Industrial security appliance for protecting equipment and networks in discrete manufacturing and the process industry for protection of industrial communication with a firewall Additional functions: Network separation according to PROFIsafe; Address translation (NAT/NAPT); Connection to SINEMA RC; 4 × port electrical 10/100/1000 Mbps RJ45 and 2 × combo port, electrical or optical; 10/100/1000 Mbps RJ45 or 1000 Mbps SFP	**6GK5626-2GS00-2AC2**		O	O	O	O	

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
HMI components							
SINUMERIK ONE panel-based Compact CNC with multi-touch panel blackline plus for installation on the machine:							
• PPU 1740-1500, with 15" multi-touch panel blackline plus, Maximum resolution: 1366 × 768 px Mounting position: horizontal Enclosure front dimensions: 398 × 257 mm Installation dimensions: 382 × 241 × 129 mm	**6FC5317-4AA00-1CA0**		O	–	–	–	O
• PPU 1740-1900, with 19" multi-touch panel blackline plus, Maximum resolution: 1920 × 1080 px Mounting position: horizontal Enclosure front dimensions: 464 × 294 mm Installation dimensions: 448 × 278 × 129 mm	**6FC5317-4AA00-1DA0**		O	–	–	–	O
Operator panels blackline plus in new neutral design SIMATIC Thin Client Operator Panel for SINUMERIK ONE: Widescreen TFT display, capacitive touch sensor							
• SIMATIC ITC1200 V3, Industrial Thin Client, 12" Maximum resolution: 1280 × 800 px Mounting position: horizontal Enclosure front dimensions: 326 × 237 mm Installation dimensions: 310 × 221 × 72 mm	NEW **6AV6646-1BA12-0NA1**		O	O	O	O	O
• SIMATIC ITC1500 V3, Industrial Thin Client, 15" Maximum resolution: 1366 × 768 px Mounting position: horizontal Enclosure front dimensions: 398 × 257 mm Installation dimensions: 382 × 241 × 73 mm	**6AV6646-1BA15-0NA1**		O	O	O	O	O
• SIMATIC ITC1500 V3, Industrial Thin Client, 15" Maximum resolution: 1920 × 1080 px Mounting position: horizontal and vertical Enclosure front dimensions: 398 × 257 mm Installation dimensions: 382 × 241 × 73 mm	NEW **6AV6646-1BA16-0NA1**		–	O	O	O	O
• SIMATIC ITC1900 V3, Industrial Thin Client, 19" Maximum resolution: 1366 × 768 px Mounting position: horizontal Enclosure front dimensions: 464 × 294 mm Installation dimensions: 448 × 278 × 73 mm	**6AV6646-1BA18-0NA1**		O[1)]	O	O	O	O
• SIMATIC ITC1900 V3, Industrial Thin Client, 19" Maximum resolution: 1920 × 1080 px Mounting position: horizontal and vertical Enclosure front dimensions: 464 × 294 mm Installation dimensions: 448 × 278 × 73 mm	NEW **6AV6646-1BA20-0NA1**		–	O	O	O	O
• SIMATIC ITC2200 V3, Industrial Thin Client, 22" Maximum resolution: 1920 × 1080 px Mounting position: horizontal and vertical Enclosure front dimensions: 529 × 331 mm Installation dimensions: 513 × 315 × 73 mm	**6AV6646-1BA22-1NA1**		–	O	O	O	O
• SIMATIC ITC2400 V3, Industrial Thin Client, 24" Maximum resolution: 1920 × 1080 px Mounting position: horizontal and vertical Enclosure front dimensions: 585 × 363 mm Installation dimensions: 569 × 347 × 73 mm	NEW **6AV6646-1BA24-0NA1**		–	O	O	O	O

[1)] Only applies for PPU 1740-1900.

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HMI components

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
HMI components (continued)							
SIMATIC Flat Panels for SINUMERIK ONE: Multi-touch display (16:9), built-in unit for 24 V DC, DisplayPort/DVI interface, can be positioned up to 100 m away with HDBaseT, USB interface on the rear Maximum resolution: 1920 × 1080 px							
• SIMATIC IFP1500 V2 extended 15" Mounting position: horizontal and vertical Enclosure front dimensions: 398 × 257 mm Installation dimensions: 382 × 241 × 63.4 mm Requirement: IPC	**6AV7863-5MA10-2NA0**		–	O	O	O	O
• SIMATIC IFP1900 V2 extended 19" Mounting position: horizontal and vertical Enclosure front dimensions: 464 × 294 mm Installation dimensions: 448.6 × 278.6 × 63.4 mm Requirement: IPC	**6AV7863-6MA10-2NA0**		–	O	O	O	O
• SIMATIC IFP2200 V2 extended 22" Mounting position: horizontal and vertical Enclosure front dimensions: 529 × 331 mm Installation dimensions: 513 × 315 × 63.4 mm Requirement: IPC	**6AV7863-4MA10-2NA0**		–	O	O	O	O
• SIMATIC IFP2400 V2 extended 24" Mounting position: horizontal and vertical Enclosure front dimensions: 585 × 363 mm Installation dimensions: 569.6 × 347.6 × 63.4 mm Requirement: IPC	**6AV7863-7MA10-2NA0**		–	O	O	O	O
Accessories for SIMATIC Flat Panel Monitor V2							
• Cable set, comprising CAT6 cable, host module V2 (HDBaseT) Length 10 m Length 15 m Length 20 m Length 30 m	**6AV7860-1EX21-0AB1** **6AV7860-1EX21-5AB1** **6AV7860-1EX22-0AB1** **6AV7860-1EX23-0AB1**		–	O	O	O	
• Extended V2 host module, HDBaseT V2 transceiver Display port, USB to HDBaseT, 24 V power supply	**6AV7860-3EH00-0AA0**		–	O	O	O	
• CAT6A cable, sold by the meter and non-pre-assembled, without plug	**6XV1878-2C**		–	O	O	O	
• CAT6A (RJ45) plug for assembly of Ethernet cable by customer without tools, 2 units	**6AV7860-1EY00-0AA0**		–	O	O	O	

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Description ✓ Basic version I O Option I – Not available		Article No. Product ID	Order code	SINUMERIK ONE				
				PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
HMI components (continued)								
SIMATIC Panel PCs with integrated IPC for SINUMERIK ONE:								
SIMATIC IPC277G, nanopanel PC, multi-touch, Microsoft Windows 10, LTSC 2021 Maximum resolution: 1920 × 1080 px								
• SIMATIC IPC277G, nanopanel PC, 15", slim Atom X6413E (4-core), 8 GB, 256 GB SSD, Mounting position: horizontal Enclosure front dimensions: 398 × 257 mm Installation dimensions: 382 × 241 × 87.3 mm Requirement: SINUMERIK PCU base software ≥ V15 SP2 (option P86)	NEW	6AV7886-0TE24-2AC0		–	O	O	O	O
• SIMATIC IPC277G, nanopanel PC, 19", slim Atom X6413E (4-core), 8 GB, 256 GB SSD, Mounting position: horizontal Enclosure front dimensions: 464 × 294 mm Installation dimensions: 448 × 278 × 87.3 mm Requirement: SINUMERIK PCU base software ≥ V15 SP2 (option P86)	NEW	6AV7886-0UE24-2AC0		–	O	O	O	O
SIMATIC IPC277G, nanopanel PC, multi-touch, Microsoft Windows 11 IoT LTSC 2024 Maximum resolution: 1920 × 1080 px								
• SIMATIC IPC277G, Nanopanel PC, 19", slim Atom X6413E (4-core), 8 GB, 256 GB SSD, Mounting position: horizontal Enclosure front dimensions: 464 × 294 mm Installation dimensions: 448 × 278 × 87.3 mm Requirements: CNC software ≥ SW version 6.25 SP1 SINUMERIK PCU base software ≥ V16 (option P86)	NEW	6AV7886-0UE54-2AD0		–	O	O	O	O
SIMATIC IPC PX-39A, multi-touch, Microsoft Windows 10 IoT LTSC 2021 Maximum resolution: 1920 × 1080 px								
• SIMATIC IPC PX-39A, 22", Intel Xeon W-11155MLE (4-core), 8 GB, 256 GB SSD, Mounting position: horizontal and vertical Enclosure front dimensions: 529 × 331 mm Installation dimensions: 513 × 315 × 101 mm Requirement: SINUMERIK PCU base software ≥ V15 SP1 (option P86)	NEW	6AV7242-3NB10-1AA0		–	O	O	O	O
• SIMATIC IPC PX-39A, 24", Intel Xeon W-11155MLE (4-core), 16 GB, 256 GB SSD, Mounting position: horizontal and vertical Enclosure front dimensions: 585 × 361 mm Installation dimensions: 569 × 347 × 101 mm Requirement: SINUMERIK PCU base software ≥ V15 SP1 (option P86)	NEW	6AV7242-3PC10-1AA0		–	O	O	O	O
SIMATIC IPC PX-39A, multi-touch, Microsoft Windows 11 IoT LTSC 2024 Maximum resolution: 1920 × 1080 px								
• SIMATIC IPC PX-39A, 22", Intel Xeon W-11155MLE (4-Core), 8 GB, 256 GB SSD, Mounting position: horizontal and vertical Enclosure front dimensions: 529 × 331 mm Installation dimensions: 513 × 315 × 101 mm Requirement: SINUMERIK PCU base software ≥ V16 (option P86)	NEW	6AV7242-3NB10-3AA0		–	O	O	O	O
• SIMATIC IPC PX-39A, 24", Intel Xeon W-11155MLE (4-Core), 16 Gbyte, 256 Gbyte SSD, Mounting position: horizontal and vertical Enclosure front dimensions: 585 × 361 mm Installation dimensions: 569 × 347 × 101 mm Requirement: SINUMERIK PCU base software ≥ V16 (option P86)	NEW	6AV7242-3PC10-3AA0		–	O	O	O	O

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HMI components

Description ✓ Basic version I O Option I – Not available		Article No. Product ID	Order code	SINUMERIK ONE				
				PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
HMI components (continued)								
SIMATIC IPC for SINUMERIK ONE								
Microsoft Windows 10 IoT LTSC 2021								
• SIMATIC IPC Standard Performance SIMATIC IPC227G, Atom X6413E (4-core), 8 GB, 256 GB SSD Mounting type: Vertical mounting Requirement: SINUMERIK PCU base software ≥ V15 SP2 (option P86)	NEW	**6ES7647-8CE21-2AC3**		–	O	O	O	O
• SIMATIC IPC High Performance SIMATIC BX-39A, Intel Xeon W-11155MLE (4-core), 8 GB, 256 GB SSD, Mounting type: Vertical mounting Requirement: SINUMERIK PCU base software ≥ V15 (option P86)	NEW	**6AG4142-3HB10-1AA0**		–	O	O	O	O
• SIMATIC IPC High Performance plus SIMATIC BX-39A, Intel Xeon W-11865MLE (8-core), 16 GB, 256 GB SSD, Mounting type: Vertical mounting Requirement: SINUMERIK PCU base software ≥ V15 (option P86)	NEW	**6AG4142-7HC10-2AA0**		–	O	O	O	O
Microsoft Windows 11 IoT LTSC 2024								
• SIMATIC IPC Standard Performance SIMATIC IPC227G, Atom X6413E (4-core), 8 GB, 256 GB SSD Mounting type: Vertical mounting Requirement: CNC software ≥ SW version 6.25 SP1 SINUMERIK PCU base software ≥ V16 (option P86)	NEW	**6ES7647-8CE51-2AD3**		–	O	O	O	O
• SIMATIC IPC SIMATIC BX-39A, Intel Xeon W-11155MLE (4-core), 8 GB, 256 GB SSD, Mounting type: Vertical mounting Requirement: SINUMERIK PCU base software ≥ V16 (option P86)	NEW	**6AG4142-3HB10-3AA0**		–	O	O	O	O
• SIMATIC IPC SIMATIC BX-39A, Intel Xeon W-11865MLE (8-core), 16 GB, 256 GB SSD, Mounting type: Vertical mounting Requirement: SINUMERIK PCU base software ≥ V16 (option P86)	NEW	**6AG4142-7HC10-4AA0**		–	O	O	O	O

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Description ✓ Basic version I O Option I – Not available		Article No. Product ID	Order code	SINUMERIK ONE				
				PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
HMI components (continued)								
Machine Control Panels								
• SINUMERIK ONE MCP 1500 for SIMATIC Operator Panel for SINUMERIK ONE - 15" - Horizontal mounting position - 2 × Powerride - Enclosure front dimensions: 398 × 225 mm Installation dimensions: 379 × 206 × 70 mm - includes the option Configured Stop (option S24) for using the Powerride function		**6FC5303-0AP01-0AA0**		O	O	O	O	O
• SINUMERIK ONE MCP 1500.2 for SIMATIC Operator Panel for SINUMERIK ONE - 15" - Horizontal mounting position - 1 × Powerride - Enclosure front dimensions: 398 × 165 mm Installation dimensions: 364 × 131 × 70 mm - includes the option Configured Stop (option S24) for using the Powerride function	NEW	**6FC5303-0AP01-0BA0**		O	O	O	O	O
• SINUMERIK ONE MCP 1900 for SIMATIC Operator Panel for SINUMERIK ONE - 19" - Horizontal mounting position - 1 × Powerride - Enclosure front dimensions: 464 × 165 mm Installation dimensions: 430 × 131 × 70 mm - includes the option Configured Stop (option S24) for using the Powerride function		**6FC5303-0AP02-0AA0**		O	O	O	O	O
• SINUMERIK ONE MCP 1900.2 for SIMATIC Operator Panel for SINUMERIK ONE - 19" - Horizontal mounting position - 2 × Powerride - Enclosure front dimensions: 464 × 165 mm Installation dimensions: 430 × 131 × 70 mm - includes the option Configured Stop (option S24) for using the Powerride function	NEW	**6FC5303-0AP02-0BA0**		O	O	O	O	O
• SINUMERIK ONE MCP 2200 for SIMATIC Operator Panel for SINUMERIK ONE - 22" - Horizontal mounting position - with QWERTY keyboard and Powerride - Enclosure front dimensions: 529 × 265 mm Installation dimensions: 495 × 231 × 55 mm - includes the option Configured Stop (option S24) for using the Powerride function		**6FC5303-0AP03-0AA0**		O	O	O	O	O
• SINUMERIK ONE MCP 2200c for SIMATIC Operator Panel for SINUMERIK ONE - 22" - Horizontal mounting position, custom made - Enclosure front dimensions: 529 × 165 mm Installation dimensions: 495 × 131 × 70 mm - includes the option Configured Stop (option S24) for using the Powerride function		**6FC5303-0AP51-0DA0**		O	O	O	O	O
• SINUMERIK ONE MCP 2400 for SIMATIC Operator Panel for SINUMERIK ONE - 24" - Horizontal mounting position - with QWERTY keyboard and Powerride - Enclosure front dimensions: 585 × 265 mm Installation dimensions: 551 × 231 × 55 mm - includes the option Configured Stop (option S24) for using the Powerride function		**6FC5303-0AP04-0AA0**		O	O	O	O	O

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HMI components

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
HMI components (continued)							
• SINUMERIK ONE KB 1500 CNC keyboard (QWERTY) for SIMATIC Operator Panel for SINUMERIK ONE - 15" - Enclosure front dimensions: 398 × 135 mm Installation dimensions: 382 × 119 × 27 mm	**6FC5303-0AP47-0AA0**		O	O	O	O	O
• SINUMERIK ONE KB 1900 CNC keyboard (QWERTY) for SIMATIC Operator Panel for SINUMERIK ONE - 19" - Enclosure front dimensions: 464 × 135 mm Installation dimensions: 448 × 119 × 27 mm	**6FC5303-0AP47-0BA0**		O	O	O	O	O
• SINUMERIK ONE EM 464 expansion module with 13 slots for equipment, version with dummy elements for customer's own components Dimensions: 464 × 95 × 38 mm	**6FC5303-1AF68-0AA0**		O	O	O	O	O
• SINUMERIK ONE EM 464/S01 expansion module with 7 slots for equipment, version with dummy elements for customer's own components Dimensions: 464 × 95 × 38 mm	**6FC5303-1AF68-0AB0**		O	O	O	O	O
• SINUMERIK ONE key caps, square, for inserting labeling plates - Set consisting of: 90 units, clear - For more information and examples, see: https://support.industry.siemens.com/cs/ww/en/view/109802036	**6FC5248-0AF21-0AA0**		O	O	O	O	O
• SINUMERIK ONE key caps, square, can be laser-labeled - Set consisting of: 90 units, iron gray - For more information and examples, see: https://support.industry.siemens.com/cs/ww/en/view/109802036	**6FC5303-0AF27-1AA0**		O	O	O	O	O
• SINUMERIK ONE key caps, square, can be laser-labeled - Set consisting of: 49 units, iron gray	**6FC5348-0AP00-3AA0**		O	O	O	O	O
• SINUMERIK ONE key caps, laser-labeled with standard symbols, 49 units, iron gray - Full set of all key caps: 15 units operating mode block 15 units customer key block 15 units axis block 2 units Feed (start & stop) 2 units Spindle (start & stop)	**6FC5348-0AP00-3AB0**		O	O	O	O	O
• SINUMERIK ONE key caps, laser-labeled with symbols of the main functions, 49 units, iron gray - Set contains 7 units each of: Reset, Cycle Stop, Cycle Start, Feed Stop, Feed Start, Spindle Stop, Spindle Start	**6FC5348-0AP00-3AC0**		O	O	O	O	O
• SINUMERIK ONE key caps, laser-labeled with symbols for TURNING, 49 units, iron gray - Set contains 6 units each of: +C, -C, +X, -X, +Y, -Y, +Z, -Z, 1 unit blank	**6FC5348-0AP00-3AD0**		O	O	O	O	O
• SINUMERIK ONE MCP connector set X515 / X516 - set consisting of 6 units, 3 × X515, 3 × X516	**6FC5348-0AP00-0AA0**		O	O	O	O	
• Connection kit for mini HHU, pre-assembled, with connection plug for ONE MCP	NEW **6FC5348-0AP00-0AB0**		O	O	O	O	

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
HMI components (continued)							
Machine Control Panels for operator panels blackline plus:							
• SINUMERIK MCP 398C Enclosure front dimensions: 398 × 225 mm Installation dimensions: 379.5 × 206.5 × 73 mm (for 15" operator panels, to be connected via X120):	**6FC5303-0AF25-0AA0**		O	O	O	O	O
- EM 66 expansion module (for 19" operator panels MCP 398C + EM 66)	**6FC5303-0AF30-0AA0**		O	O	O	O	O
- EM 131 expansion module (for 22" operator panels MCP 398C + EM 131)	**6FC5303-0AF31-0AA0**		–	O	O	O	O
- EM 187 expansion module (for 24" operator panels MCP 398C + EM 187)	**6FC5303-0AF28-0AA0**		–	O	O	O	O
Accessories for operator panels:							
• Terminal kit for PPU 1740-1500 and PPU 1740-1900, (1 × 24 V connector, 4 × I/O connectors, 4 × dust protection blanking plugs for DRIVE-CLiQ ports)	**6SL3064-4CB00-0AA0**		O	–	–	–	–
• Hinged cover for PPU 1740-1500, PPU 1740-1900 (spare part)	**6FC5348-0AA30-4AA0**		O	–	–	–	–
• Service package mounting clips for SIMATIC IFP, SIMATIC IPC 477E	**6AV6881-6AD10-0AA0**		O	O	O	O	O
Accessories for MCP 398C machine control panels:							
• Cable set for additional machine control panel control devices, length 500 mm (1 set = 60 units)	**6FC5247-0AA35-0AA0**		O	O	O	O	O
• Spindle/rapid traverse override rotary switch, 1×16G, T=24, cap, button, pointer, rapid-traverse and spindle dials	**6FC5247-0AF12-1AA0**		O	O	O	O	O
• Feedrate/rapid traverse override rotary switch, 1×23G, T=32, cap, button, pointer, rapid traverse and feedrate dials	**6FC5247-0AF13-1AA0**		O	O	O	O	O
• Connection kit for mini HHU, pre-assembled, with connection plug for MCP 398C	**6FX2006-1BG11**		O	O	O	O	O
MPP Machine Push Button Panel with machine control panel functions:							
• SINUMERIK MPP 398 IEH - with connection for HT 8/HT 2 - with feedrate override	**6FC5303-1AF50-8AA0**		O	O	O	O	O
• SINUMERIK MPP 464 IEH - with connection for HT 8/HT 2 - with feedrate override	**6FC5303-1AF60-8AA0**		O	O	O	O	O
• SINUMERIK MPP 1500E - with connection for HT 8 - with EKS	**6FC5303-1AF31-0AA0**		O	O	O	O	O
Interface for customer machine control panel SINUMERIK MCP Interface PN	**6FC5303-0AF03-0AA0**		O	O	O	O	O

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HMI components

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
HMI components (continued)							
Electronic handwheels: Note: Third handwheel can be operated as contour handwheel			2 (3)[1]	2 (3)[2]	2 (3)[2]	2 (3)[2]	✓
• With 120 mm × 120 mm front panel, 5 V DC, RS422	**6FC9320-5DB01**		O	O	O	O	O
• With 76.2 mm × 76.2 mm front panel, 5 V DC, RS422	**6FC9320-5DC01**		O	O	O	O	O
• With 76.2 mm × 76.2 mm front panel, 24 V DC, HTL	**6FC9320-5DH01**		O	O	O	O	O
• Without front panel, without setting wheel, 5 V DC, RS422	**6FC9320-5DF01**		O	O	O	O	O
• Without front panel, with setting wheel, 5 V DC, RS422	**6FC9320-5DM00**		O	O	O	O	O
• Portable in housing, 2.5 m spiral cable, 5 V DC, RS422	**6FC9320-5DE02**		O	O	O	O	O
Accessories for handwheels:							
• Signal cable, pre-assembled for connection of a handwheel to the COM board fitted in MCP 398C							
- Length 0.2 m	**6FX8002-2CP00-1AA2**		O	O	O	O	O
- Length 0.3 m	**6FX8002-2CP00-1AA3**		O	O	O	O	O
- Length 0.7 m	**6FX8002-2CP00-1AA7**		O	O	O	O	O
- Length 1 m	**6FX8002-2CP00-1AB0**		O	O	O	O	O
- Length 2 m	**6FX8002-2CP00-1AC0**		O	O	O	O	O
- Length 3 m	**6FX8002-2CP00-1AD0**		O	O	O	O	O
- Length 5 m	**6FX8002-2CP00-1AF0**		O	O	O	O	O
- Length 7 m	**6FX8002-2CP00-1AH0**		O	O	O	O	O
- Length 10 m Other lengths available	**6FX8002-2CP00-1BA0**		O	O	O	O	O
• Signal cable, pre-assembled for connection of a handwheel to the COM board fitted in SINUMERIK ONE MCP							
- Length 0.2 m	**6FX8002-2CP01-1AA2**		O	O	O	O	O
- Length 0.3 m	**6FX8002-2CP01-1AA3**		O	O	O	O	O
- Length 0.7 m	**6FX8002-2CP01-1AA7**		O	O	O	O	O
- Length 1 m	**6FX8002-2CP01-1AB0**		O	O	O	O	O
- Length 2 m	**6FX8002-2CP01-1AC0**		O	O	O	O	O
- Length 3 m	**6FX8002-2CP01-1AD0**		O	O	O	O	O
- Length 5 m	**6FX8002-2CP01-1AF0**		O	O	O	O	O
- Length 7 m	**6FX8002-2CP01-1AH0**		O	O	O	O	O
- Length 10 m Other lengths available	**6FX8002-2CP01-1BA0**		O	O	O	O	O
• Flange socket for portable handwheel	**6FC9341-1AQ**		O	O	O	O	O
• Adapter set for three-point fixing of the handwheel	**6FC9320-5DN00**		O	O	O	O	O
• Holder for mini handheld unit 6FX2007-1AD03 / -1AD13 and electronic handwheel in housing 6FC9320-5DE02	**6FX2006-1BG70**		O	O	O	O	O
• Signal connector, 9-pin	**6FX2003-0SA01**		O	O	O	O	O
Siemens control panel solutions Customized Operator Panels (cOP) and customized Machine Control Panels (cMCP)	See SiePortal		O	O	O	O	O
Electrical equipment for machines and plants Panel building for machine tools at the Siemens Systems Engineering Plant Chemnitz (WKC)	See SiePortal		O	O	O	O	O

[1] On PPU on-board.

[2] For NCU via MCP or connection box.

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
HMI components (continued)							
Handheld units:							
SINUMERIK HT 10 handheld terminal Screen size 10.1", capacitive multi-touch, 16:10 format, resolution 1280 × 800 px, 7 mechanical keys, 1 non illuminated EMERGENCY STOP button, 1 three-step enabling button, IP65	**6FC5403-0AA21-0AA1**		O	O	O	O	O
SINUMERIK HT 10 handheld terminal additionally equipped with handwheel	**6FC5403-0AA21-1AA1**		O	O	O	O	O
• Connection cable for HT 10			O	O	O	O	O
- Length 2 m	**6XV1440-4BH20**						
- Length 5 m	**6XV1440-4BH50**						
- Length 8 m	**6XV1440-4BH80**						
- Length 10 m	**6XV1440-4BN10**						
- Length 15 m	**6XV1440-4BN15**						
- Length 20 m	**6XV1440-4BN20**						
- Length 25 m	**6XV1440-4BN25**						
• Wall holder for HT 10 handheld terminal	**6FC5348-0AA20-0AA0**		O	O	O	O	O
• Touch pen for HT 10 ELO touch pen system, including adhesive holder	**6AV6881-0AV21-0AA0**		O	O	O	O	O
• Touch pen caps for ELO touch pen system	**6AV2181-8AV60-0AX0**		O	O	O	O	O
SINUMERIK HT 2 handheld terminal with 128 × 64 pixel display	**6FC5303-0AA00-2AA0**		O	O	O	O	O
• Magnetic clamp for HT 2	**6FC5348-0AA08-0AA0**		O	O	O	O	O
• Holder for HT 2	**6FC5348-0AA08-1AA0**		O	O	O	O	O
• Slide-in labels for HT 2, inscribable (3 A4 sheets)	**6FC5348-0AA08-2AA0**		O	O	O	O	O
• Connection module/connection cable for SINUMERIK HT 2/HT 8 handheld terminals			O	O	O	O	O
- Basic PN connection module for HT 2/HT 8 without EMERGENCY STOP override for mounting in the control cabinet with switch	**6FC5303-0AA01-1AA0**		O	O	O	O	O
- Spiral connection cable For SINUMERIK HT 2/HT 8 Length 1.5 m, extendable to 3.5 m	**6FC5348-0AA08-3AA0**		O	O	O	O	O
Accessories for Handheld Terminals:							
• SIMATIC HMI connection box standard for SINUMERIK HT 10, SINUMERIK HT 2 (>FS:10) and SIMATIC Mobile Panels without emergency stop circuit bridging, wall mounting, PROFINET and PROFIsafe, connection of safety relay, terminal detection, integrated switch, degree of protection IP65	**6AV2125-2AE13-0AX0**		O	O	O	O	O
• SIMATIC HMI connection box advanced for SINUMERIK HT 10, SINUMERIK HT 2 (>FS:10) and SIMATIC Mobile Panels with emergency stop circuit bridging, wall mounting, PROFINET and PROFIsafe, connection of safety relay, terminal detection, integrated switch, MRP-Client, IRT, degree of protection IP65	**6AV2125-2AE23-0AX0**		O	O	O	O	O
• SIMATIC HMI connection box distributed for SINUMERIK HT 10, SINUMERIK HT 2 and SIMATIC Mobile Panels with emergency stop circuit bridging, with Panel Present signal, wall mounting, PROFINET and PROFIsafe, connection of safety relay, terminal detection, integrated switch, MRP-Client, IRT, degree of protection IP65	NEW **6AV2185-5AE23-1AA0**		O	O	O	O	O

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HMI components

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
HMI components (continued)							
Mini handheld unit EMERGENCY STOP button (2-channel), enabling button (3-step), metal connector, handwheel with magnetic latching							
• With coiled connecting cable Length 2.1 m, extendable to 3.5 m	**6FX2007-1AD03**		O	O	O	O	O
• With straight cable, length 5 m	**6FX2007-1AD13**		O	O	O	O	O
Connection kit for mini HHU, non-assembled, with connection plug	**6FX2006-1BG03**		O	O	O	O	O
Connection kit for mini HHU, pre-assembled, with connection plug for SINUMERIK PPU 1740	**6FX2006-1BG20**		O	O	O	O	O
Adapter plate for connection kit for mini handheld unit non-assembled, plastic on metal version	**6FX2006-1BG45**		O	O	O	O	O
90 degree angle socket for connection kit, non-assembled, metal version	**6FX2006-1BG56**		O	O	O	O	O

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Operation							
SINUMERIK Operate /NCU Machine operation with SINUMERIK Operate on a SINUMERIK ONE NCU or PPU Single license, without data storage medium for approved SINUMERIK/SIMATIC Panels	**6FC5800-0BS00-0YB0**	**S00**	O	O	O	O	O
SINUMERIK Operate /PCU Machine operation with SINUMERIK Operate on a SINUMERIK or SIMATIC IPC Software option, supply of an electronic license (PDF) Requirement: SINUMERIK PCU base software /IPC (option P86) Note: SINUMERIK Operate, see CNC software DVD-ROM	**6FC5800-0BP88-0YB0**	**P88**	–	O	O	O	O
SINUMERIK PCU base software /IPC version ≤ 14 for SIMATIC IPC with Microsoft Windows 10 Software option, supply of an electronic license (PDF) Requirement: SIMATIC IPC for SINUMERIK SINUMERIK Operate (option P88) with CNC software ≤ 6.15 SP2/6.20/6.21	**6FC5800-0BP86-0YB0**	**P86**	–	O	O	O	O
SINUMERIK PCU base software /IPC version ≥ 15 for SIMATIC IPC with Microsoft Windows 10 / 11 Software option, supply of an electronic license (PDF) Requirement: SIMATIC IPC for SINUMERIK Microsoft Windows 10 LTSC 2021 (21H2) / Microsoft Windows 11, SINUMERIK Operate (option P88) with CNC software ≥ 6.15 SP3/6.21	**6FC5800-0BP86-1YB0**	**P86**	–	O	O	O	O
SINUMERIK Operate /PC Software for machine operation with SINUMERIK Operate on PC License key on SD card of NCU/PPU, Single license without data storage medium for general PCs, Note: SINUMERIK Operate on CNC software DVD-ROM Option P87 cannot be used in combination with EES (option P75) and CNC user memory expanded (option P77)	**6FC5800-0BP87-0YB0**	**P87**	–	O	O	O	O
SINUMERIK Operate /universal client Machine operation with SINUMERIK Operate on universal thin clients (SINUMERIK Operate on NCU/PPU) Single license, without data storage medium for general panels	**6FC5800-0BS87-0YB0**	**S87**	–	O	O	O	O
SINUMERIK Operate /PC (**A**utomation **L**icence **M**anager) Software for machine operation with PC: Machine operation with SINUMERIK Operate /PC, License key locally on PC, Single license without data storage medium for general PCs, Alternative to option P87 Note: SINUMERIK Operate, see CNC software DVD-ROM	**6FC5860-2YF00-1YB0**		–	O	O	O	O
SINUMERIK Operate Display Manager Multi-window technology for SINUMERIK Operate Single license, without data storage medium Requirement: SINUMERIK Operate /universal client (option S87) or SINUMERIK Operate /NCU (option S00) or SINUMERIK Operate /PCU (option P88) or SINUMERIK Operate /PC (option P87) Note: Resolution > 1920 × 1080 px	**6FC5800-0BP81-0YB0**	**P81**	–	O	O	O	O

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Operation

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Operation (continued)							
Sidescreen			✓	✓	✓	✓	✓
Tooltips			✓	✓	✓	✓	✓
Control unit management:							
• Identical display on all OPs with TCU:			✓	✓	✓	✓	✓
- Simultaneous operation interlock			✓	✓	✓	✓	✓
- Activate/deactivate MCP/MPP			✓	✓	✓	✓	✓
- Different resolutions			✓	✓	✓	✓	✓
Number of operator panel fronts with TCU on an NCU/PPU			2	2	4	4	✓
Number of operator panel fronts with TCU on a SIMATIC IPC for SINUMERIK (including 1 operator panel front directly on the IPC, e.g. SIMATIC IFP):			–	5	5	5	✓
• From 2/4 operator panel fronts, as many operator panel fronts as required due to intelligent suppression			✓	✓	✓	✓	✓
• One integrated HMI and one SINUMERIK Operate simultaneously on one NCU/PPU (only for loading/unloading tools)			✓	✓	✓	✓	✓
• One or several TCUs which can be switched over via several NCUs/PPUs and SIMATIC IPCs for SINUMERIK			–	–	–	–	–
• One HMI switchable via several NCUs/PPUs Requirement: CNC SW ≥ 6.22			✓	✓	✓	✓	✓
Operation via VNC viewer SINUMERIK VNC Viewer for PC/PG on DVD	**6FC5852-1.C..-0YA8**		✓	✓	✓	✓	✓
Plain text display of user variables			✓	✓	✓	✓	✓
Multi-channel display, number of machining channels			4	4	4	4	✓
Workpiece-related actual value system			✓	✓	✓	✓	✓
CNC program messages			✓	✓	✓	✓	✓
Automatic, time-controlled screen blanking The function can be used to save energy. Requirement: SINUMERIK Operate			✓	✓	✓	✓	✓
Access protection			7–level	7–level	7–level	7–level	✓
Electronic Key System EKS Single license, without data storage medium	**6FC5800-0BP53-0YB0**	**P53**	O	O	O	O	O

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Operation (continued)							
Languages of the operating software SINUMERIK Operate and SINUMERIK ONE: • Chinese Simplified, English, French, German, Italian, Spanish			✓	✓	✓	✓	✓
• Language switchover online			✓	✓	✓	✓	✓
• Pre-installed languages (for IPC without limitation)			✓	✓	✓	✓	✓
Additional languages of the operating software SINUMERIK Operate and SINUMERIK ONE [1]	**6FC5800-0BN00-0YB0**	**N00**	O	O	O	O	O
Language extensions for the operating software SINUMERIK Operate and SINUMERIK ONE, on DVD-ROM without license: • Languages for CNC-SW 6.15 SPx/6.23: Bulgarian, Chinese Traditional, Croatian, Czech, Danish, Dutch, Finnish, Greek, Hindi, Hungarian, Indonesian, Japanese, Korean, Malay, Polish, Portuguese/Brazilian, Romanian, Russian, Slovak, Slovenian, Swedish, Tamil, Thai, Turkish, Vietnamese	**6FC5860-0YC45-0YB0**		O	O	O	O	O
SINUMERIK Collection User, manufacturer, and service documentation for download and displayed in the browser, from CNC SW 6.25 Language: German	**6FC5397-0AE16-2AA0**		O	O	O	O	O
SINUMERIK Collection User, manufacturer, and service documentation for download and displayed in the browser, from CNC SW 6.25 Language: English	**6FC5397-0AE16-2BA0**		O	O	O	O	O
SINUMERIK compile cycles General notes and documentation Languages: English, French, German, Italian, Portuguese (BR), Spanish	See Siemens Industry Online Support		O	O	O	O	O
SINUMERIK documentation Provision of documentation for further processing (copyright) Delivery of a license Valid for use on one control (see license conditions)	**6FC5395-0BP00-0XB0**		O	O	O	O	O

[1] Available languages and SW versions on request.

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Monitoring functions

Description ✓ Basic version I O Option I – Not available		Article No. Product ID	Order code	SINUMERIK ONE				
				PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Monitoring functions								
Working area limitation				✓	✓	✓	✓	✓
Limit switch monitoring Software and hardware limit switches				✓	✓	✓	✓	✓
Position monitoring				✓	✓	✓	✓	✓
Standstill monitoring				✓	✓	✓	✓	✓
Clamping monitoring				✓	✓	✓	✓	✓
2D/3D protection areas				✓	✓	✓	✓	✓
Contour monitoring				✓	✓	✓	✓	✓
Axis limitation from the PLC				✓	✓	✓	✓	✓
Spindle speed limitation				✓	✓	✓	✓	✓
Run MyCC /GPM [1] Geometry monitoring interface Requirement: Loadable compile cycle		**6FC5800-0BM83-0YB0**	**M83**	O	O	O	O	–
Protect MyMachine /CC Axis Monitor Axis collision protection Requirement: Loadable compile cycle		**6FC5800-0BN06-0YB0**	**N06**	O	O	O	O	–
Protect MyMachine /3D Primitives Control-integrated function for collision avoidance on machine tools Protection area elements are primitive bodies (cube, cylinder, sphere, ...). Monitoring for collisions for up to 64 protection elements, 64 protection areas and 20 collision pairs in the 1st channel		**6FC5800-0BS03-0YB0**	**S03**	O	O	O	O	O
Protect MyMachine /3D Primitives, additional channel Software options for the 2nd to 10th channel, can be used in each module up to the max. number of channels (see Maximum configuration of machining channels) Monitoring of collisions across multiple channels Requirements: Protect MyMachine /3D Primitives (option S03), Machining channel, each additional (option C11 ... C19)		**6FC5800-0BE00-0YB0**	**E01 ... E09**	O	O	O	O	O
Protect MyMachine /3D STL Control-integrated function for collision avoidance on machine tools Protection area elements are primitive bodies (cube, cylinder, sphere, ...) or bodies in STL format in the 1st channel.	NEW	**6FC5800-0BS02-0YB0**	**S02**	O [2]	O [2]	O	O	O
Protect MyMachine /3D STL, additional channel Software options for the 2nd to 10th channel, can be used in each module up to the max. number of channels (see Maximum configuration of machining channels) Monitoring of collisions across multiple channels Requirements: Protect MyMachine /3D STL (option S02), Machining channel, each additional (option C11 ... C19)	NEW	**6FC5800-0BG00-0YB0**	**G01 ... G09**	O [2]	O [2]	O	O	O

[1] Compile cycle that is not generally released. Requires machine-specific release process. Please contact your local Siemens sales office or Regional Company.

[2] From CNC software version 6.25, combination with Protect MyMachine /3D STL is possible. Excluded with CNC software version ≤ 6.24.

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Monitoring functions (continued)							
Protect MyMachine /Open Control-integrated interface for connecting the external collision avoidance, such as Protect MyMachine /3D Twin, or CAS from ModuleWorks Software option for 1st channel Requirements: SIMATIC IPC, ModuleWorks package (ModuleWorks GmbH), OPC UA	6FC5800-0BS04-0YB0	S04	O	O	O	O	O
Protect MyMachine /Open, additional channel Software options for the 2nd to 10th channel, can be used in each module up to the max. number of channels (see Maximum configuration of machining channels) Monitoring of collisions across multiple channels Requirements: Protect MyMachine /Open (option S04), Machining channel, each additional (option C11 ... C19)	6FC5800-0BF00-0YB0	F01 ... F09	O	O	O	O	O
Protect MyMachine /3D Twin NEW Function for collision avoidance on machine tools, executable on an external IPC Requirements: Industrial PC with Microsoft Windows operating system, Minimum system requirements: 4 processor cores, Intel Core i5 class or comparable, 8 GB RAM, 6 GB on SSD, integrated or external GPU Protect MyMachine /Open (option S04 or F0.) Access MyMachine /OPC UA (option P67) CNC software ≥ SW version 6.25, CNC software ≥ SW version 6.15 SP5	MCS4260N		–	O	O	O	O
Protect MyMachine /3D Twin - Pro NEW Extended functionality for collision avoidance on machine tools, executable on an external IPC (Pro Package) Requirements: Industrial PC with Microsoft Windows operating system, Minimum system requirements: 6 processor cores, Intel Core i7 class or comparable, 8 GB RAM, 6 GB on SSD, integrated or external GPU Protect MyMachine /3D Twin (MCS4260N) CNC software ≥ SW version 6.25, CNC software ≥ SW version 6.15 SP5	MCS4260PN		–	O	O	O	O
Contour monitoring with tunnel function	6FC5800-0BM52-0YB0	M52	O	O	O	O	O
Path length evaluation	6FC5800-0BM53-0YB0	M53	O	O	O	O	O
ESR drive-autonomous stop and retract Software option Supply of an electronic license (PDF)	6FC5800-0BM60-0YB0	M60	O	O	O	O	O
Extended stop and retract ESR NC-controlled and drive-autonomous Software option Supply of an electronic license (PDF)	6FC5800-0BM61-0YB0	M61	O	O	O	O	O

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Compensations

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Compensations							
Backlash compensation			✓	✓	✓	✓	✓
Leadscrew error compensation			✓	✓	✓	✓	✓
Measuring system error compensation			✓	✓	✓	✓	✓
Feedforward control:							
• Velocity-dependent			✓	✓	✓	✓	✓
• Acceleration-dependent			✓	✓	✓	✓	✓
Weight counterbalance, electronic (basic function of SINAMICS S120)			✓	✓	✓	✓	✓
Temperature compensation			✓	✓	✓	✓	✓
Quadrant error compensation			✓	✓	✓	✓	✓
Circularity test			✓	✓	✓	✓	✓
Friction compensation with adaptive characteristics (Friction Compensation)	**6FC5800-0BS06-0YB0**	**S06**	O	O	O	O	O
Leadscrew error compensation, bidirectional	**6FC5800-0BM54-0YB0**	**M54**	O	O	O	O	O[1]
Sag compensation, multi-dimensional	**6FC5800-0BM55-0YB0**	**M55**	O	O	O	O	O[1]
Run MyCC /KPXT [2] Drive current measurement Requirement: Loadable compile cycle	**6FC5800-0BM82-0YB0**	**M82**	O	O	O	O	–
Run MyCC /LECO [2] Learning error compensation Requirement: Loadable compile cycle	**6FC5800-0BN77-0YB0**	**N77**	O	O	O	O	–
Nodding compensation ECO	**6FC5800-0BS20-0YB0**	**S20**	O	O	O	O	O
Nodding compensation ADVANCED	**6FC5800-0BS21-0YB0**	**S21**	O	O	O	O	O
Nodding compensation Run MyCC /NOCO [2] Requirement: Loadable compile cycle	**6FC5800-0BN63-0YB0**	**N63**	O	O	O	O	–
Run MyCCI /UCI [2] Universal spatial compensation interface Requirement: Loadable compile cycle	**6FC5800-0BN75-0YB0**	**N75**	O	O	O	O	–
Multi Error Compensation Eco Compensation in the plane with 2 input axes Requirement: CNC software ≥ SW version 6.22	**6FC5800-0BS80-0YB0**	**S80**	O	O	O	O	O
Multi Error Compensation Basic Spatial compensation with 3 input axes Requirement: CNC software ≥ SW version 6.22	**6FC5800-0BS81-0YB0**	**S81**	O	O	O	O	O
Multi Error Compensation Advanced Spatial compensation with 5 input axes Requirement: CNC software ≥ SW version 6.22	**6FC5800-0BS82-0YB0**	**S82**	–	–	O	O	O
Run MyCC /VCS-A3 Spatial compensation for 3 axes Requirement: Loadable compile cycle	**6FC5800-0BN15-0YB0**	**N15**	O	O	O	O	–
Run MyCC /VCS-A5 Spatial compensation for 5 axes Requirement: Loadable compile cycle	**6FC5800-0BN16-0YB0**	**N16**	O	O	O	O	–
Run MyCC /VCS-A5 plus Spatial compensation for 5 axes plus Requirement: Loadable compile cycle	**6FC5800-0BN17-0YB0**	**N17**	O	O	O	O	–
Run MyCC /VCS-ECO Spatial compensation for 5 axes Requirement: Loadable compile cycle	**6FC5800-0BN18-0YB0**	**N18**	O	O	O	O	–
Run MyCC /VCS-ROT [2] Spatial compensation for 2 rotary axes Requirement: Loadable compile cycle	**6FC5800-0BN31-0YB0**	**N31**	O	O	O	O	–

[1] Restricted functionality of export version:
The tolerance band that can be corrected is limited to 1 mm.

[2] Compile cycle that is not generally released. Requires machine-specific release process.
Please contact your local Siemens sales office or Regional Company.

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Compensations (continued)							
Run MyCC /VIBX Vibration extinction Requirement: Loadable compile cycle	6FC5800-0BN11-0YB0	N11	O	O	O	O	–
Run MyCC /COCO [1] Magnetic cogging torque compensation Requirement: Loadable compile cycle	6FC5800-0BN46-0YB0	N46	O	O	O	O	–
Cogging torque compensation axis/spindle, each additional (alternative to option N46)	6FC5800-0BD50-0YB0	D51...D59	O	O	O	O	O
Run MyCC /AXCO Compensation of a forced mechanical coupling in the machine coordinate system Requirement: Loadable compile cycle	6FC5800-0BM81-0YB0	M81	O	O	O	O	–
Run MyCC /SEC-KT Spatial compensation for kinematic transformations Requirement: Loadable compile cycle	6FC5800-0BM57-0YB0	M57	O	O	O	O	–
Run MyCC /ROCO Accuracy [1] RObot COmpensation Accuracy Requirement: Loadable compile cycle	6FC5800-0BN83-0YB0	N83	–	–	O	O	–
Run MyCC /ROCO Productivity [1] RObot COmpensation Productivity Requirement: Loadable compile cycle	6FC5800-0BN84-0YB0	N84	–	O	O	O	–
Intelligent Load Control (ILC) Intelligent load adjustment	6FC5800-0BS11-0YB0	S11	O	O	O	O	O
Intelligent Dynamic Control (IDC) Intelligent dynamic response adaptation	6FC5800-0BS23-0YB0	S23	O	O	O	O	O
Run MyCC /EMC [1] Model-based engineered motion control Requirement: Loadable compile cycle Contains: Option N63 Nodding compensation Run MyCC /NOCO	6FC5800-0BN47-0YB0	N47	O	O	O	O	–
Programmable logic controller (PLC)							
SIMATIC S7-1500F integrated [*]			✓	✓	✓	✓	✓
PROFINET IO controller			✓	✓	✓	✓	✓
PROFINET IO device			✓	✓	✓	✓	✓
Fieldbus communication via PROFINET IO RT/IRT (X150):			✓	✓	✓	✓	✓
Number of interfaces / Number of ports (RJ45)			1 / 2	1 / 2	1 / 2	1 / 2	✓
Distributed I/O via PROFINET (RT/IRT):							
• Via integrated interface, data transfer rate X150, max.			100 Mbps	100 Mbps	100 Mbps	100 Mbps	✓
• Minimum update time for IRT (min. update time is determined by the time slice set for PROFINET IO communication, by the number of IO devices used, and by the amount of configured user data)			1 ms	1 ms	1 ms	1 ms	✓
• Minimum update time for RT (min. update time is determined by the time slice set for PROFINET IO communication, by the number of IO devices used, and by the amount of configured user data)			1 ms	1 ms	1 ms	1 ms	✓

[1] Compile cycle that is not generally released. Requires machine-specific release process.
Please contact your local Siemens sales office or Regional Company.

[*] PLC performance scales with hardware performance of the respective module.

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Programmable logic controller (PLC)

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Programmable logic controller (PLC) (continued)							
Distributed PROFINET IO devices, maximum number	See SiePortal		255	255	255	255	✓
• including IO devices with IRT, max.			64	64	64	64	✓
Plant communication via PROFINET IO RT (X160):			✓	✓	✓	✓	✓
Number of interfaces / Number of ports (RJ45)			1 / 1	1 / 1	1 / 1	1 / 1	✓
Distributed I/O via PROFINET (RT):							
• Via integrated interface, data transfer rate X160, max.			100 Mbps	100 Mbps	100 Mbps	100 Mbps	✓
• Update time for RT (for a send clock of 1 ms) (min. update time is determined by the time slice set for PROFINET IO communication, by the number of IO Devices used, and by the amount of configured user data)			1 ms … 512 ms	1 ms … 512 ms	1 ms … 512 ms	1 ms … 512 ms	✓
Distributed PROFINET IO devices for RT, max. number	See SiePortal		128	128	128	128	✓
Fieldbus communication via PROFIBUS DP (X126):							
Number of interfaces:			–	1	1	1	✓
• Data transfer rate X126, max.			–	12 Mbps	12 Mbps	12 Mbps	✓
Distributed PROFIBUS DP devices, max. number			–	125	125	125	✓
I/O address range Inputs – all inputs are in the process image (16 KB via integrated PROFINET IO interface X150, 8 KB via integrated PROFINET IO interface X160 and 8 KB via integrated PROFIBUS DP interface X126) [1]			32 KB	32 KB	32 KB	32 KB	✓
I/O address range Outputs – all outputs are in the process image (16 KB via integrated PROFINET IO interface X150, 8 KB via integrated PROFINET IO interface X160, 8 KB via integrated PROFIBUS DP interface X126) [1]			32 KB	32 KB	32 KB	32 KB	✓
Bit memory			16 KB	16 KB	16 KB	16 KB	✓
S7 timers			2048	2048	2048	2048	✓
IEC timers			Any	Any	Any	Any	✓
S7 counters			2048	2048	2048	2048	✓
IEC counters			Any	Any	Any	Any	✓
Retentive data area			768 KB	768 KB	768 KB	768 KB	✓
Usable retentive memory for bit memories, timers, counters, DBs and technological data (axes)			700 bytes	700 bytes	700 bytes	700 bytes	✓
PLC blocks Total number of elements (blocks OB, FB, FC, DB) and UDTs			12000	12000	12000	12000	✓
FBs:							
• max. size			1 MB	1 MB	1 MB	1 MB	✓
• Largest number (usable nos. 30 … 999, 1024 … 24999)			65535	65535	65535	65535	✓
FC:							
• max. size			1 MB	1 MB	1 MB	1 MB	✓
• Largest number (usable nos. 30 … 999, 1024 … 24999)			65535	65535	65535	65535	✓

[1] Note:
PROFIBUS DP interface X126 not available on PPU 1740.

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Programmable logic controller (PLC) (continued)							
DB:							
• max. size (optimized/non-optimized block access)			16 MB / 64 KB	16 MB / 64 KB	16 MB / 64 KB	16 MB / 64 KB	✓
• Largest number (usable nos. 62 … 70, 1100 … 24999)			59999	59999	59999	59999	✓
Know-how protection:							
• User program protection			✓	✓	✓	✓	✓
• Copy protection			✓	✓	✓	✓	✓
• Block protection			✓	✓	✓	✓	✓
• Access protection			✓	✓	✓	✓	✓
• "Write protection" protection level			✓	✓	✓	✓	✓
• "Read/write protection" protection level			✓	✓	✓	✓	✓
• "Complete protection" protection level			✓	✓	✓	✓	✓
Standard motion control functionality: Extended number of motion control resources, maximum Requirement: from CNC SW 6.20			✓ 640	✓ 640	✓ 720	✓ 1520	✓
SINUMERIK PP 72/48D PN I/O module	**6FC5311-0AA00-0AA0**		O	O	O	O	O
SINUMERIK PP 72/48D 2/2A PN I/O module	**6FC5311-0AA00-1AA0**		O	O	O	O	O
• Terminal strip converter for SINUMERIK I/O module PP 72/48D from 50-pin IDC connector to screw terminal for 1-to-1 connection	**6EP5406-5AA0**		O	O	O	O	O
• Set of cables for SINUMERIK I/O module PP 72/48D for self-assembly, consisting of: - 6 m ribbon cable, 50-pin - 8 insulation displacement connectors, 50-pin - Strain relief clamp	**6EP5306-5BG00**		O	O	O	O	O

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Safety functions

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Safety functions							
SINUMERIK ONE Safety Integrated							
Safety Integrated acceptance test with SINUMERIK Operate			✓	✓	✓	✓	✓
Basic Safety Integrated functions:							
• Safe Torque Off (STO) [1]			✓	✓	✓	✓	✓
• Safe Stop 1 (SS1) (time-controlled) [1]			✓	✓	✓	✓	✓
• Safe Brake Control (SBC)			✓	✓	✓	✓	✓
Safety Integrated F-PLC Enabling of the F-PLC integrated in the SINUMERIK for failsafe programming, axis monitoring not included	**6FC5800-0BS60-0YB0**	**S60**	O	O	O	O	O
Safety-related communication via standard bus (PROFIsafe with SIMATIC ET 200 range)	See SiePortal		O	O	O	O	O
Safe integration of sensors via SIMATIC DP / AS-i F-Link	See SiePortal		O	O	O	O	O
TIA Portal library SafeUserData Entry of safe configuration data via SINUMERIK Operate that is then made available to the safety program for evaluation (safe signals as Bool, Int, DInt)			✓	✓	✓	✓	✓
Safety Integrated Extended Functions:							
Safety Integrated axis/spindle Safe function for motion monitoring 1 axis/spindle Requirement: Safety Integrated F-PLC (option S60)	**6FC5800-0BK00-0YB0**	**K01...K09**	O	O	O	O	O
Safety Integrated multi-axis package Safe function for motion monitoring for all axes/spindles activated in the SINUMERIK Requirement: Safety Integrated F-PLC (option S60)	**6FC5800-0BS61-0YB0**	**S61**	O	O	O	O	O
Safe shutdown (stops) via PROFIsafe:			O	O	O	O	O
• Safe Torque Off (STO)			O	O	O	O	O
• Safe Stop 1 (SS1), time or speed-controlled shutdown			O	O	O	O	O
• Safe Stop 2 (SS2), safe shutdown combined with Safe Operating Stop (SOS)			O	O	O	O	O
• Safe Stop 2e (SS2e), safe shutdown via external stop			O	O	O	O	O
• Safe Stop 2 (SS2 ESR), extended stop and retract, e.g. in the event of a power failure			O	O	O	O	O
• Stop F, internal system reaction			O	O	O	O	O

[1] Via on-board terminals on NCU/PPU or via PROFIsafe.

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Safety functions (continued)							
Safe monitoring functions:							
• Safe Operating Stop (SOS)			O	O	O	O	O
• Safely-Limited Speed (SLS) incl. SLS override with closed-loop control between 0-100%			O	O	O	O	O
• Safe Speed Monitor (SSM)			O	O	O	O	O
• Safe Acceleration Monitor (SAM)			O	O	O	O	O
• Safe Brake Ramp (SBR)			O	O	O	O	O
• Safely-Limited Position (SLP)			O	O	O	O	O
• Safe Cam (SCA)			O	O	O	O	O
• Safe Position (SP)			O	O	O	O	O
• Safely-Limited Speed (SLS), encoderless			O	O	O	O	O
• Safe Direction (SDI)			O	O	O	O	O
• Safely-Limited Acceleration (SLA)			O	O	O	O	O
• Safe Brake Management (SBM) Contains the functions:			O	O	O	O	O
- Safe Brake Control (SBC)			O	O	O	O	O
- Safe Brake Test (SBT) – diagnostic function			O	O	O	O	O
SIMATIC Safe Kinematics Safety function for Run MyRobot /Direct Control and Run MyRobot/ Direct handling ≥ V2.0	**6ES7823-0FE04-1AA0**		O	O	O	O	O

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Engineering

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Engineering							
Development, commissioning and simulation of virtual machine tools with SINUMERIK ONE							
Create MyVirtual Machine /Operate Virtual test environment for SINUMERIK ONE	**MCS4250f** [2] **MCS4250n** [3] **MCS4250a** [4]		O[1]	O[1]	O[1]	O[1]	O
Create MyVirtual Machine /Open Virtual test environment for SINUMERIK ONE Extension with an open communications interface	**MCS4255f** [2] **MCS4255n** [3] **MCS4255a** [4]		O[1]	O[1]	O[1]	O[1]	O
Create MyVirtual Machine /3D Virtual test environment for SINUMERIK ONE Extension comprising a Virtual Machine Creator for JobShop applications	**MCS4210** [2] **MCS4210n** [3] **MCS4210a** [4]		O[1]	O[1]	O[1]	O[1]	O
Create MyVirtual Machine /SIMIT Connector Virtual test environment for SINUMERIK ONE Extension for coupling the Closed Loop Engineering	**MCS4212** [2] **MCS4212n** [3] **MCS4212a** [4]		O[1]	O[1]	O[1]	O[1]	O
Commissioning the SINUMERIK ONE CNC and SINAMICS S120 drive system integrated in SINUMERIK Operate Requirement: SINUMERIK Operate (option P88, S00 or S87)			✓	✓	✓	✓	✓
Alarms and messages			✓	✓	✓	✓	✓
Action log can be activated for diagnostic purposes: • Logbook for alarms and keys.			✓	✓	✓	✓	✓
Machine logbook, electronic			✓	✓	✓	✓	✓
Trace: CNC, PLC and drives			✓	✓	✓	✓	✓
PLC status (generally possible via SIMATIC STEP 7 on PG/PC)			✓	✓	✓	✓	✓
SIMATIC STEP 7 & STEP 7 Safety In TIA Portal V 17 for SINUMERIK ONE hardware for service functions Requirement: SIMATIC IPC with SINUMERIK PCU base software (option P86)							
• Single license on DVD-ROM	**6FC5862-1YC17-0YA0**		–	O	O	O	O
• Single license without data storage medium	**6FC5862-1YC17-0YH0**		–	O	O	O	O
SIMATIC STEP 7 & STEP 7 Safety In TIA Portal V 18 for SINUMERIK ONE hardware for service functions Requirement: SIMATIC IPC with SINUMERIK PCU base software (option P86)							
• Single license on DVD-ROM	**6FC5862-1YC18-0YA0**		–	O	O	O	O
• Single license without data storage medium	**6FC5862-1YC18-0YH0**		–	O	O	O	O
SIMATIC STEP 7 & STEP 7 Safety In TIA Portal V 19 for SINUMERIK ONE hardware for service functions Requirement: SIMATIC IPC with SINUMERIK PCU base software (option P86)							
• Single license, including software download	**6FC5862-1YC23-0YA0**		–	O	O	O	O
• Single license	**6FC5862-1YC23-0YH0**		–	O	O	O	O

[1] In Create MyVirtual Machine, no distinctions are mapped between the hardware types/classes. Projects created with Create/Run MyVirtual Machine can be loaded into SINUMERIK ONE NCU/PPU, taking into consideration the HW configurations and constraints.

[2] Floating: Server-based licenses, can be used on multiple workstations (not simultaneously); order recommendation.

[3] Node-locked Server: Server license that is connected to particular clients (work station/PC)
Node-locked Standalone: Local license file that is connected to a particular client (work station/PC)

[4] Academics: For non-commercial use.

For more information, request the detailed description of licenses.

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Description ✓ Basic version I O Option I – Not available		Article No. Product ID	Order code	SINUMERIK ONE				
				PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Engineering (continued)								
SIMATIC STEP 7 & STEP 7 Safety in TIA Portal V 20 for SINUMERIK ONE hardware for service functions Requirement: SIMATIC IPC with SINUMERIK PCU base software (option P86)	NEW							
• Single license, including software download	NEW	6FC5862-1YC24-0YA0		–	O	O	O	O
• Single license without data storage medium	NEW	6FC5862-1YC24-0YH0		–	O	O	O	O
Trace Requirement: SINUMERIK Operate				✓	✓	✓	✓	✓
Circularity test Requirement: SINUMERIK Operate				✓	✓	✓	✓	✓
Measuring functions for manual drive optimization				✓	✓	✓	✓	✓
Auto Servo Tuning AST				✓	✓	✓	✓	✓
Commissioning functions for the SINAMICS S120 drive system with fully automatic speed and position controller optimization Requirement: SINUMERIK Operate:								
• Single-axis optimization, incl. gantry axes				✓	✓	✓	✓	✓
• Speed controller optimization incl. current setpoint filter setting				✓	✓	✓	✓	✓
• Position controller optimization incl. speed setpoint filter setting				✓	✓	✓	✓	✓
• Setting of feedforward control				✓	✓	✓	✓	✓
• Overview of optimization results				✓	✓	✓	✓	✓
• Closed circuit can be manually tuned using a Bode plot				✓	✓	✓	✓	✓
• Optimization of path interpolation				✓	✓	✓	✓	✓
• Assignment of torque feedforward control				✓	✓	✓	✓	✓
• Complete user strategy, freely parameterizable				✓	✓	✓	✓	✓
• Backward and forward during optimization, re-optimization of the speed controller				✓	✓	✓	✓	✓
• Generating reports, individual axis and path interpolation				✓	✓	✓	✓	✓
• Optimization project can be loaded and optimized offline on the PC				✓	✓	✓	✓	✓
AST from part program Call Auto Servo Tuning from the part program		6FC5800-0BS10-0YB0	S10	O	O	O	O	O
AST addon license For shorter integral times and greater interference immunity of the speed controller Frequency-selective, interference-optimized speed controller setting (to eliminate known periodic interfering frequencies) Requirement: CNC software ≥ SW version 6.24	NEW	6FC5800-0BS12-0YB0	S12	O	O	O	O	O
SINUMERIK Expert Trace Operate As an additional option to the free Diagnostic Trace with the range of functions: • FFT analysis (Fast Fourier Transfom) • Mathematical functions • Up to 10 measuring cursors • Csv export Requirement: CNC software ≥ SW version 6.24	NEW	6FC5800-0BS30-0YB0	S30	O	O	O	O	O

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Engineering

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Engineering (continued)							
Creation of PLC programs for SINUMERIK ONE							
SINUMERIK Toolbox Tools for SINUMERIK ONE • CNC software 6.1x for SINUMERIK ONE NCU/PPU • CNC software 6.1x with SINUMERIK Operate • STEP 7 toolbox in TIA Portal for SINUMERIK ONE • SINUMERIK Operate for PCU • SINUMERIK Operate for PC • PCU base software • SINUMERIK Operate commissioning tool • Eboot system • Protector - cycle protection tool • SINUMERIK VNC viewer • WKonvert-Assistant - Tool for parameterizing Tool Ident Connection • Access MyMachine P2P - Tool for communication with SINUMERIK ONE • Documentation for SINUMERIK ONE in En/De Note: Supplied on DVD-ROM for the CNC software (Article No. 6FC5852-1.C..-0YA8)			O	O	O	O	O
SIMATIC STEP 7 Professional (TIA Portal) Software on PC/PG for programming and commissioning the PLC S7-1500F of the SINUMERIK ONE	See SiePortal		O	O	O	O	O
• Ladder diagram (LAD)			O	O	O	O	O
• Function block diagram (FBD)			O	O	O	O	O
• Statement list (STL)			O	O	O	O	O
• Structured Control Language (SCL)			O	O	O	O	O
• GRAPH for sequential processes			O	O	O	O	O
TIA Portal Openness	Free option package for TIA Portal		O	O	O	O	O
SIMATIC STEP 7 Safety (TIA Portal) Software option for programming and commissioning the F-functions of the PLC S7-1500F of the SINUMERIK ONE	See SiePortal		O	O	O	O	O
SIMATIC S7-PLCSIM Advanced V4.0, floating license	See SiePortal		O	O	O	O	O
Series commissioning via:							
• USB interface with storage medium, e.g. memory stick			✓	✓	✓	✓	✓
• Network drive			✓	✓	✓	✓	✓
Create MyWorkflow /ESW (Automated Engineering & Service Workflow) Data workflow in engineering for SINUMERIK ONE using Openness Clients: • Drive-axis assignment • Configuration data • Simplified commissioning of Failsafe modules ("initialization") • Creating a DSF archive Supplied on DVD-ROM for the CNC software (Article No. 6FC5852-1.C..-0YA8), functions contained in Toolbox Delivery of a software license, SW version 1.2	**6FC5862-2YF60-0YA0**		O	O	O	O	

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Engineering (continued)							
Create MyConfig Software for series production and a software upgrading of a SINUMERIK ONE							
• Delivery of the software package via download	**6FC5862-2YC60-0YA0**		O	O	O	O	
• Delivery of ALM license (Automation License Manager)	**6FC5862-2YC60-0YB0**		O	O	O	O	
Create MyConfig /Diff Functions for identifying differences by comparison	**6FC5862-2YC65-0YB0**		O	O	O	O	
Create MyConfig ADDON /ECAD Import (option for Automation License Manager ALM) "EPLAN Add/In" Requirement: ePlan Professional EADN270 license	**6FC5862-2YF44-1YA0**		O	O	O	O	
Create MyConfig ADDON /DIFF SCRIPTING (option for Automation License Manager ALM) "EPLAN Add/In" Requirement: ePlan Professional EADN270 license	**6FC5862-2YF44-2YA0**		O	O	O	O	
Analyze MyMachine /SINUMERIK Tuning Center Software for path and axis optimization of a machine tool with SINUMERIK ONE including diagnostic trace			O	O	O	O	
• Software download of OSD + ALM license on USB flash drive	**6FC5862-4YC64-0YA0**		O	O	O	O	
• 30-day trial version	**6FC5862-4YC67-0YA0**		O	O	O	O	
Analyze MyMachine /SINUMERIK Expert Trace Contains the software Analyze MyMachine /SINUMERIK Tuning Center with Analyze MyMachine /SINUMERIK Expert Trace license for recording and analyzing SINUMERIK ONE data	**6FC5862-4YC65-0YA0**		O	O	O	O	
STARTER Commissioning tool for SINAMICS and MICROMASTER (for topology and diagnostics) As alternative functionality to SINUMERIK Operate for: • S120 drives integrated in SINUMERIK ONE • SINAMICS S120 CU3x0-2 PN (NC-controlled and HMI-assigned) As the sole commissioning function for: • External SINAMICS S120 CU320-2 (PLC-controlled; SMC function)	**6SL3072-0AA00-0AG0**		O	O	O	O	

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Engineering

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Engineering (continued)							
Access MyMachine /P2P Service tool for PC/PG with Microsoft Windows 10, single license, current SW version Note: Supplied on DVD-ROM for the CNC software (Article No. 6FC5852-1.C..-0YA8)	**6FC5860-7YC00-0YA0**		O	O	O	O	
Development of customer-specific user interfaces for SINUMERIK ONE							
Run MyScreens: • Integrate screens in SINUMERIK Operate, Max. number of screens than can be freely created			5	5	5	5	✓
Run MyScreens: • Integrate screens in SINUMERIK Operate > 5 screens, extended functions	**6FC5800-0BP64-0YB0**	**P64**	O	O	O	O	O
Create MyHMI /3GL: • Program screens, operating areas, and user interfaces in high-level languages Single license and for download, without data storage medium, 3 languages (en, de, zh)	**MCS1200**		O[1)]	O	O	O	
Run MyHMI /3GL: Use extended HMI applications Contains: Run MyScreens (option P64) Special note: Option P64 is only active if a 3GL application is used.	**6FC5800-0BP60-0YB0**	**P60**	O[1)]	O	O	O	O
Create MyHMI /WinCC: • Configure screens and operating areas with WinCC Software for PC, Single license without data storage medium	**6FC5861-3YP00-0YB0**		–	O	O	O	
Run MyHMI /WinCC: • Single license, electronic license as PDF	**6FC5800-0BP61-0YB0**	**P61**	–	O	O	O	O
Run MyHMI /WinCC Frame IPC: Configure screens and user interfaces Contains the options: • Run MyHMI /WinCC (option P61) • SINUMERIK PCU base software /IPC (option P86) • SINUMERIK Operate /PCU (option P88) Requirement: SIMATIC IPC for SINUMERIK	**6FC5800-0BP62-0YB0**	**P62**	–	O	O	O	O
Run MyHMI /SIMATIC OP: • Utilize the user interface on SIMATIC Panel Requirement: SIMATIC Comfort or Mobile Panels	**6FC5800-0BP03-0YB0**	**P03**	–	O	O	O	O

[1)] Applies exclusively to programming the Embedded HMI with Qt. You can obtain the Qt license key directly from: The Qt Company: https://www.qt.io

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Description ✓ Basic version I O Option I – Not available		Article No. Product ID	Order code	SINUMERIK ONE				
				PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Engineering (continued)								
Create MyHMI /WinCC Unified Configure screens and operating areas with WinCC Single license, SW version V20 SP1 Requirements: CNC software ≥ SW version 6.25 For multi-window display on displays ≥ 19" with full HD resolution (1920 × 1080), option SINUMERIK Operate Display Manager (option P81) is additionally required. Note: The remaining HMI user memory on SD card (option P12) is restricted to 5 GB.		**6FC5872-3YE24-0YA0**		–	O[1]	O	O	
Run MyHMI /WinCC Unified for CNC machines with SINUMERIK ONE NCU (embedded)		**6FC5800-0BU01-0YB0**		–	–	O	O	O
Run MyHMI /WinCC Unified for CNC machines with SINUMERIK ONE and IPC		**6FC5800-0BU03-0YB0**		–	O	O	O	O
Run MyHMI /WinCC Unified for CNC machines with SINUMERIK ONE and Unified Comfort Panel		**6FC5800-0BU04-0YB0**		–	O	O	O	O
Create MyHMI /PRO: • Integrate screens with variable layout in SINUMERIK Operate Single license, SW version 9.5		**6FC5867-3YC85-0YA8**		O	O	O	O	
Run MyHMI /PRO: • Use standardized and freely configurable operator screens for machine operation in SINUMERIK Operate		**6FC5800-0BP47-0YB0**	**P47**	O	O	O	O	O
Create MyHMI /AUTOMOTIVE Configure screens and operating areas with WinCC Unified Use standardized and freely configurable operator screens for machine operation in SINUMERIK Operate SW version V20 SP1 Requirements: CNC software ≥ SW version 6.25 For multi-window display on displays ≥ 19" with full HD resolution (1920 × 1080), option SINUMERIK Operate Display Manager (option P81) is additionally required. Note: The remaining HMI user memory on SD card (option P12) is restricted to 5 GB.	NEW	**6FC5873-3YE24-0YA0**		–	O[1]	O	O	
Run MyHMI /AUTOMOTIVE for CNC machines on SINUMERIK ONE NCU (embedded)	NEW	**6FC5800-0BU06-0YB0**	**U06**	–	–	O	O	O
Run MyHMI /AUTOMOTIVE for CNC machines with SINUMERIK ONE and IPC	NEW	**6FC5800-0BU08-0YB0**	**U08**	–	O	O	O	O
Run MyHMI /AUTOMOTIVE for PLC machines on Unified Comfort Panels	NEW	**6FC5800-0BU11-0YB0**		–	–	–	–	–
Run MyHMI /AUTOMOTIVE for PLC machines on IPC	NEW	**6FC5800-0BU13-0YB0**		–	–	–	–	–
For industry-specific solutions (assembly station in automotive industry plants):								
Software for SIMATIC Comfort Panel KTP900F Mobile and TP1200 for machine operation with HMI Lite Single license, current SW version		**6FC5263-0PY11-0AG0**		–	–	–	–	–
Software for SIMATIC Comfort Panel KTP900F Mobile and TP1200 for machine operation with HMI Lite Single license, SW version V17 on DVD-ROM		**6FC5263-1PY11-7AG0**		–	–	–	–	–
Software for SIMATIC Comfort Panel KTP900F Mobile and TP1200 for machine operation with HMI Lite Single license, without data storage medium		**6FC5263-0PY11-0AG1**		–	–	–	–	–
TIA Selection Tool Tool for selecting motors, converter and controller		See https://www.siemens.com/global/en/products/automation/topic-areas/tia/tia-selection-tool.html		✓	✓	✓	✓	✓

[1] Requirement: HMI on PC.

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Boost productivity

Description ✓ Basic version I O Option I – Not available		Article No. Product ID	Order code	SINUMERIK ONE				
				PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Boost productivity								
SINUMERIK Ctrl-Energy Functionalities that can be used to save energy		See SiePortal		✓	✓	✓	✓	✓
Ctrl-E Profile Saving and selection of switch-off profiles				✓	✓	✓	✓	✓
Export energy values from SENTRON PAC 3200/4200 via PROFINET Requirement: CNC software ≥ SW version 6.22				✓	✓	✓	✓	✓
Export energy values from SENTRON PAC 2200 via Modbus TCP (via Modbus/Profinet gateway in ET200) Requirement: CNC software ≥ SW version 6.23	NEW			✓	✓	✓	✓	✓
Flux reduction for asynchronous motors Function for reducing magnetization losses in motors at standstill or during operations with reduced power				✓	✓	✓	✓	✓
Ctrl-E Analysis Visualization of the electrical energy consumption of the entire machine via the CNC user interface as a basis for the development of energy-efficient machining strategies Requirement: CNC- software ≥ SW version 6.24				✓	✓	✓	✓	✓
Ctrl-E /Advanced Analysis Advanced evaluation options for energy analysis Requirement: CNC- software ≥ SW version 6.24	NEW	**6FC5800-0BQ51-0YB0**	**Q51**	O	O	O	O	O
Ctrl-E Balanced Spindle Acceleration CNC function for adjusting the spindle acceleration to the respective rapid traverse profile/feed profile in the CNC program Requirement: CNC software ≥ SW version 6.22		**6FC5800-0BQ52-0YB0**	**Q52**	O	O	O	O	O
Connect MyMachines (Mcenter) Functions: • Application management: Filtering, tool statistics • AMP: Generic data, data collection/acquisition and buffering • Reset password • Usability: Keep width of table columns		**MCS3150**		O	O	O	O	
Manage MyResources /Tools Optimization of the entire tool cycle Functions: • Tool master data revisioning • OEM data updates • Tool statistics offline handling • Support for turning machines (without magazine configuration) Requirement: Mcenter must be installed for operation.		**MCS5140**		O	O	O	O	
Manage MyResources /Tools OA		**MCS5150**		O	O	O	O	
Manage MyResources /Programs Functions: • Factory-wide management of CNC packages • Direct and secure data transfer to and from the machine • Seamless communication • Revisioning • Lifecycle Management package Requirement: Mcenter must be installed for operation.		**MCS5180**		O	O	O	O	
Manage MyResources /Programs OA		**MCS5190**		O	O	O	O	

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Description ✓ Basic version I O Option I – Not available		Article No. Product ID	Order code	SINUMERIK ONE				
				PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Boost productivity (continued)								
Analyze MyPerformance /OEE Transparency about machine utilization instead of a "black box" Functions: • Reveal important indicators for improving machine utilization • Increase productivity by eliminating bottlenecks • Continuous monitoring of the measures taken and their efficiency Requirement: Mcenter must be installed for operation.		MCS2500		O	O	O	O	
Analyze MyPerformance /OA Transparency about machine utilization instead of a "black box" Functions: • Reveal important indicators for improving machine utilization • Increase productivity by eliminating bottlenecks • Continuous monitoring of the measures taken and their efficiency Requirement: Mcenter must be installed for operation.		MCS2505		O	O	O	O	
Optimize MyMachining /AdaptiveControl Standard Real-time dynamic feedrate control to optimize machine throughput without changing the NC code For standard machining processes and single-spindle machines Requirement: CNC software ≥ SW version 6.15 SP4	NEW	6FC5800-0BT00-0YB0	T00	O	O	O	O	O
Optimize MyMachining /AdaptiveControl Pro Advanced dynamic feedrate control for complex machining processes and/or complex machines, protects cutting tools and prolongs their service life Requirement: CNC software ≥ SW version 6.15 SP4 Optimize MyMachining /AdaptiveControl Standard (option T00)	NEW	6FC5800-0BT02-0YB0	T02	O	O	O	O	O
Optimize MyMachining /ToolMonitor Pro Advanced, reliable breakage detection for smallest tools and short cutting times Requirement: CNC software ≥ SW version 6.15 SP4	NEW	6FC5800-0BT03-0YB0	T03	O	O	O	O	O
Optimize MyMachining /Multi-spindle extension Activate Optimize MyMachining /AdaptiveControl Pro and/or Optimize MyMachining /ToolMonitor Pro on machines with multiple spindles Requirement: CNC software ≥ SW version 6.15 SP4 Optimize MyMachining /AdaptiveControl Pro (option T02) or Optimize MyMachining /ToolMonitor Pro (option T03)	NEW	6FC5800-0BT04-0YB0	T04	O	O	O	O	O
Optimize MyMachining /Multi-channel extension Activate Optimize MyMachining /AdaptiveControl Pro and/or Optimize MyMachining /ToolMonitor Pro on machines with multiple channels Requirement: CNC software ≥ SW version 6.15 SP4 Optimize MyMachining /AdaptiveControl Pro (option T02) or Optimize MyMachining /ToolMonitor Pro (option T03)	NEW	6FC5800-0BT05-0YB0	T05	O	O	O	O	O
Optimize MyMachining /Single part extension Activate Optimize MyMachining /AdaptiveControl Pro (option T02) for non-series production, such as mold & die machining Requirement: CNC software ≥ SW version 6.15 SP4 Optimize MyMachining /AdaptiveControl Pro (option T02)	NEW	6FC5800-0BT06-0YB0	T06	O	O	O	O	O

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Machine Services

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Machine Services							
Liability for material defects for SINUMERIK ONE products and software Duration: 12 months For the currently applicable conditions, please refer to the terms and conditions of sale and delivery in the respective countries. To do this, use the adjacent link to the terms and conditions of sale and delivery in SiePortal. The duration of the liability for material defects can be extended by the following additional offers: • Registration of the SINUMERIK ONE system • Repair Service Contract • Local Service Contract	See terms and conditions of sale and delivery in SiePortal		✓	✓	✓	✓	✓
Extended liability for material defects on SINUMERIK ONE system through registration Valid for all Siemens components in the machine, including the hardware components SINUMERIK ONE, SIMATIC, SINAMICS and SIMOTICS. Exclusion: Fans of SINUMERIK and SINAMICS modules, wearing parts, such as bearings and brakes of motors, as well as software. Requirement: Registration of the Siemens components in myRegistration	See myRegistration		✓	✓	✓	✓	✓
Registration of SINUMERIK ONE Note, request and functionality for registration, e.g. via the Easy Registration function (standard function in SINUMERIK Operate for registering the SINUMERIK NCU/PPU via QR code). Benefit: With a registration of the machine and its equipment, service and support by Siemens can be provided faster and better. Requirement: SINUMERIK Operate			✓	✓	✓	✓	✓
Repair Service Contract Repair service by Siemens at the installation site of the machine within the warranty period Exception: Fans of SINUMERIK and SINAMICS modules, wearing parts, such as bearings and brakes of motors, and their mechanical removal and reinstallation Requirement: Registration of the Siemens components in myRegistration (e.g. via IdentSnapshot)	See SiePortal		O	O	O	O	O
IdentSnapshot Software for registration of machines and plants with products of the most varied product families, e.g. SINUMERIK, SINAMICS, and SIMATIC in myRegistration	See Siemens Industry Online Support		✓	✓	✓	✓	✓
Siemens Industry Online Support Free access to Siemens Industry Online Support (SIOS), where you will find product information, documentation, downloads, application examples, FAQs, forums, etc.	See Siemens Industry Online Support		✓	✓	✓	✓	✓
Siemens Industry Service Card The Siemens Industry Service Card is a prepaid card for technical support via telephone from the Siemens hotline. It offers a combination of prioritized callback and expert technical support for products in the SINUMERIK ONE system. The card is available as digital version for immediate use and is valid for three years from the date of purchase. • Priority call-back • Extended technical support • 24/5 available Mon – Fri around the clock • Mature products	See SiePortal		O	O	O	O	O

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE				
			PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Machine Services (continued)							
SINUMERIK Optimization Check Optimization of machine parameter settings and axes to increase productivity Analysis of the machine and advice to improve the machine dynamic response as a parameter list plus indications This analysis can also be performed via a remote connection. Requirement: SINUMERIK ONE from SW 6.13	See SiePortal		O	O	O	O	O
SINUMERIK Optimization Check – Mechanics For early detection of machine wear and quality checks after hardware has been replaced (e.g. ball screw). The frequency response and trace measurements are recorded from which a machine fingerprint is created. The machine fingerprint, i.e. the mechanical reference data, is repeatedly compared with new measurements in order to detect changes on the machine. A result report is generated and discussed with the customer.	See SiePortal		O	O	O	O	O
Remote Services for Machine Tools Remote Services offer the option of digitizing the reactive customer service. This service comprises remote fault analysis and support for fault rectification on a Siemens SINUMERIK control system by a remote expert from Siemens AG. Benefits • Reduction of unscheduled downtimes thanks to fast remote support and fault analysis by Siemens Service experts • Reduced service costs due to rapid service support and increased service availability • High safety standards using state-of-the-art remote technology Requirement: The availability of a remote infrastructure is a prerequisite for using the Remote Services for Machine Tools offering.	See SiePortal		O	O	O	O	O
Spare parts for the SINUMERIK ONE system Our spare parts supply service is available worldwide, ensuring both smooth and fast delivery of spare parts – which in turn helps secure optimum system availability. Most original spare parts from Siemens and for the SINUMERIK ONE system are available for ten years or more after product discontinuation. However, spare parts for certain products (in particular industrial PCs) are only kept in stock for shorter periods. You can click on the link opposite for a detailed description of the spare part offering and a list of the assemblies that are kept in stock for fewer than 10 years.	See SiePortal		O	O	O	O	O
Local service contract With the Local Service Contract (LSC), we offer machine users an individual, modular preventive program that ensures the availability of machine tools and manufacturing systems to a considerable extent and therefore makes an important contribution to efficient production.	See SiePortal		O	O	O	O	O
Retrofit with SINUMERIK ONE For example, retrofitting a machine tool with an older CNC to SINUMERIK ONE and SINAMICS drives to maintain or increase productivity, to increase IT security, and in preparation for digitalization of the machine.	See SiePortal		O	O	O	O	O

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Machine Services

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Machine Services (continued)							
Courses and training for SINUMERIK ONE SITRAIN - Digital Industry Academy provides an extensive training offering for Siemens industrial products. You can find more information at www.siemens.com/sitrain. A range of courses specifically for SINUMERIK ONE is offered.	See SITRAIN		O	O	O	O	
Training rack for SINUMERIK ONE The training rack is suitable for practical exercises in commissioning, operation and programming, and maintenance tasks. Components: • SINUMERIK ONE NCU 1750 with SD card • CNC SW export version, current SW version • SINAMICS S120 Active Line Module 16 kW • SINAMICS S120 Active Interface Module 16 kW • Motor Module • Numeric Control Extension NX10.3 • SIMATIC IPC427E • ITC2200 neutral version • MCP2200c machine control panel • SIMATIC PROFIsafe I/Os Note: Option licenses must be ordered separately.	**6ZB2410-0BQ00**		–	–	O	–	O
Training case SINUMERIK OP Touch The SINUMERIK OP Touch training case for training and demonstration is suitable for operation of the training case SINUMERIK ONE (6ZB2410-0BN00). The training case contains a ready-to-use operating unit comprising: • ITC2200 • IPC427E with the current SINUMERIK Operate software • MCP 398C machine control panel with EM131 Note: Option licenses must be ordered separately.	**6ZB2410-0BL00**		–	–	O	–	O
Training case SINUMERIK ONE with NCU 1750 For training, testing, and demonstration • Connection: 230 V 1 AC • SINUMERIK ONE NCU 1750 with current CNC software with license and SINUMERIK Operate Export • SINAMICS Smart Line Module 5 kW, modified for 230 V • Double Motor Module 2 × 3 A • 2 SIMOTICS S-1FK7022 motors, DRIVE-CLiQ encoder absolute / incremental • Stroboscope for testing synchronous operation • SIMATIC ET 200SP IM 155-6PN with 4 modules DI, DQ, TM, AI • S7 mounting rail 85 mm with 24 V and PN connection • Operator box, can be removed for I/Os of the NCU 1750 or ET200 Note: Option licenses must be ordered separately.	**6ZB2410-0BN00**		–	–	O	–	O

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Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Product Partner SINUMERIK Systems							
Under the name Product Partner SINUMERIK Systems, leading companies present acknowledged know-how in specific fields. Their products and solutions, that they develop, produce and sell themselves, are a useful complement to our comprehensive portfolio. Our partner network also makes a crucial contribution toward you benefiting from maximally complete, high quality and efficient solutions, enabling you to sustainably improve your competitiveness. This partnership allows intensive cooperation, and ensures a high degree of compatibility and reliability in productive use with SINUMERIK CNCs.	See SiePortal						
AfM Technology GmbH							
• Volumetric compensation Requirements: Run MyCC /VCS-A3 (option N15) or Run MyCC /VCS-A5 (option N16 for NCU 1750/1760) or Run MyCC /VCS-A5 plus (option N17 for NCU 1750/1760)	See SiePortal		O*)	O*)	O*)	O*)	
AUVESY GmbH							
• versiondog	See SiePortal		O*)	O*)	O*)	O*)	
Balance Systems S.r.l.							
• VM25 system Requirement: IPC	See SiePortal		–	O*)	O*)	O*)	
Conoptica AS NEW							
• CU2 Tool	See SiePortal		–	O*)	O*)	O*)	
DDX S.r.l. NEW							
• Easy-Suite	See SiePortal		O*)	O*)	O*)	O*)	
HeMuTec GmbH NEW							
• 5X-Analyzer - Kinematics check Requirements: Sag compensation, multi-dimensional (option M55) Machining package 5 axes (option M30)	See SiePortal		O*)	O*)	O*)	O*)	
Hexagon AICON ETALON GmbH							
• LaserTRACER/LaserTRACER-MT Requirements: Sag compensation, multi-dimensional (option M55) and Run MyCC /VCS-A3 (option N15) or Run MyCC /VCS-A5 (option N16 for NCU 1750/1760) or Run MyCC /VCS-A5 plus (option N17 for NCU 1750/1760)	See SiePortal		O*)	O*)	O*)	O*)	O*)
Hexagon Metrology GmbH NEW							
• m&h LS-R-4.8 Wireless laser scanner	See SiePortal		O*)	O*)	O*)	O*)	
• m&h RWP220.50-G-UTP Ultrasonic wall thickness measurement	See SiePortal		O*)	O*)	O*)	O*)	
Hofmann Mess- und Auswuchttechnik GmbH & Co. KG							
• AB 9000 Requirement: IPC	See SiePortal		–	O*)	O*)	O*)	
i4.0 – Wotton NEW							
• i4.0 Box for data connectivity	See SiePortal		O*)	O*)	O*)	O*)	

*) Siemens accepts no liability for the information in this table.
For release and compatibility of the displayed product for module types and SW versions SINUMERIK ONE, ask the Product Partner. For more information, contact the product partner listed in the "Article No." column directly.
You can find more information on the internet at:
siemens.com/product-partner

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Product Partner SINUMERIK Systems

Description ✓ Basic version I O Option I – Not available	Article No. Product ID	Order code	SINUMERIK ONE PPU 1740	NCU 1740	NCU 1750	NCU 1760	in CNC SW export version
Product Partner SINUMERIK Systems (continued)							
IBS Precision Engineering NEW							
• Rotary Inspector Requirements: Sag compensation, multi-dimensional (option M55) and Run MyCC /VCS-A3 (option N15) or Run MyCC /VCS-A5 (option N16 for NCU 1750/1760) or Run MyCC /VCS-A5 plus (option N17 for NCU 1750/1760)	See SiePortal		O*)	O*)	O*)	O*)	
Innomotics GmbH							
• 2SP1 motor spindles	See SiePortal		O*)	O*)	O*)	O*)	O*)
iSiOS GmbH							
• Robot calibration	See SiePortal		O*)	O*)	O*)	O*)	
Juxtum Inc. NEW							
• Viz-Adapter MTConnect	See SiePortal		O*)	O*)	O*)	O*)	
MARPOSS GmbH							
• Laser Tool Setter	See SiePortal		O*)	O*)	O*)	O*)	
MARPOSS GmbH (ARTIS)							
• CTM-tool and process monitoring system	See SiePortal		O*)	O*)	O*)	O*)	
• Genior Modular Fully automatic tool and process monitoring system	See SiePortal		O*)	O*)	O*)	O*)	
MARPOSS GmbH (DITTEL)							
• DS6000 UP Balancing and process monitoring Requirement: IPC	See SiePortal		–	O*)	O*)	O*)	
MCU GmbH & Co. KG							
• Toolinspect II	See SiePortal		O*)	O*)	O*)	O*)	
ModuleWorks GmbH							
• Real-time Collision Avoidance Requirement: IPC	See SiePortal		–	O*)	O*)	O*)	
Nordmann GmbH & Co. KG NEW							
• Tool monitoring system SEM-MODUL-e2	See SiePortal		O*)	O*)	O*)	O*)	
Renishaw							
• XM-60 multi-axis laser measurement system Requirements: Run MyCC /VCS-A3 (option N15) or Run MyCC /VCS-A5 (option N16 for NCU 1750/1760) or Run MyCC /VCS-A5 plus (option N17 for NCU 1750/1760)	See SiePortal		O*)	O*)	O*)	O*)	
TDM Systems GmbH							
• iCut	See SiePortal		O*)	O*)	O*)	O*)	
VibroCut NEW							
• VibroCut oscillate	See SiePortal		O*)	O*)	O*)	O*)	
• VibroCut ultrasonic	See SiePortal		O*)	O*)	O*)	O*)	
Volumetric Accuracy Research Institute Co., Ltd. NEW							
• Measurement service for SINUMERIK VCS Requirements: Run MyCC /VCS-A3 (option N15) or Run MyCC /VCS-A5 (option N16 for NCU 1750/1760) or Run MyCC /VCS-A5 plus (option N17 for NCU 1750/1760)	See SiePortal		O*)	O*)	O*)	O*)	

*) Siemens accepts no liability for the information in this table.
For release and compatibility of the displayed product for module types and SW versions SINUMERIK ONE, ask the Product Partner. For more information, contact the product partner listed in the "Article No." column directly.
You can find more information on the internet at:
siemens.com/product-partner

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Overview

The SINUMERIK ONE CNC software is available for order in various different versions.

In general, two variants are distinguished for SINUMERIK ONE:

CNC software as export version

This CNC software does not require an export license. If you order the export version, be aware that certain CNC functions cannot be used.

CNC software with full functional scope

Note that an export license is required to export this CNC software. For information on which CNC functions require an export license, see the function overview.

The types of delivery of the CNC software can be selected using different article numbers:

- **Standard order**
  CNC software in the standard or export version, pre-installed on SD card with Single License
- **Replacement orders if the SD card fails, for example**
  CNC software in standard or export version on SD card without Single License
- **For customers who install the CNC software and license on the SD card themselves**
  CNC software in standard or export version on DVD and Single License for CNC software can be ordered separately
- **Bundle order**
  CNC software in standard or export version with Single License and additional licenses for CNC options, pre-installed on SD card can be ordered using order codes

Ordering examples for SINUMERIK ONE

- CNC software 31-3 with SINUMERIK Operate on SD card
- 3 × additional axis/spindle
- 1 × additional machining channel
- Travel to fixed stop with Force Control

Ordering individually

Order using the complete article number

License key via download

Item	Article No.	Quantity (Qty)	Article description
1	**6FC5852-1YG25-1YA0**	1	SINUMERIK ONE CNC software 31-3 export software 6.25 SP1 6 languages (EN, DE, FR, IT, SP, CHS) on SD card with electronic license ********** email address required for delivery
2	**6FC5800-0BA00-0YB0**	3	1 axis/spindle in addition
3	**6FC5800-0BC10-0YB0**	1	Additionally 1 machining channel
4	**6FC5800-0BM01-0YB0**	1	Travel to fixed stop with Force Control

Ordering as a bundle

Order using article number with order codes

License key included in delivery

Item	Article No.	Quantity (Qty)	Article description
1	**6FC5852-1YG25-1YA0-Z A03+C11+M01**	1	SINUMERIK ONE CNC software 31-3 export software 6.25 SP1 6 languages (EN, DE, FR, IT, SP, CHS) on SD card with electronic license ********** email address required for delivery A03: License for 3 axes/spindles in addition C11: Additionally 1 machining channel M01: Travel to fixed stop with Force Control

Explanation of ordering example in a bundle

If the article number 6FC5852-1YG25-1YA0 is ordered with the additional identifier "Z" and the order codes "A03+C11+M01", delivery will be made as a bundle. This bundle includes the CNC software, the license for the CNC software, and all other licenses, which are already pre-installed on the SD card.

The following are supplied:

- 1 × SD card containing:
  - 1 × CNC software
  - 1 × license for CNC software
  - 3 × "1 axis/spindle in addition" licenses ’ (order code A03)
  - 1 × "Additional 1 machining channel" license (order code C11)
  - 1 × "Travel to fixed stop with Force Control" license (order code M01)

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Restricted functionality of export versions

Overview

Description – Not available I ⌘ Restricted functionality of the export version	Article No.	Order code	CNC SW export version
Multi-axis interpolation > 4 interpolating axes	6FC5800-0BM15-0YB0	M15	–
Run MyCC /2TRA [1] Transformation: DOUBLETRANSMIT	6FC5800-0BM25-0YB0	M25	–
Machining package 5 axes	6FC5800-0BM30-0YB0	M30	–
Run MyCC /RCTRA Transformation: Handling	6FC5800-0BM31-0YB0	M31	–
Synchronized actions stage 2	6FC5800-0BM36-0YB0	M36	⌘ [2]
Run MyCC /PACO Transformation: PARACOP 3 axes	6FC5800-0BM44-0YB0	M44	–
Run MyCC /TRIC [1] Transformation: TRICEPT 5-axis transformation	6FC5800-0BM46-0YB0	M46	–
Run MyCC /SCIS [1] Transformation: Pantograph kinematics	6FC5800-0BM51-0YB0	M51	–
Leadscrew error compensation, bidirectional	6FC5800-0BM54-0YB0	M54	⌘ [3]
Sag compensation, multi-dimensional	6FC5800-0BM55-0YB0	M55	⌘ [3]
Run MyCC /SEC-KT Spatial compensation for kinematic transformations	6FC5800-0BM57-0YB0	M57	–
Run MyCC /CLC-FD 1D/3D clearance control in position control cycle with free direction	6FC5800-0BM65-0YB0	M65	⌘ [4]
Run MyCC /SCRA [1] Transformation SCARA 2/3 axes	6FC5800-0BM68-0YB0	M68	–
Run MyCC /HEX [1] Transformation: HEXAPOD 6-axis transformation	6FC5800-0BM71-0YB0	M71	–
Generic couplings:			
• CP-Basic	6FC5800-0BM72-0YB0	M72	⌘ [5]
• CP-Comfort	6FC5800-0BM73-0YB0	M73	⌘ [5]
• CP-Expert	6FC5800-0BM74-0YB0	M74	⌘ [5]
Electronic transfer CP	6FC5800-0BM76-0YB0	M76	⌘ [5]
Run MyCC /SKID [1] Transformation: Double slide	6FC5800-0BM80-0YB0	M80	–
Run MyCC /AXCO Compensation of a forced mechanical coupling in the machine coordinate system	6FC5800-0BM81-0YB0	M81	–
Run MyCC /KPXT [1] Drive current measurement	6FC5800-0BM82-0YB0	M82	–
Run MyCC /GPM [1] Geometry monitoring interface	6FC5800-0BM83-0YB0	M83	–
Run MyCC /DSTT [1] Transformation: Dynamic Swivel TRIPOD Basis 5 axes	6FC5800-0BM84-0YB0	M84	–
Run MyCC /CRIP [1] Crank interpolation	6FC5800-0BN04-0YB0	N04	–
Protect MyMachine /CC Axis Monitor Axis collision protection	6FC5800-0BN06-0YB0	N06	–
Run MyCC /SUTI [1] Utility programs for the servo	6FC5800-0BN10-0YB0	N10	–
Run MyCC /VIBX Vibration extinction	6FC5800-0BN11-0YB0	N11	–

[1] Compile cycle that is not generally released. Requires machine-specific release process.
Please contact your local Siemens sales office or Regional Company.

[2] The number of simultaneously traversing path and positioning axes is restricted to 4.

[3] The tolerance band that can be corrected is limited to 1 mm.

[4] Clearance control can be applied to only one axis because 5-axis transformation is not available.
The number of interpolating axes is restricted to 4 without Run MyCC /CLC and to 3 with Run MyCC /CLC.

[5] The number of simultaneously traversing axes is restricted to 4.

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Restricted functionality of export versions

Description – Not available I ⌘ Restricted functionality of the export version	Article No.	Order code	CNC SW export version
Run MyCC /VCS-A3 Spatial compensation for 3 axes	6FC5800-0BN15-0YB0	N15	–
Run MyCC /VCS-A5 Spatial compensation for 5 axes	6FC5800-0BN16-0YB0	N16	–
Run MyCC /VCS-A5 plus Spatial compensation for 5 axes plus	6FC5800-0BN17-0YB0	N17	–
Run MyCC /VCS-ECO Spatial compensation for 5 axes	6FC5800-0BN18-0YB0	N18	–
Run MyCC /WZTR [1] Transformation: W-Z Tracking	6FC5800-0BN22-0YB0	N22	–
Run MyCC /RDCC [1] Transformation: Redundant and spatially distributed axes at workpiece in Cartesian coordinate system	6FC5800-0BN26-0YB0	N26	–
Run MyCC /VCS-ROT [1] Spatial compensation for 2 rotary axes	6FC5800-0BN31-0YB0	N31	–
Run MyCC /THYK [1] Transformation: TRIPOD hybrid kinematics	6FC5800-0BN36-0YB0	N36	–
Run MyCC /ROTE [1] Transformation: Rotating eccentric	6FC5800-0BN37-0YB0	N37	–
Run MyCC /FXSC [1] Cartesian travel to fixed stop	6FC5800-0BN38-0YB0	N38	–
Run MyCC /ECCE [1] Transformation: Eccentric	6FC5800-0BN41-0YB0	N41	–
Run MyCC /MSPZ [1] Metal spinning protection zone	6FC5800-0BN42-0YB0	N42	–
Run MyCC /2RPT [1] Transformation: Rotating workpiece and tool	6FC5800-0BN43-0YB0	N43	–
Run MyCC /ECCA [1] Transformation: Eccentric axis	6FC5800-0BN44-0YB0	N44	–
Run MyCC /SW2A [1] Transformation: Swivel by 2 linear axes	6FC5800-0BN45-0YB0	N45	–
Run MyCC /COCO [1] Magnetic cogging torque compensation	6FC5800-0BN46-0YB0	N46	–
Run MyCC /EMC [1] Model-based engineered motion control	6FC5800-0BN47-0YB0	N47	–
Run MyCC /SANS [1] Scalable analog setpoint	6FC5800-0BN48-0YB0	N48	–
Run MyCC /XOUT [1] Extrapolated switching signals	6FC5800-0BN51-0YB0	N51	–
Run MyCC /ROBX [1] Transformation robotic extended	6FC5800-0BN54-0YB0	N54	–
Run MyCC /AMOV [1] Variable-based axis movement	6FC5800-0BN62-0YB0	N62	–
Nodding compensation Run MyCC /NOCO [1]	6FC5800-0BN63-0YB0	N63	–
Run MyCC /SYMO [1] Synchronous motion	6FC5800-0BN66-0YB0	N66	–
Run MyCC /DTOW [1] Differential Tower Transformation	6FC5800-0BN68-0YB0	N68	–
Run MyCCI /UCI [1] Universal spatial compensation interface	6FC5800-0BN75-0YB0	N75	–
Run MyCC /LECO [1] Learning error compensation	6FC5800-0BN77-0YB0	N77	–
Run MyCC /AJET Setting of the jerk filter time	6FC5800-0BN82-0YB0	N82	–

[1] Compile cycle that is not generally released. Requires machine-specific release process. Please contact your local Siemens sales office or Regional Company.

[2] The number of simultaneously traversing path and positioning axes is restricted to 4.

[3] The tolerance band that can be corrected is limited to 1 mm.

[4] Clearance control can be applied to only one axis because 5-axis transformation is not available. The number of interpolating axes is restricted to 4 without Run MyCC /CLC and to 3 with Run MyCC /CLC.

[5] The number of simultaneously traversing axes is restricted to 4.

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Restricted functionality of export versions

Description – Not available I ⌘ Restricted functionality of the export version	Article No.	Order code	CNC SW export version
Run MyCC /ROCO Accuracy [1] RObot COmpensation Accuracy	6FC5800-0BN83-0YB0	N83	–
Run MyCC /ROCO Productivity [1] RObot COmpensation Productivity	6FC5800-0BN84-0YB0	N84	–
SINUMERIK Run MyRobot /Direct Handling	6FC5800-0BP84-0YB0	-	–
Run MyCC /SCTR SC [1] Run MyCC /SCTR SC transformation	6FC5800-0BR03-0YB0	R03	–
Run MyCC /DIFA+FASTIPO [1] Drive-interpolating following axis	6FC5800-0BR04-0YB0	R04	–
Run MyRobot /Direct control ROBX AR Transformation for articulated arm robot	6FC5800-0BR05-0YB0	R05	–
Run MyCC /TPWA [1] Transformation with programmable wrist axes	6FC5800-0BR10-0YB0	R10	–
Machining package 5 axes Additional function 7th axis	6FC5800-0BS01-0YB0	S01	–
Handling package	6FC5800-0BS31-0YB0	S31	–
SINUMERIK ONE Dynamics 5 Axis Milling	6FC5800-0BS43-0YB0	S43	–

[1] Compile cycle that is not generally released. Requires machine-specific release process.
Please contact your local Siemens sales office or Regional Company.

[2] The number of simultaneously traversing path and positioning axes is restricted to 4.

[3] The tolerance band that can be corrected is limited to 1 mm.

[4] Clearance control can be applied to only one axis because 5-axis transformation is not available.
The number of interpolating axes is restricted to 4 without Run MyCC /CLC and to 3 with Run MyCC /CLC.

[5] The number of simultaneously traversing axes is restricted to 4.

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Appendix



3/2 **Partners**

3/3 **Industrial Software Store**
3/3 Machine Tool Software Store

3/4 **Notes on software**
3/4 Software licenses
3/6 Setup texts and software update services

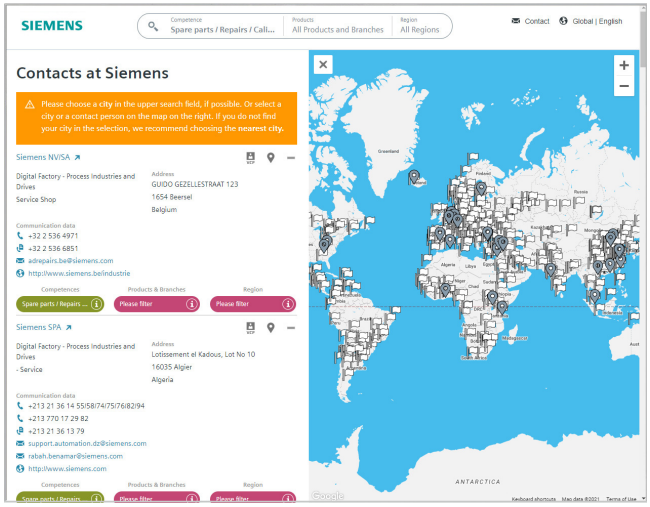
3/7 **Metal surcharges**

3/11 **Conditions of sale and delivery**

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Overview

Partners at Siemens



At your service locally, around the globe for consulting, sales, training, service, support, spare parts on the entire portfolio of Digital Industries.

Your partner can be found in our Personal Contacts Database at:
www.siemens.com/automation-contact

You start by selecting

- the required competence,
- products and branches,
- a country and a city

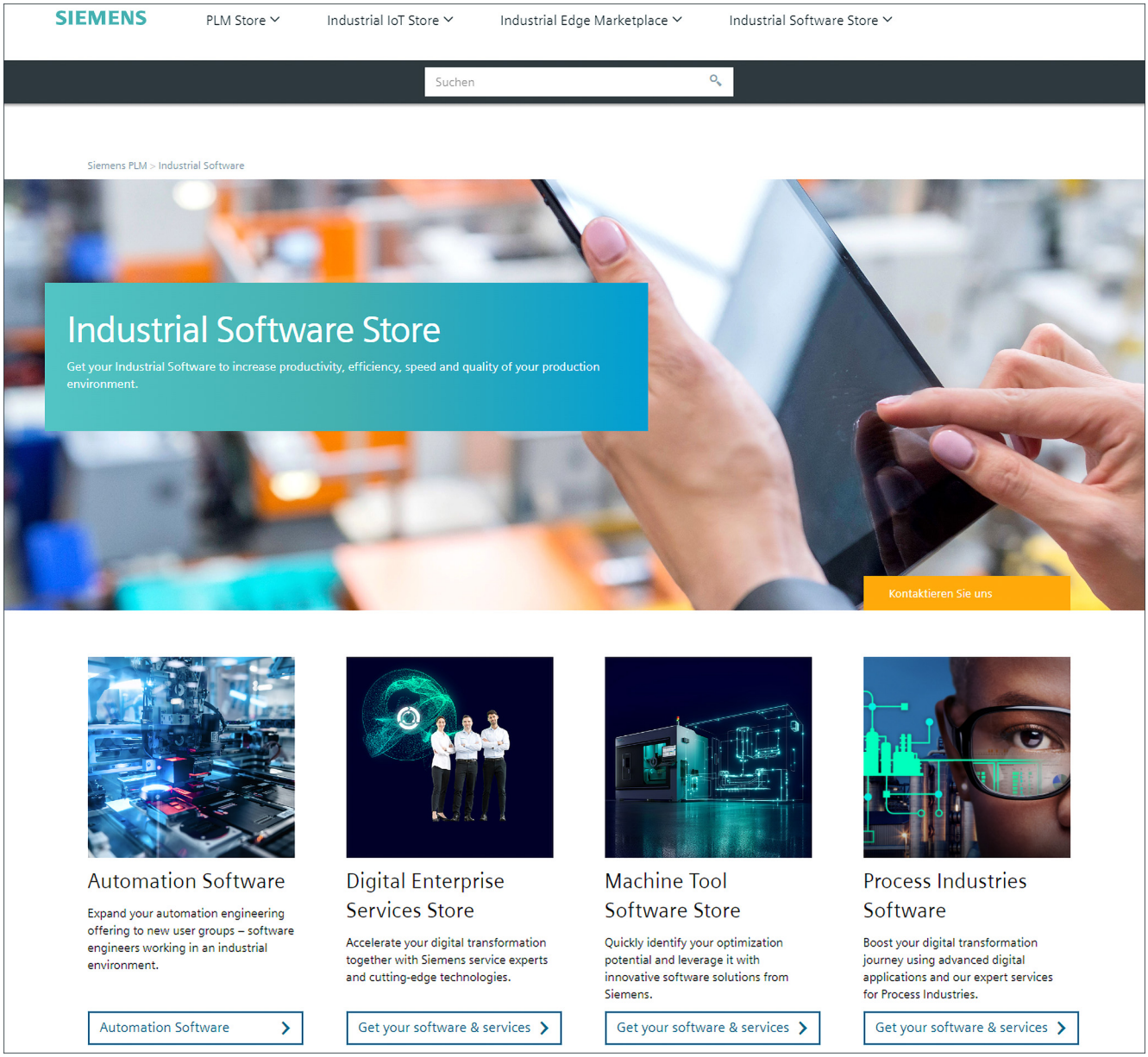
or by a

- location search or free text search.

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Overview

Path to greater machine tool productivity



Machine Tool Software Store

Our comprehensive and unique digitalization portfolio of application-specific software is tailored for machine tool builders and users to increase their productivity. Included are applications based upon a scaled IT infrastructure with functions for engineers and manufacturers.

Our software portfolio facilitates the management, analysis and optimization of CNC machines in both greenfield and brownfield projects.

dex.siemens.com/?selected=industrialsoftware

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Software licenses

Overview

Software types

Software requiring a license is categorized into types.
The following software types have been defined:

- Engineering software
- Runtime software

Engineering software

This includes all software products for creating (engineering) user software, e.g. for configuring, programming, parameterizing, testing, commissioning or servicing.

Data generated with engineering software and executable programs can be duplicated for your own use or for use by third-parties free-of-charge.

Runtime software

This includes all software products required for plant/machine operation, e.g. operating system, basic system, system expansions, drivers, etc.

The duplication of the runtime software and executable programs created with the runtime software for your own use or for use by third-parties is subject to a charge.

You can find information about license fees according to use in the ordering data (e.g. in the catalog). Examples of categories of use include per CPU, per installation, per channel, per instance, per axis, per control loop, per variable, etc.

Information about extended rights of use for parameterization/configuration tools supplied as integral components of the scope of supply can be found in the readme file supplied with the relevant product(s).

License types

Siemens Digital Industries and Smart Infrastructure offers various types of software license:

- Floating license
- Single license
- Rental license
- Rental floating license
- Trial license
- Demo license
- Demo floating license

Floating license

The software may be installed for internal use on any number of devices by the licensee. Only the concurrent user is licensed. The concurrent user is the person using the program. Use begins when the software is started.
A license is required for each concurrent user.

Single license

Unlike the floating license, a single license permits only one installation of the software per license.

The type of use licensed is specified in the ordering data and in the Certificate of License (CoL). Types of use include for example per instance, per axis, per channel, etc.

One single license is required for each type of use defined.

Rental license

A rental license supports the "sporadic use" of engineering software. Once the license key has been installed, the software can be used for a specific period of time (the operating hours do not have to be consecutive).
One license is required for each installation of the software.

Rental floating license

The rental floating license corresponds to the rental license, except that a license is not required for each installation of the software. Rather, one license is required per object (for example, user or device).

Trial license

A trial license supports "short-term use" of the software in a non-productive context, e.g. for testing and evaluation purposes. It can be transferred to another license.

Demo license

The demo license support the "sporadic use" of engineering software in a non-productive context, for example, use for testing and evaluation purposes. It can be transferred to another license. After the installation of the license key, the software can be operated for a specific period of time, whereby usage can be interrupted as often as required.

One license is required per installation of the software.

Demo floating license

The demo floating license corresponds to the demo license, except that a license is not required for each installation of the software. Rather, one license is required per object (for example, user or device).

Certificate of License (CoL)

The CoL is the licensee's proof that the use of the software has been licensed by Siemens. A CoL is required for every type of use and must be kept in a safe place.

Downgrading

The licensee is permitted to use the software or an earlier version/release of the software, provided that the licensee owns such a version/release and its use is technically feasible.

Delivery versions

Software is constantly being updated.
The following delivery versions

- PowerPack
- Upgrade

can be used to access updates.

Existing bug fixes are supplied with the ServicePack version.

PowerPack

PowerPacks can be used to upgrade to more powerful software. The licensee receives a new license agreement and CoL (Certificate of License) with the PowerPack. This CoL, together with the CoL for the original product, proves that the new software is licensed.
A separate PowerPack must be purchased for each original license of the software to be replaced.

Upgrade

An upgrade permits the use of a new version of the software on the condition that a license for a previous version of the product is already held.
The licensee receives a new license agreement and CoL with the upgrade. This CoL, together with the CoL for the previous product, proves that the new version is licensed.
A separate upgrade must be purchased for each original license of the software to be upgraded.

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Overview

ServicePack

ServicePacks are used to debug existing products. ServicePacks may be duplicated for use as prescribed according to the number of existing original licenses.

License key

Siemens Digital Industries and Smart Infrastructure supplies software products with and without license keys.

The license key serves as an electronic license stamp and is also the "switch" for activating the software (floating license, rental license, etc.).

The complete installation of software products requiring license keys includes the program to be licensed (the software) and the license key (which represents the license).

Software Update Service (SUS)

As part of the SUS contract, all software updates for the respective product are made available to you free of charge for a period of one year from the invoice date. The contract will automatically be extended for one year if it is not canceled three months before it expires.

The possession of the current version of the respective software is a basic condition for entering into an SUS contract.

You can download explanations concerning license conditions from https://mall.industry.siemens.com/legal/ww/en/terms_of_trade_en.pdf

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Setup texts and software update services

Overview

The "General License Conditions for Software Products for Automation and Drives" are applicable for supplies and deliveries of DI software products.

Legal notes during setup for new software products

All software products feature a uniform reference to the license conditions. The license conditions are enclosed either with the documentation or in the software pack. When software is downloaded from the Internet, the license contract is displayed before the ordering procedure and must be accepted by the user before downloading can continue.

Notice:
This software is protected by German and/or US copyright laws and the regulations of international agreements. Unauthorized reproduction or sale of this software or parts of it is a criminal offense. This will lead to criminal and civil prosecution, and may result in significant fines and/or claims for damages. Prior to installing and using the software, please read the applicable license conditions for this software. You will find these in the documentation or packaging.

If you have received this software on a CD-ROM that is marked Trial version, or accompanying software that is licensed for your use, the software is only permitted to be used for test and validation purposes in accordance with the accompanying conditions for the trial license. To this end, it is necessary for programs, software libraries, etc. are installed on your computer. We therefore urgently recommend that installation is performed on a single-user computer or on a computer that is not used in the production process or for storing important data, since it cannot be completely excluded that existing files will be modified or overwritten. We accept no liability whatsoever for damage and/or data losses that result from this installation or the non-observance of this warning. Every other type of use of this software is only permitted if you are in possession of a valid license from Siemens is obtained.

If you are not in possession of a valid license that can be proven by presenting an appropriate Certificate of License/software product certificate, please abort installation immediately and contact a Siemens office without delay to avoid claims for damages.

Software update services

Order
To order the software update service, an article number must be specified. The software update service can be ordered when the software products are ordered or at a later date. Subsequent orders require that the ordering party is in possession at least of a single license.

Note:
It is recommended that the software update service is ordered as early as possible. If a new software version of a software product is released for delivery by Siemens, only those customers will receive it automatically who are entered in the appropriate delivery list at Siemens at this time. Previous software versions, or the current software version are not supplied when the software update service is ordered. The software update service requires that the software product is up-to-date at the time of completion of the contract for the software update service.

Delivery
When a software update service is ordered, you will be sent the contractual conditions of this service and the price is due for payment. At the same time, you will be included in a delivery list for the software product to be updated. If Siemens releases a new software version for the corresponding software product for general sale (function version or product version), it will be delivered automatically to the goods recipient specified in the delivery address within the contract period.

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Explanation of the raw material/metal surcharges[1)]

Surcharge calculation

To compensate for variations in the price of the raw materials silver, copper, aluminum, lead, gold, dysprosium[2)] and/or neodym[2)], surcharges are calculated on a daily basis using the so-called metal factor. These apply to products containing these raw materials and are calculated per raw material. These surcharges are added to the price of a product if the basic official price (BOP) of the raw material in question is exceeded.

Surcharges are calculated in accordance with the following criteria:

Basic official price (BOP) of the raw material

- BOP of the workday prior to receipt of the order or prior to release order (Daily Price) for:
  - Silver (processed) [3)]
  - Gold (processed) [3)]
  - Aluminum per 100 kg (LME-notation/10, converted from USD to EUR using LME-FX-Rate [MTLE] +1.2%) [4)]
  - Copper per 100 kg (LME-notation/10, converted from USD to EUR using LME-FX-Rate [MTLE] +1.2%) + 1% [4)]
  - Lead (constant 199.50 EUR per 100 kg)
- If BOP is suspended, the last one is used.

Metal factor of the products

Certain products are displayed with a metal factor. The metal factor determines the official price (for those raw materials concerned) as of which the metal surcharges are applied and the calculation method used (weight or percentage method). An exact explanation is given below.

Structure of the metal factor

Metal factor consists of several digits; the first digit indicates whether the percentage method of calculation refers to the list price or a discounted price (customer net price) (L = list price / N = customer net price).

The remaining digits indicate the method of calculation used for the respective raw material. If no surcharge is added for a raw material, a "-" is used.

1st digit	List or customer net price using the percentage method
2nd digit	for silver (AG)
3rd digit	for copper (CU)
4th digit	for aluminum (AL)
5th digit	for lead (PB)
6th digit	for gold (AU)
7th digit	for dysprosium (Dy)[2)]
8th digit	for neodym (Nd)[2)]

Weight method

The weight method uses the BOP, the daily price and the raw material weight. In order to calculate the surcharge, the BOP must be subtracted from the daily price. The difference is then multiplied by the raw material weight.

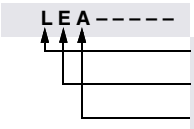
The BOP can be found in the table below using the number (1 to 9) of the respective digit of the metal factor. The raw material weight can be found in the respective product descriptions.

Percentage method

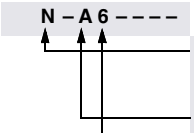
Use of the percentage method is indicated by the letters A-Z at the respective digit of the metal factor.

The surcharge is increased - dependent on the deviation of the daily price compared with the BOP - using the percentage method in "steps" and consequently offers surcharges that remain constant within the framework of this "step range". A higher percentage rate is charged for each new step. The respective percentage level can be found in the table below.

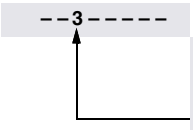
Metal factor examples



Basis for % surcharge: List price
Silver Basis 150 €, Step 50 €, 0.5 %
Copper Basis 150 €, Step 50 €, 0.1 %
No surcharge for aluminum
No surcharge for lead
No surcharge for gold
No surcharge for dysprosium
No surcharge for neodym



Basis for % surcharge: Customer net price
No surcharge for silver
Copper Basis 150 €, Step 50 €, 0.1 %
Aluminum acc. to weight, basic offic. price 225 €
No surcharge for lead
No surcharge for gold
No surcharge for dysprosium
No surcharge for neodym



No basis necessary
No surcharge for silver
Copper acc. to weight, basic official price 150 €
No surcharge for aluminum
No surcharge for lead
No surcharge for gold
No surcharge for dysprosium
No surcharge for neodym

1) Refer to the separate explanation on the next page regarding the raw materials dysprosium and neodym (= rare earths).

2) For a different method of calculation, refer to the separate explanation for these raw materials on the next page.

3) Source: Umicore, Hanau (https://pmm.umicore.com/en).

4) Source: The London Metal Exchange – an HKEX Company (https://www.lme.com/).
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Explanation of the raw material/metal surcharges for dysprosium and neodym (rare earths)

Surcharge calculation

To compensate for variations in the price of the raw materials silver[1], copper[1], aluminum[1], lead[1], gold[1], dysprosium and/or neodym, surcharges are calculated on a daily basis using the so-called metal factor. This applies to products containing at least one of these raw materials. The surcharge for dysprosium and neodym is calculated as a supplement to the price of a product if the basic official price (BOP) of the raw material in question is exceeded.

The surcharge is calculated in accordance with the following criteria:

- Basic official price (BOP) of the raw material[2]
  Three-month basic average price (see below) in the period before the quarter in which the order was received or the release order took place (average official price) for
  - Dysprosium (Dy metal, 99 %min. FOB China; USD/kg)
  - Neodym (Nd metal, 99 %min. FOB China; USD/kg)
- Metal factor of the products
  Certain products are displayed with a metal factor. The metal factor indicates (for those raw materials concerned) the BOP as of which the surcharges for dysprosium and neodym are calculated using the weight method. An exact explanation of the metal factor is given below.

Three-month average price

The prices of rare earths vary according to the foreign currency, and there is no freely accessible stock exchange listing. This makes it more difficult for all parties involved to monitor changes in price. In order to avoid continuous adjustment of the surcharges, but to still ensure fair, transparent pricing, an average price is calculated over a three-month period using the average monthly foreign exchange rate from USD to EUR (source: European Central Bank). Since not all facts are immediately available at the start of each month, a one-month buffer is allowed before the new average price applies.

Examples of calculation of the average official price:

Period for calculation of the average price:	Period during which the order/release order is effected and the average price applies:
Sep 2012 - Nov 2012	Q1 in 2013 (Jan - Mar)
Dec 2012 - Feb 2013	Q2 in 2013 (Apr - Jun)
Mar 2013 - May 2013	Q3 in 2013 (Jul - Sep)
Jun 2013 - Aug 2013	Q4 in 2013 (Oct - Dec)

Structure of the metal factor

The metal factor consists of several digits; the first digit is not relevant to the calculation of dysprosium and neodym.

The remaining digits indicate the method of calculation used for the respective raw material. If no surcharge is added for a raw material, a "-" is used.

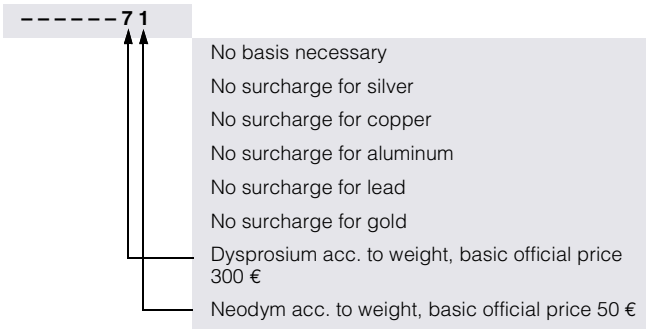
1st digit	List or customer net price using the percentage method
2nd digit	for silver (AG)[1]
3rd digit	for copper (CU)[1]
4th digit	for aluminum (AL)[1]
5th digit	for lead (PB)[1]
6th digit	for gold (AU)[1]
7th digit	for dysprosium (Dy)
8th digit	for neodym (Nd)

Weight method

The weight method uses the basic official price, the average price and the raw material weight. In order to calculate the surcharge, the BOP must be subtracted from the average price. The difference is then multiplied by the raw material weight.

The basic official price can be found in the table below using the number (1 to 9) of the respective digit of the metal factor. Your Sales contact can inform you of the raw material weight.

Metal factor examples



[1] For a different method of calculation, refer to the separate explanation for these raw materials on the previous page.
[2] Source: Asian Metal Ltd (www.asianmetal.com)

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Values of the metal factor

Percentage method	Basic official price in €	Step range in €	% surcharge 1st step	% surcharge 2nd step	% surcharge 3rd step	% surcharge 4th step	% surcharge per additional step
			Price in €	Price in €	Price in €	Price in €	
			150.01 - 200.00	200.01 - 250.00	250.01 - 300.00	300.01 - 350.00	
A	150	50	0.1	0.2	0.3	0.4	0.1
B	150	50	0.2	0.4	0.6	0.8	0.2
C	150	50	0.3	0.6	0.9	1.2	0.3
D	150	50	0.4	0.8	1.2	1.6	0.4
E	150	50	0.5	1.0	1.5	2.0	0.5
F	150	50	0.6	1.2	1.8	2.4	0.6
G	150	50	1.0	2.0	3.0	4.0	1.0
H	150	50	1.2	2.4	3.6	4.8	1.2
I	150	50	1.6	3.2	4.8	6.4	1.6
J	150	50	1.8	3.6	5.4	7.2	1.8
			175.01 - 225.00	225.01 - 275.00	275.01 - 325.00	325.01 - 375.00	
O	175	50	0.1	0.2	0.3	0.4	0.1
P	175	50	0.2	0.4	0.6	0.8	0.2
R	175	50	0.5	1.0	1.5	2.0	0.5
			225.01 - 275.00	275.01 - 325.00	325.01 - 375.00	375.01 - 425.00	
S	225	50	0.2	0.4	0.6	0.8	0.2
U	225	50	1.0	2.0	3.0	4.0	1.0
V	225	50	1.0	1.5	2.0	3.0	1.0
W	225	50	1.2	2.5	3.5	4.5	1.0
			150.01 - 175.00	175.01 - 200.00	200.01 - 225.00	225.01 - 250.00	
Y	150	25	0.3	0.6	0.9	1.2	0.3
			400.01 - 425.00	425.01 - 450.00	450.01 - 475.00	475.01 - 500.00	
Z	400	25	0.1	0.2	0.3	0.4	0.1
	Price basis (1st digit)						
L	Calculation based on the list price						
N	Calculation based on the customer net price (discounted list price)						
Weight method	**Basic official price in €**						
1	50	Calculation based on raw material weight					
2	100						
3	150						
4	175						
5	200						
6	225						
7	300						
8	400						
9	555						
Miscellaneous							
-	No metal surcharge						

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Notes

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1. General Provisions

1.1 Scope

The purchase of hardware, software, cloud services and other services (collectively “Offerings”) through this catalog from Siemens Aktiengesellschaft (“Siemens”) is subject to the following General Terms and Conditions for the sale of Siemens Offerings ("Catalog T&C"). These Catalog T&C apply only for orders placed with Siemens in Germany. Customer means the entity that places an order via Siemens.

1.2 Order Process

If Customer writes something in the "free choice text" (F), it is solely for Customer’s internal order processing and is not reviewed or taken into consideration by Siemens and will not be legally binding.

1.3 Formation of contract

Any information provided in this catalog or in SiePortal does not constitute a binding offer and may contain errors. In particular, illustrations contained in the Offering descriptions, or in Siemens’ catalogs are not binding. Any data, statement to dimensions and weights in our Offering descriptions and catalogs are subject to change without prior notice and become only binding upon contract conclusion.

An individual contract between Siemens and Customer is established when Siemens expressly accepts an order from Customer electronically or in another form, or if such acceptance does not occur, when Siemens fulfills Customer's order through delivery or performance. When the order is placed in SiePortal Siemens will send Customer an electronic confirmation of the receipt of the order, which, however, does not constitute an acceptance of the order. Neither is the Customer obligated to place an order, nor is Siemens obligated to accept any orders placed by Customer.

1.4 Additional Terms

Subordinately to these Catalog T&C, the specific terms and conditions in the text of the product description (if available) apply. If the text of the product description says that the specific terms and conditions mentioned there apply exclusively, only those specific terms and conditions apply. If the text of the product description does not contain any specific terms and conditions or if the specific terms and conditions mentioned there do not apply exclusively, the following terms and conditions apply subordinately in the version valid at the time of contract conclusion:

1.4.1 Hardware

a. "General Conditions for the Supply of Products and Services of the Electrical and Electronics Industry (“Grüne Lieferbedingungen”- GL)"[1] and
b. the Siemens' "Terms on Export Control and Electronically Order Processing"[1].

If the hardware contains open-source software those terms prevail over the GL, we will provide a notification with the hardware specifying the applicable terms. The same applies with respect to any third-party software components in the hardware.

1.4.2 Other Offerings than hardware

a. for software on premise including any associated maintenance and support services:
the "General Software and Cloud Supplemental Terms (AGB Germany)"[2] and the ""Specific Supplemental Terms for on-premises Software of Infrastructure & Industry Business"[2];
b. for cloud services:
the "General Software and Cloud Supplemental Terms (AGB Germany)"[2] and the cloud services product specific terms stated in the product description;
c. for services:
the "Services Supplemental Terms (AGB Germany)"[2]; and
d. generally for all Offerings under a. - c.:
"Base Terms Germany (AGB Germany)"[2].

1.5 Customers with master or frame agreement

To the extent Offerings purchased are covered by an existing master or frame agreement, the terms and conditions of that agreement shall apply instead of these Catalog T&C.

1.6 Payment agreement and data sharing

If Customer can select and selects that a payment agreement shall apply to the order, upon conclusion of the order the payment agreement will apply and prevail over these Catalog T&C.

For the assessment of financing options, Siemens shares information and data in relation to Customer's company obtained within the scope of its business relationship with its affiliates (as defined in the Base Terms Germany (AGB Germany)).

[1] Download under www.siemens.com/sts-base-terms-deu
[2] Download under www.siemens.com/sts-st-deu

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2. Prices, payment terms and price escalation

2.1 Prices

The prices are in € (Euro), EXW INCOTERMS®2020, exclusive packaging.

2.2 Taxes

All prices and expenses stated in the order are exclusive of any taxes, duties, tariffs and any other charges ("Tax"). Customer will pay or refund Siemens for any applicable Tax imposed by any government authority for Customer's use or receipt of the Offerings.

If Customer is exempt from value-added or sales tax, or similar taxes, it must provide a valid, timely, and executed exemption certificate, direct pay permit, or other such government-approved documentation.

If Customer is required by law to deduct or withhold Tax, Customer will increase the amount it pays to Siemens so that Siemens still receives the amount originally invoiced. Customer will promptly provide all tax receipts confirming it has paid Tax or has withheld Tax.

2.3 Price escalation

2.3.1 Price escalation for metal surcharges

To account for fluctuations commodity prices (silver, copper, aluminum, lead, gold, dysprosium and/or neodymium), surcharges will apply to Offerings containing these metals. Surcharges are calculated daily using the Offering's Metal Factor, which defines the applicable commodity and pricing methods. Details are provided in the exhibit "Metal surcharges"[3)]. The surcharges calculated by using these pricing methods will be added to the Offering's price.

If applicable, the Metal Factor is shown with the Offering's price information or listed on SiePortal (https://sieportal.siemens.com).

2.3.2 General price escalation

If, between ordering and delivery, there are any new or modified taxes, duties, tariffs, or equivalent measures that are directly or indirectly applicable to Siemens' offering, including any hardware, software, or service components contained therein, or a price increase of more than 20% for relevant commodities (e.g. electronic components, semiconductors) applies, Siemens reserves the right to adjust the price accordingly.

2.4 Price Changes

Prices stated in the SiePortal or in Siemens' catalogs are subject to change without prior notice.

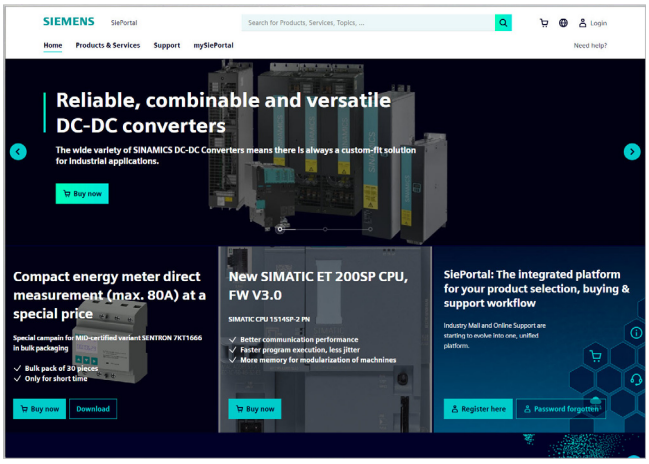
3. Units of measurement and weight

The dimensions and weight of all Offerings are given in meters and kilograms. If the dimensions or weight of the Offerings are specified in other units of measurement, such as inches or pounds, this indicates that these Offerings are intended for export.

3

[3)] Download under https://mall.industry.siemens.com/legal/de-en/LKB002966.pdf

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Easy product selection and ordering with SiePortal

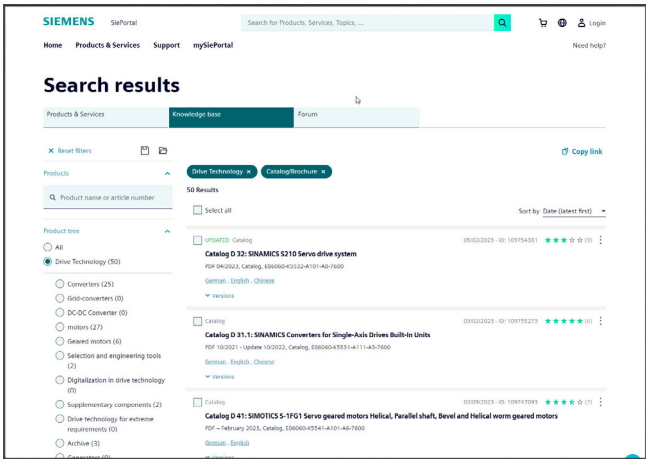
SiePortal > Products & Services

The internet ordering platform of Siemens AG is located in SiePortal. It provides you with online access to a comprehensive product spectrum that is presented in an informative, well-organized way.

Powerful search functions help you select the required products, while configurators enable you to configure complex product and system components quickly and easily. CAx data are also available for you to use.

Data transfer allows the entire procedure, from selection through ordering to tracking and tracing, to be carried out online. Availability checks, individual customer discounting, and quotation preparation are also possible.

https://sieportal.siemens.com



Downloading catalogs

SiePortal > Support > Knowledge base

You can download catalogs and brochures in PDF format from Siemens Industry Online Support without having to register.

The filter box makes it possible to perform targeted searches.

https://sieportal.siemens.com

Cybersecurity information

Siemens provides products and solutions with industrial cybersecurity functions that support the secure operation of plants, systems, machines and networks.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial cybersecurity concept. Siemens' products and solutions constitute one element of such a concept.

Customers are responsible for preventing unauthorized access to their plants, systems, machines and networks. Such systems, machines and components should only be connected to an enterprise network or the internet if and to the extent such a connection is necessary and only when appropriate security measures (e.g. firewalls and/or network segmentation) are in place.

For additional information on industrial cybersecurity measures that may be implemented, please visit **www.siemens.com/cybersecurity-industry**

Siemens' products and solutions undergo continuous development to make them more secure. Siemens strongly recommends that product updates are applied as soon as they are available and that the latest product versions are used. Use of product versions that are no longer supported, and failure to apply the latest updates may increase customer's exposure to cyber threats.

To stay informed about product updates, subscribe to the Siemens Industrial Cybersecurity RSS Feed under **www.siemens.com/cert**

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